数电期末复习

林平

第一章 数字逻辑基础

1.1 基本概念

1.1.2 数制转化

任意进制 ⟶ 十进制

$$(N)_r = \sum_{i=-m}^{n-1} k_i (r)^i = k_{n-1} r^{n-1} + k_{n-2} r^{n-2} + \cdots + k_1 r^1 + k_0 r^0 + \cdots + k_{-m} r^{-m}$$

如二进制数 $(1101.001)_2 = 1\times2^3+1\times2^2+0\times2^1+1\times2^0+0\times2^{-1}+0\times2^{-2}+1\times2^{-3}$

八进制数 $(357.61)_8 = 3\times8^2+5\times8^1+7\times8^0+6\times8^{-1}+1\times8^{-2}$

十进制 ⟶ r进制

整数部分 取r取余，余数依次往前写 ← 1 0 奇偶

如

2|175 …1
2|87 …1
2|43 …1
2|21 …1
2|10 …0
2|5 …1
2|2 …0
2|1 …1
0

8|175 …7
8|21 …5
8|2 …2
0

16|175 …F
16|10 …A
0

$(175)_{10} = (10101111)_2 = (257)_8 = (AF)_{16}$

小数部分 乘r看个位数字

如

0.125 × 2 = 0.25 …0
0.25 × 2 = 0.5 …0
0.5 × 2 = 1 …1

0.125 × 8 = 1 …1

0.125 × 16 = 2 …2

$(0.125)_{10} = (0.001)_2 = (0.1)_8 = (0.2)_{16}$

二、八、十六进制转化

二进制作桥梁

如 $(0A)_{16} = (1010)_2 = (12)_8$

$(22)_8 = (010010)_2 = (12)_{16}$

1.1.3 数字电路中的编码

原码：$(N)_{原} = \begin{cases}[0]\ (原数) & 正\\ [1]\ (原数) & 负\end{cases}$

$(+10010101)_2 = (0\ 10010101)_{原}$

$(-10010101)_2 = (1\ 10010101)_{原}$

反码：$(N)_{反} = \begin{cases}[0]\ (原数) & 正\\ [1]\ (原数按位取反) & 负\end{cases}$

$(+10010101)_2 = (0\ 10010101)_{反}$

$(-10010101)_2 = (10110\ 1010)_{反}$

补码 $(N)_{补}=\begin{cases}[0](原数) & 正\\ [1](原数按位取反加1)\ 反码+1 & 负\end{cases}$

$(+1001010)_2=(01001010)_{补}$

$(-1001010)_2=(11001110)_{补}$

$(X_1+X_2)_{补}=(X_1)_{补}+(X_2)_{补}$

$((X_1)_{补})_{补}=(X)_{原}$

$X_1-X_2=((X_1-X_2)_{补})_{补}=(X_1+(-X_2)_{补})_{补}$

$1001-1101=((1001)_{补}+(-1100)_{补})_{补}=(01001+10100)_{补}=(11101)_{补}=-3$

补：小数

反码 $\begin{cases}符号位+原数\\ 符号位+原数反码\end{cases}$

补码 $\begin{cases}符号位+原小数\\ 符号位+原小数补码\end{cases}$

x	$[x]_{原}$	$[x]_{反}$	$[x]_{补}$	x	$[x]_{原}$	$[x]_{反}$	$[x]_{补}$
+1010	01010	01010	01010	-1001	11001	10110	10111
+0100	00100	00100	00100	-0011	10011	11100	11101
+0.1001	0.1001	0.1001	0.1001	-0.1001	1.1001	1.0110	1.0111
+0.0000	0.0000	0.0000	0.0000	-0.0000	1.0000	1.1111	1.0000

编码

8421码+3

十进制数	有权码 8421	5421	2421	2421*	5211	无权码 余三码
0	0000	0000	0000	0000	0000	0011
1	0001	0001	0001	0001	0001	0100
2	0010	0010	0010	0010	0100	0101
3	0011	0011	0011	0011	0101	0110
4	0100	0100	0100	0100	0111	0111
5	0101	1000	0101	1011	1000	1000
6	0110	1001	0110	1100	1001	1001
7	0111	1010	0111	1101	1100	1010
8	1000	1011	1110	1110	1101	1011
9	1001	1100	1111	1111	1111	1100

$(120)_{10}=(0001\ 0010\ 0000)_{8421BCD}$

格雷码

1.1.4 数字电路基本功能

与 $L=A\cdot B$

或 $L=A+B$

非 $L=\overline{A}$

异或 相同 0 相异 1

A	B	L
0	0	0
0	1	1
1	0	1
1	1	0

$L=A\oplus B$

同或 相同 1 相异 0

A	B	L
0	0	1
0	1	0
1	0	0
1	1	1

$L=A\odot B$

1.2 逻辑代数

只有 加、乘、求反，没有 除/减

1.2.1 逻辑运算

(1) 运算定律

或	与	
$A+0=A$	$A\cdot 1=A$	0-1律
$A+1=1$	$A\cdot 0=0$	0-1律
$A+A=A$	$A\cdot A=A$	重叠率
$A+\bar{A}=1$	$A\cdot\bar{A}=0$	互补律
$A=\bar{\bar{A}}$		否定之否定
$A+B=B+A$	$A\cdot B=B\cdot A$	交换率
$(A+B)+C=A+(B+C)$	$(A\cdot B)\cdot C=A\cdot(B\cdot C)$	结合率
$A\cdot(B+C)=A\cdot B+A\cdot C$	$A+(B\cdot C)=(A+B)\cdot(A+C)$	分配律
$\overline{A+B+C}=\bar{A}\cdot\bar{B}\cdot\bar{C}$	$\overline{A\cdot B\cdot C}=\bar{A}+\bar{B}+\bar{C}$	摩根定律

(2) 常用公式

$A+AB=A$

$AB+\bar{A}B=B$

$A+\bar{A}B=A+B$ (分配率)

(3) 运算规律

① 代入规律　$A\to AB$

② 对偶规则　·换+，+换·　0换1，1换0　变换后 F'

③ 反演规则　·改+，+改·　0改1，1改0　原变量→反变量

若 $F_1=F_2$ 则 $F_1'=F_2'$

若 $L=A\cdot\bar{B}+CD$ 则 $\bar{L}=(A+B)\cdot(\bar{C}+\bar{D})$

最小项	A	B	C	最小项	最大项	A	B	C	最大项
$\bar{A}\bar{B}\bar{C}$	0	0	0	m_0	$A+B+C$	0	0	0	M_0
$\bar{A}\bar{B}C$	0	0	1	m_1	$A+B+\bar{C}$	0	0	1	M_1
$\bar{A}B\bar{C}$	0	1	0	m_2	$A+\bar{B}+C$	0	1	0	M_2
$\bar{A}BC$	0	1	1	m_3	$A+\bar{B}+\bar{C}$	0	1	1	M_3
$A\bar{B}\bar{C}$	1	0	0	m_4	$\bar{A}+B+C$	1	0	0	M_4
$A\bar{B}C$	1	0	1	m_5	$\bar{A}+B+\bar{C}$	1	0	1	M_5
$AB\bar{C}$	1	1	0	m_6	$\bar{A}+\bar{B}+C$	1	1	0	M_6
ABC	1	1	1	m_7	$\bar{A}+\bar{B}+\bar{C}$	1	1	1	M_7
为1					为0				

$M_i=\bar{m}_i$

$m_i\cdot m_j=0$　　$M_i+M_j=1$　　$(i\neq j)$

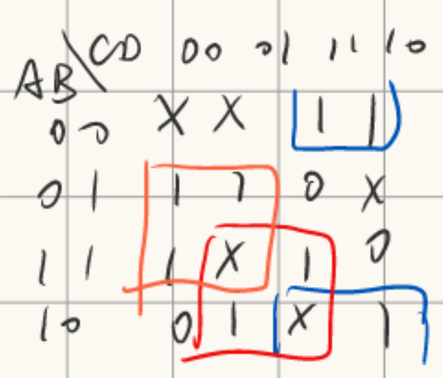
$\sum_{i=0}^{2^n-1} m_i=1$　　$\prod_{i=0}^{2^n-1} M_i=0$

1.3 逻辑函数化简

① 代数式化简

② 卡诺图化简

相邻方格最多只能一个变量不同



$Z = f(A, B, C, D) = B\bar{C} + \bar{B}C + CD$ （最简形式）

补充：1 bit 比特/位 0/1

1 Byte 字节 = 8 bit 8位二进制

$1kB = 2^{10} B = 1024B$

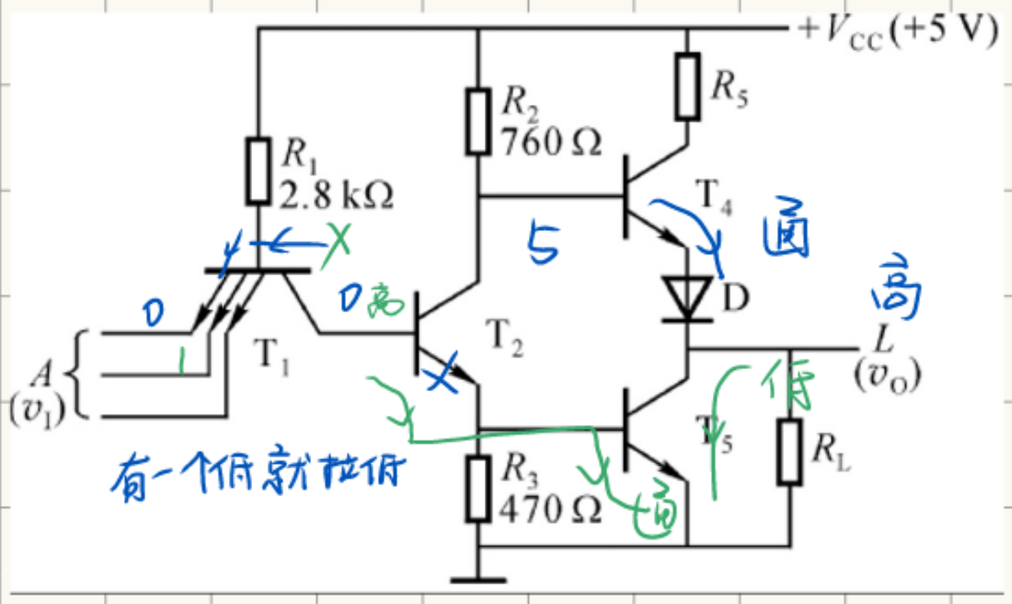
$1MB = 2^{20} B = 2^{10} kB$

bit → B → kB → MB → GB → TB → PB

第二章 集成逻辑门电路

2.1 集成门电路

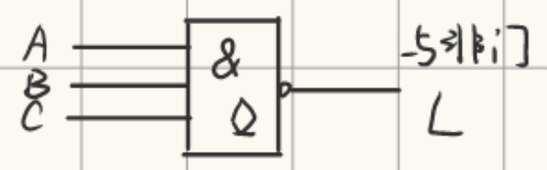
2.1.1 TTL



有一个低就拉低

① 输入有1个0 输出1

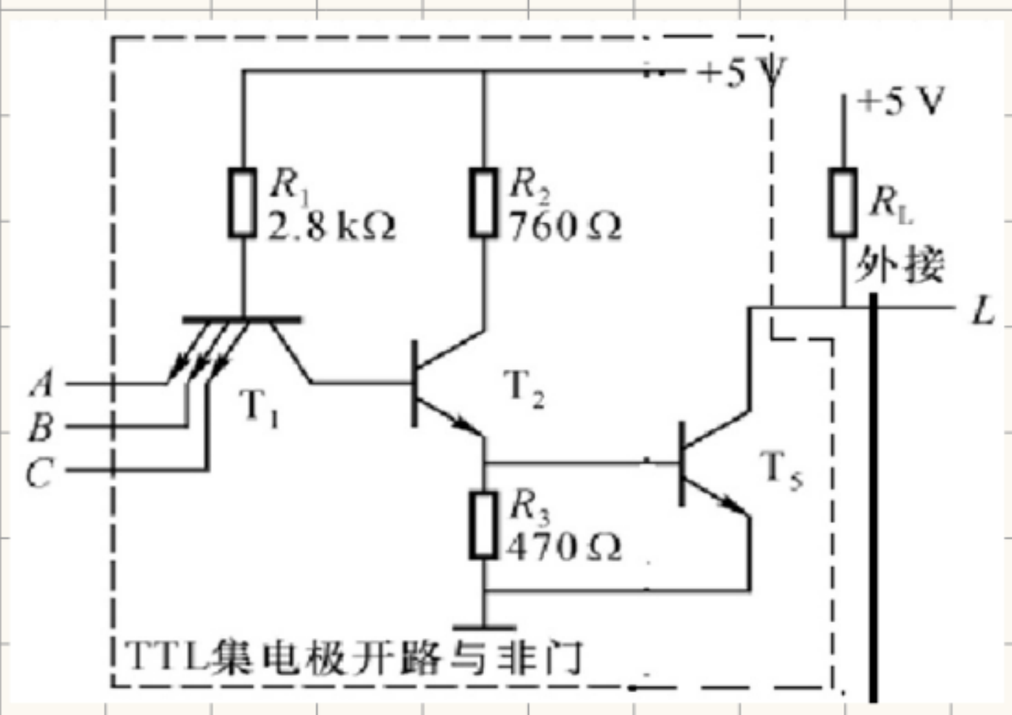
② 输入都为1 输出0



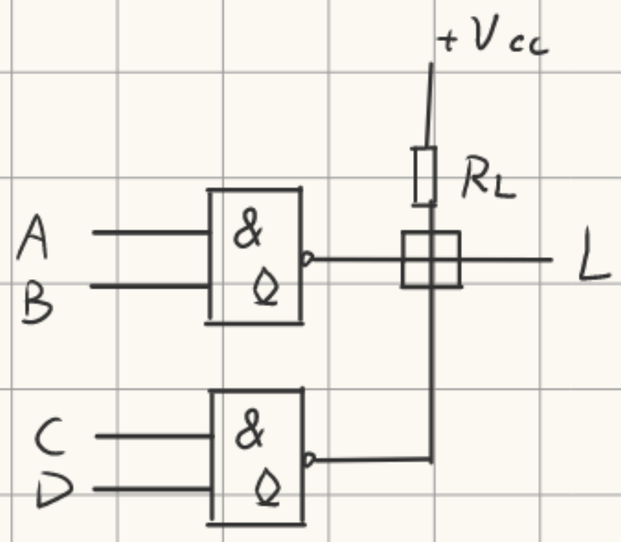
与非门

$L = \overline{ABC}$

去掉 R_5 T_4、D，形成 OC门。



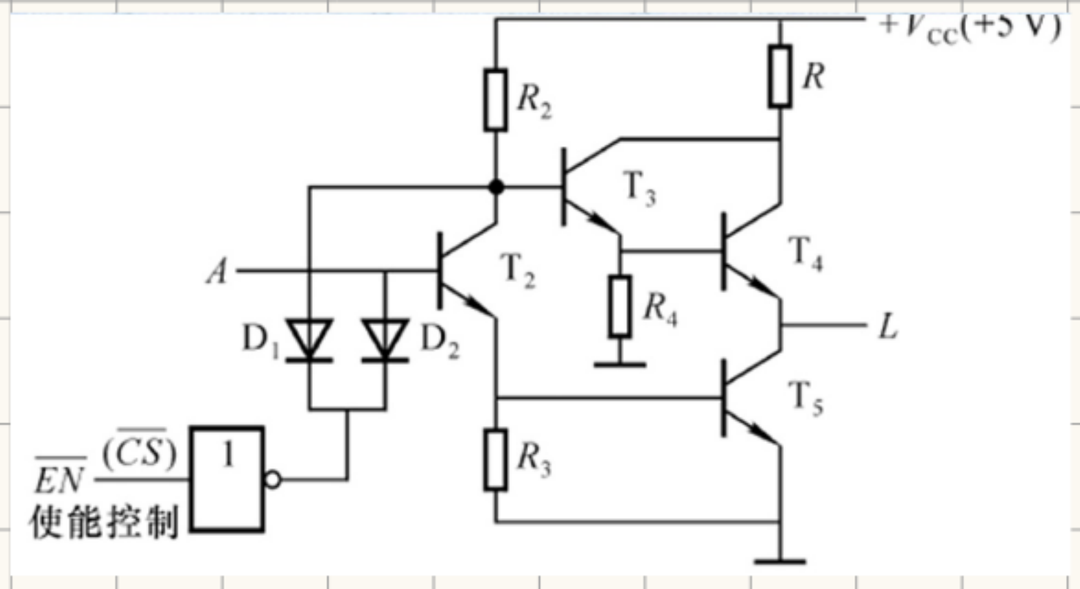
外接上 拉电阻 实现 线与(都为1才1)

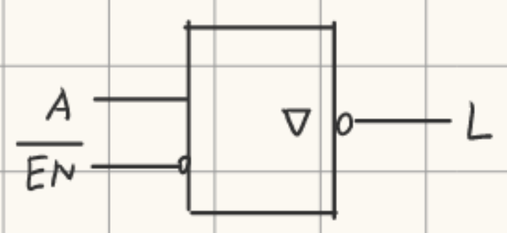


$L = \overline{A \cdot B} \cdot \overline{C \cdot D}$

$= \overline{AB + CD}$

TTL三态输出门



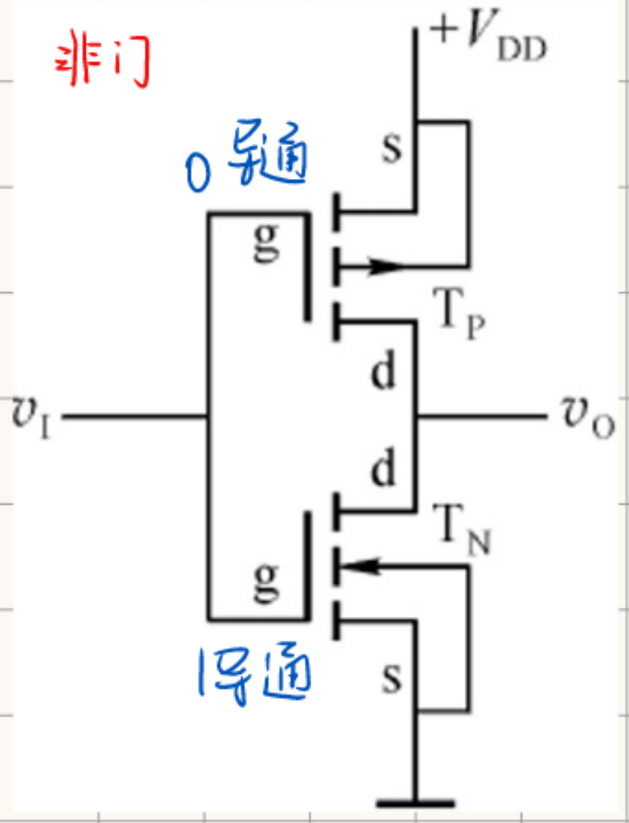


$\overline{EN}$	输入A	输出L
0	1	0
	0	1
1	0或1	高阻态

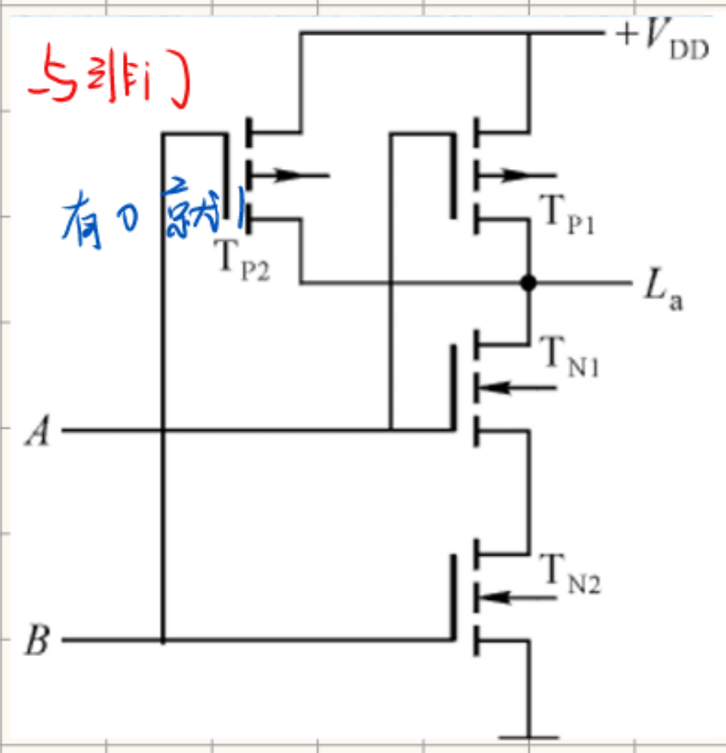
电位被拉低，都不导通

2.1.2 CMOS 集成逻辑门电路

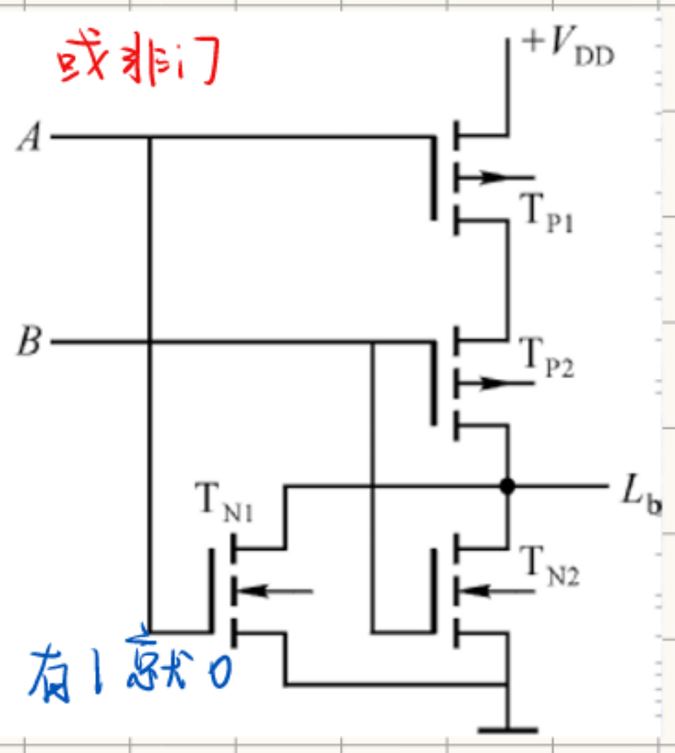
非门



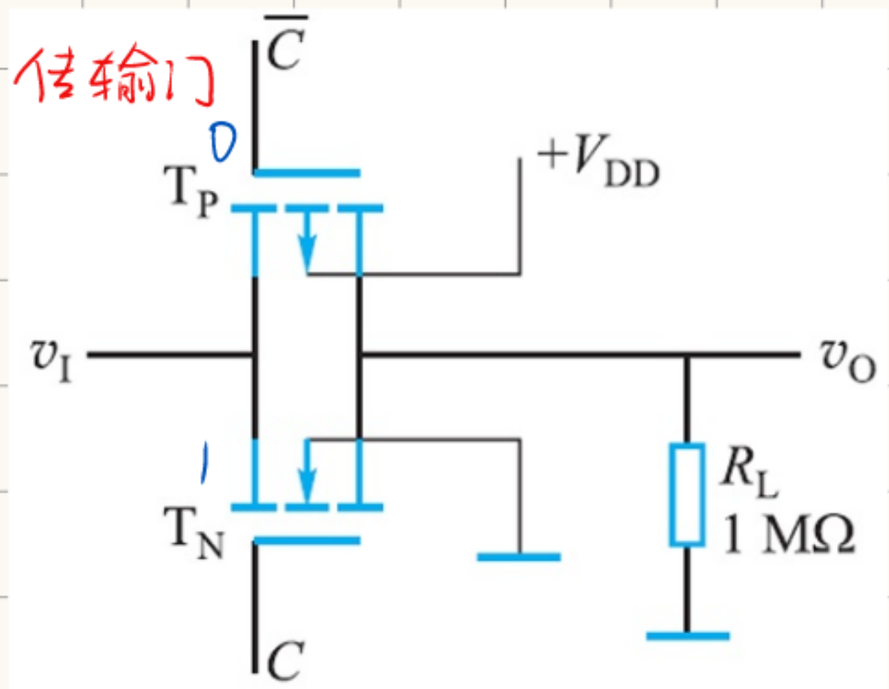
与非门

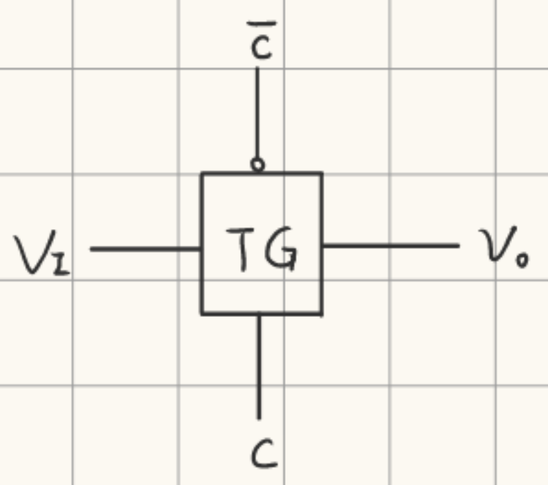


或非门



CMOS 传输门

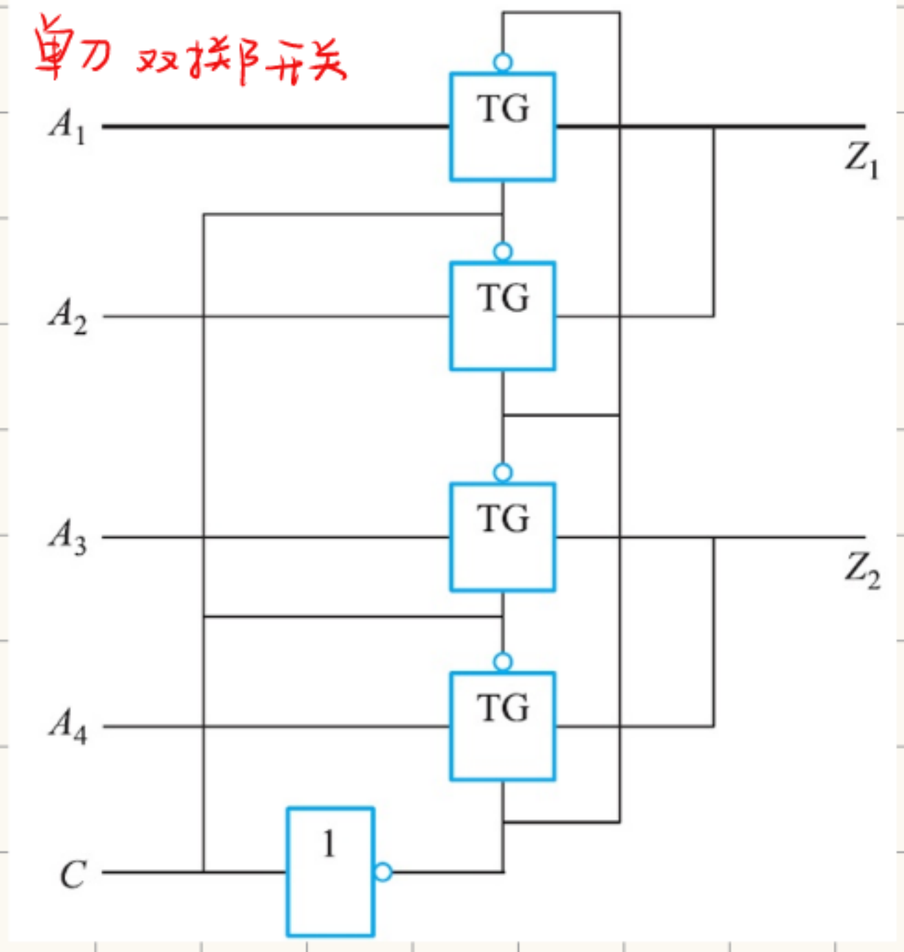




C=1 $V_o \approx V_I$

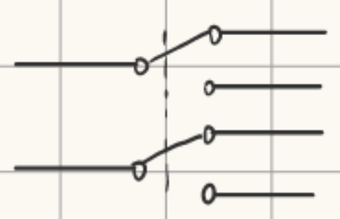
C=0 不传输

单刀双掷开关



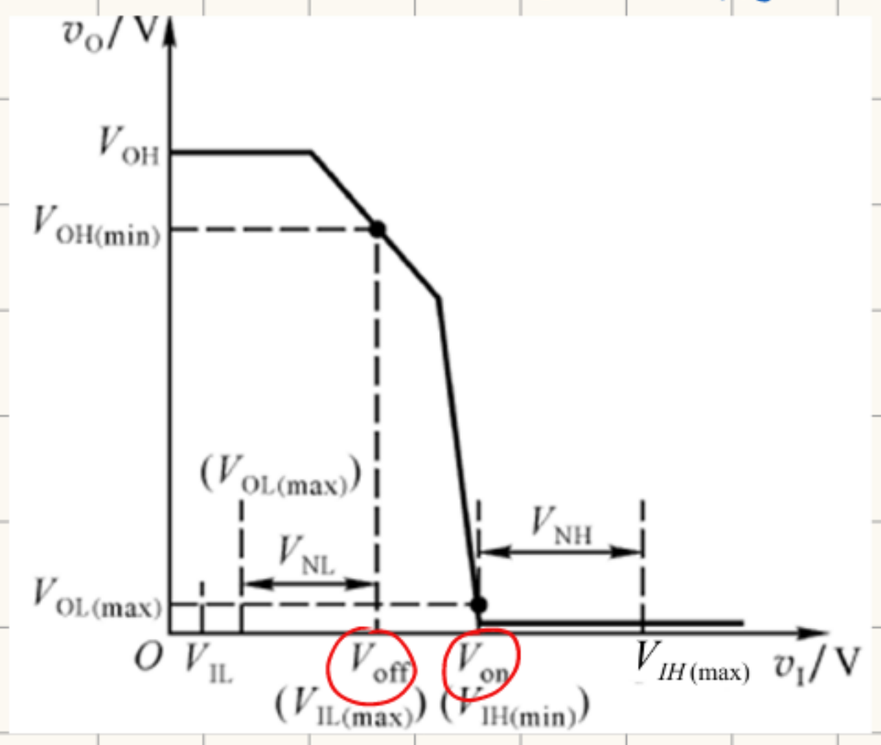
C=0 ⟶ $Z_1 = A_2$ $Z_2 = A_4$;

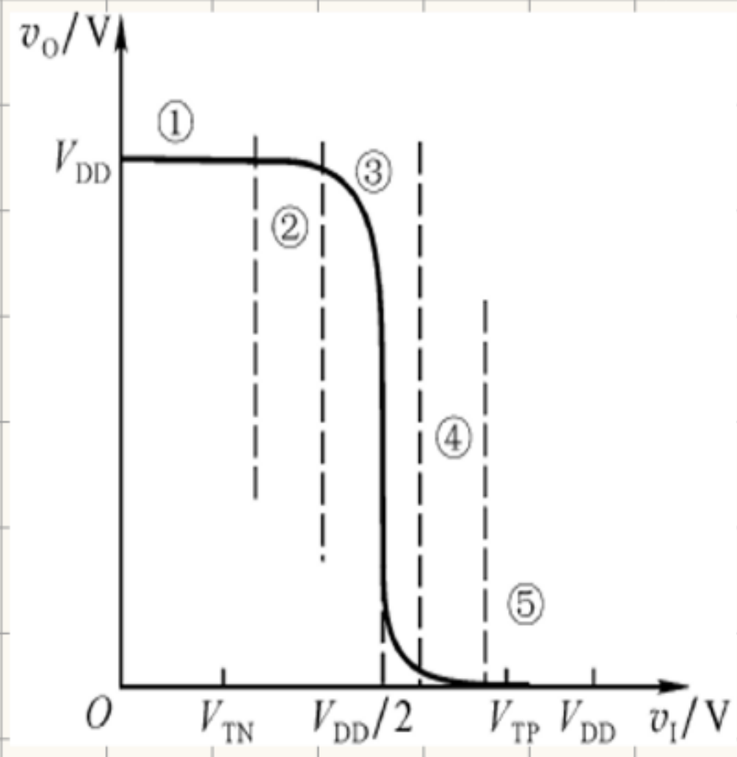
C=1 ⟶ $Z_1 = A_1$ $Z_2 = A_3$ 。



2.2 外部特性与指标

TTL 和 CMOS 电压传输特性曲线



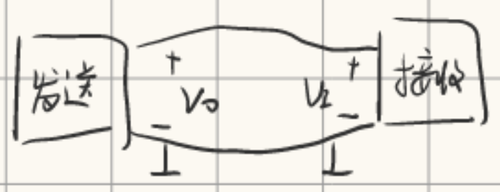


2.2.1 静态电平原则

输入低电平上限 $V_{IL\,max}$ 开门电平 V_{on}

输入高电平下限 $V_{IH\,min}$ 关门电平 V_{off}

中间值 V_T

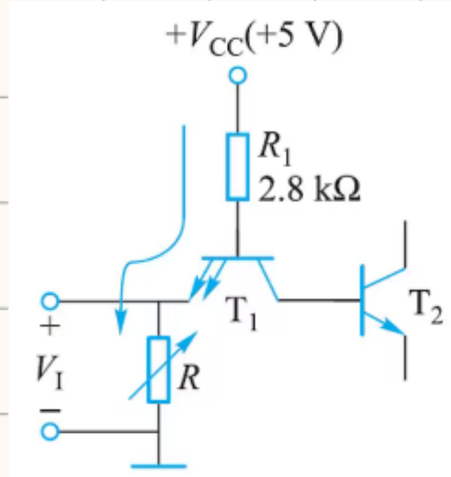


① 低电平输入噪声容限 $V_{NL} = V_{IL\,max} - V_{OL\,max}$ （认为低更高，发出低）

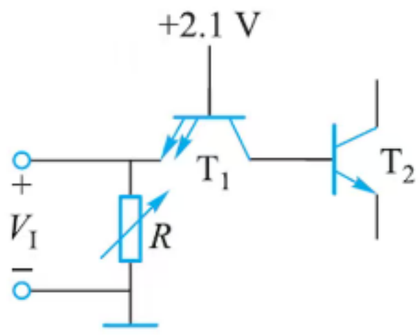
② 高电平输出噪声容限 $V_{NH} = V_{OH\,min} - V_{IH\,min}$ （发出高，认为高更低）

2.2.2 外部特性

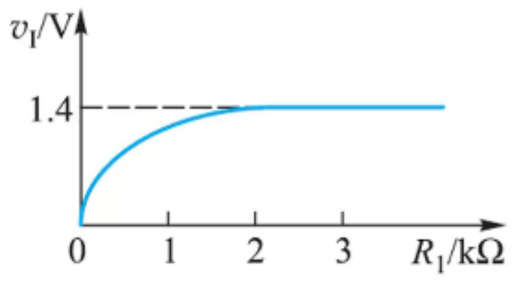
①输入输



(a) T_1管饱和导通

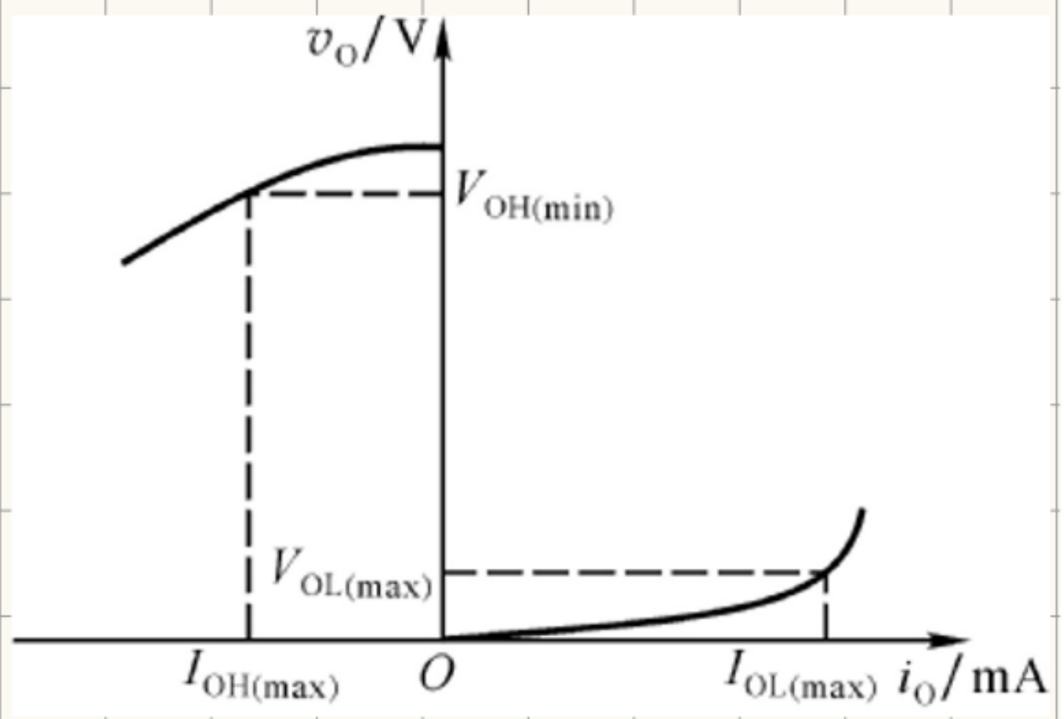


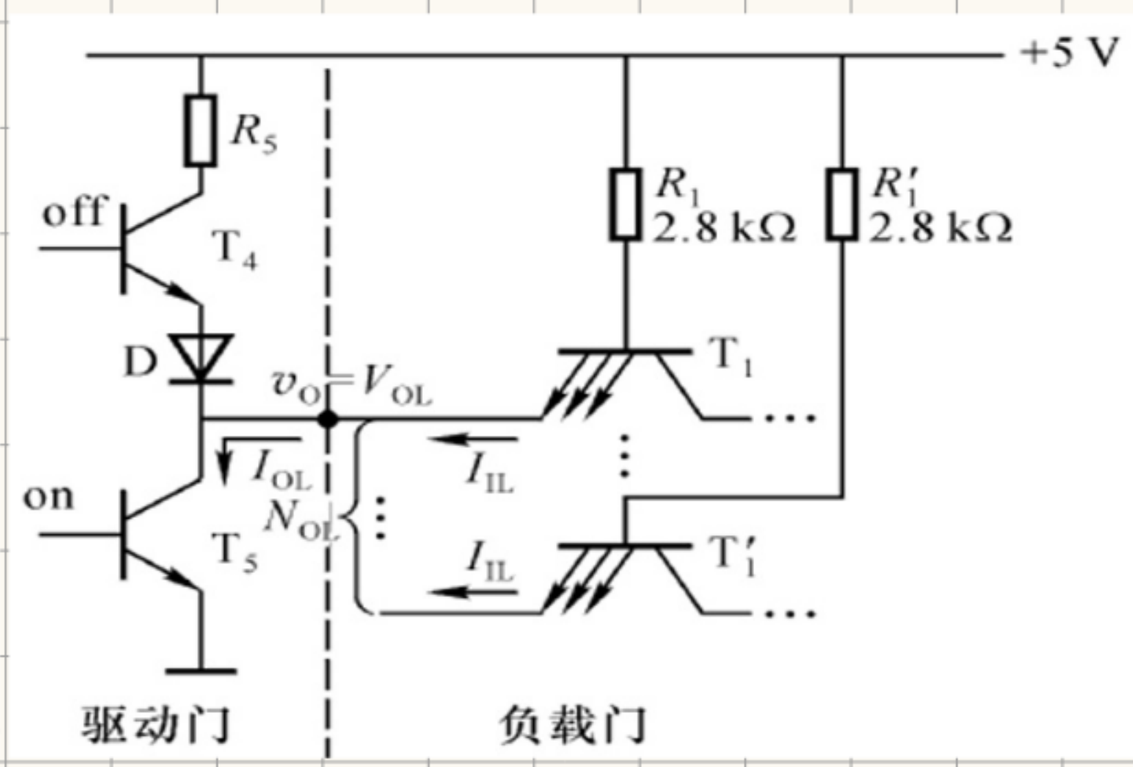
(b) T_1管倒置工作



(c) 输入端负载特性曲线

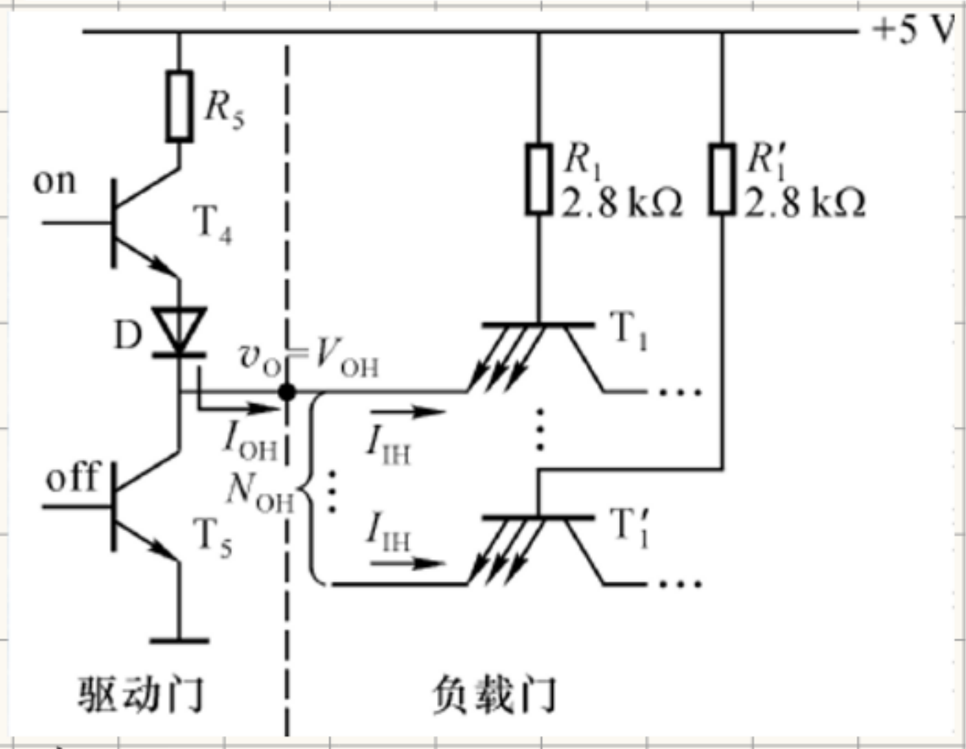
②输出负载





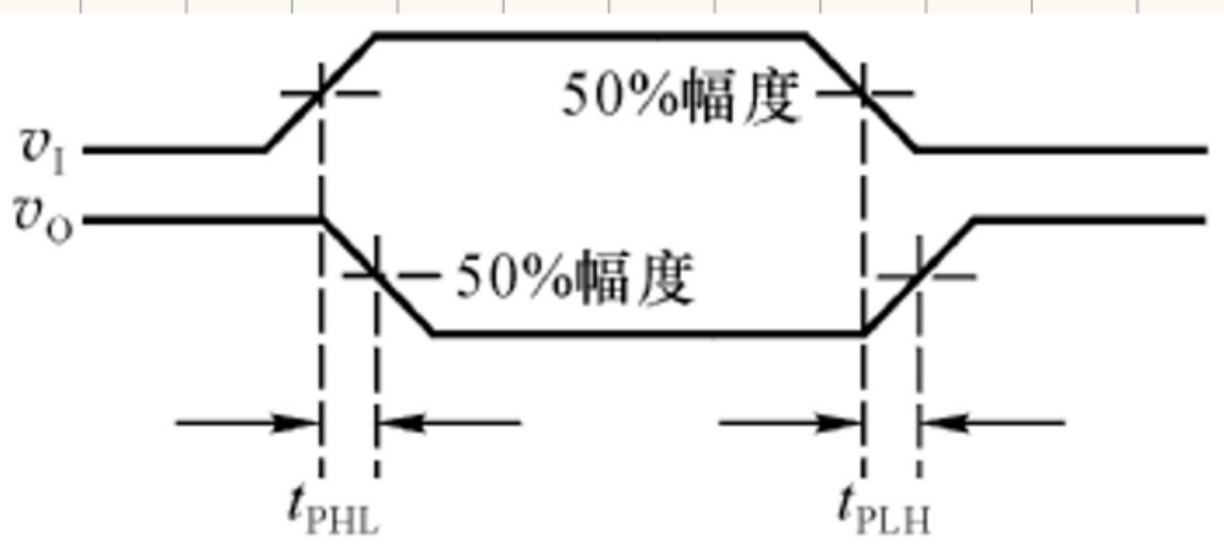
低电平输出特性 ——— 灌电流负载

扇出系数 $N_{OL} = \left.\frac{I_{OL\,max}}{I_{IL}}\right|_{V_{OL}\ max}$



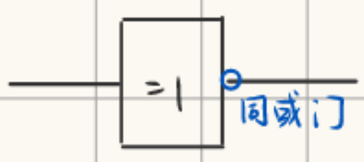
高电平输出特性 ——— 拉电流负载

$$N_{OH} = \left.\frac{I_{OH\,max}}{I_{IH}}\right|_{V_{OL}\ max}$$

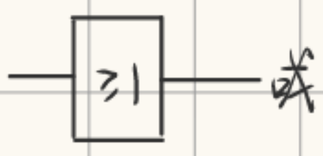


$$t_{pd} = \frac{1}{2}(t_{PHL} + t_{PLH})$$

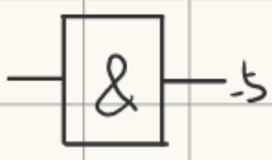
参数名称	CMOS（4000系列）	TTL（74LS系列）
$V_{OH(min)}$/V	4.6	2.7
$V_{OL(max)}$/V	0.05	0.5
$I_{OH(max)}$/mA	-0.51　4（HC系列列）	-0.4
$I_{OL(max)}$/mA	0.51　4（HC系列）	8
$V_{IH(min)}$/V	3.5	2
$V_{IL(max)}$/V	1.5	0.8
$I_{IH(max)}$/uA	0.1	20
$I_{IL(max)}$/mA	-0.0001	-0.4　-1.6(74系列)



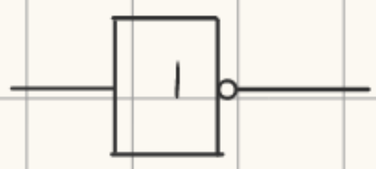
同或门　异或门



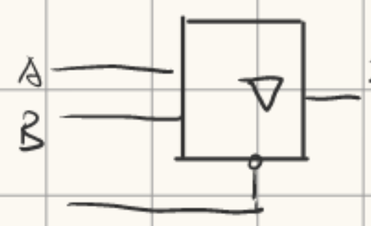
或



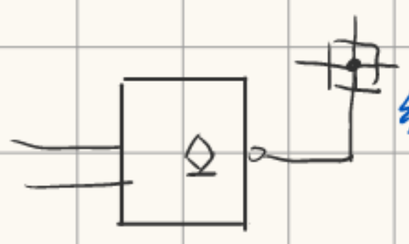
与



非



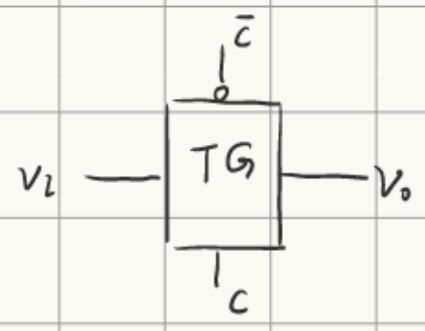
三态门 { TTL　接地 { 大电阻 1 / 小电阻 0
CMOS　输入电阻无穷，接地就是0



线与

与非门，输出可直接接一起，实现线与

OC/OD

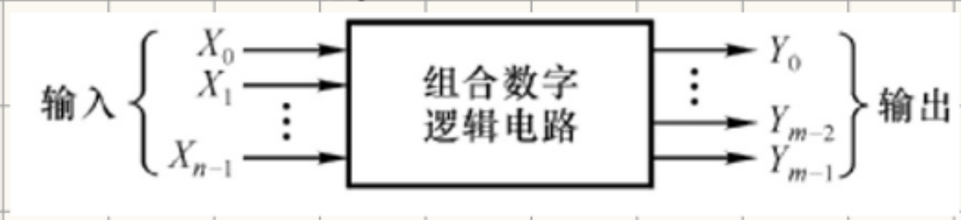


CMOS传输门

第三章 组合逻辑电路

3.1 基本组合逻辑电路分析与设计

3.1.1 结构功能



$$Y_0 = f_0(X_0, X_1, \cdots X_{n-1})$$

$$\vdots$$

$$Y_{m-1} = f_{m-1}(X_0, X_1, \cdots, X_{n-1})$$

3.1.2 分析

(1) 写逻辑表达式

(2) 真值表化简

(3) 确定逻辑功能

设计

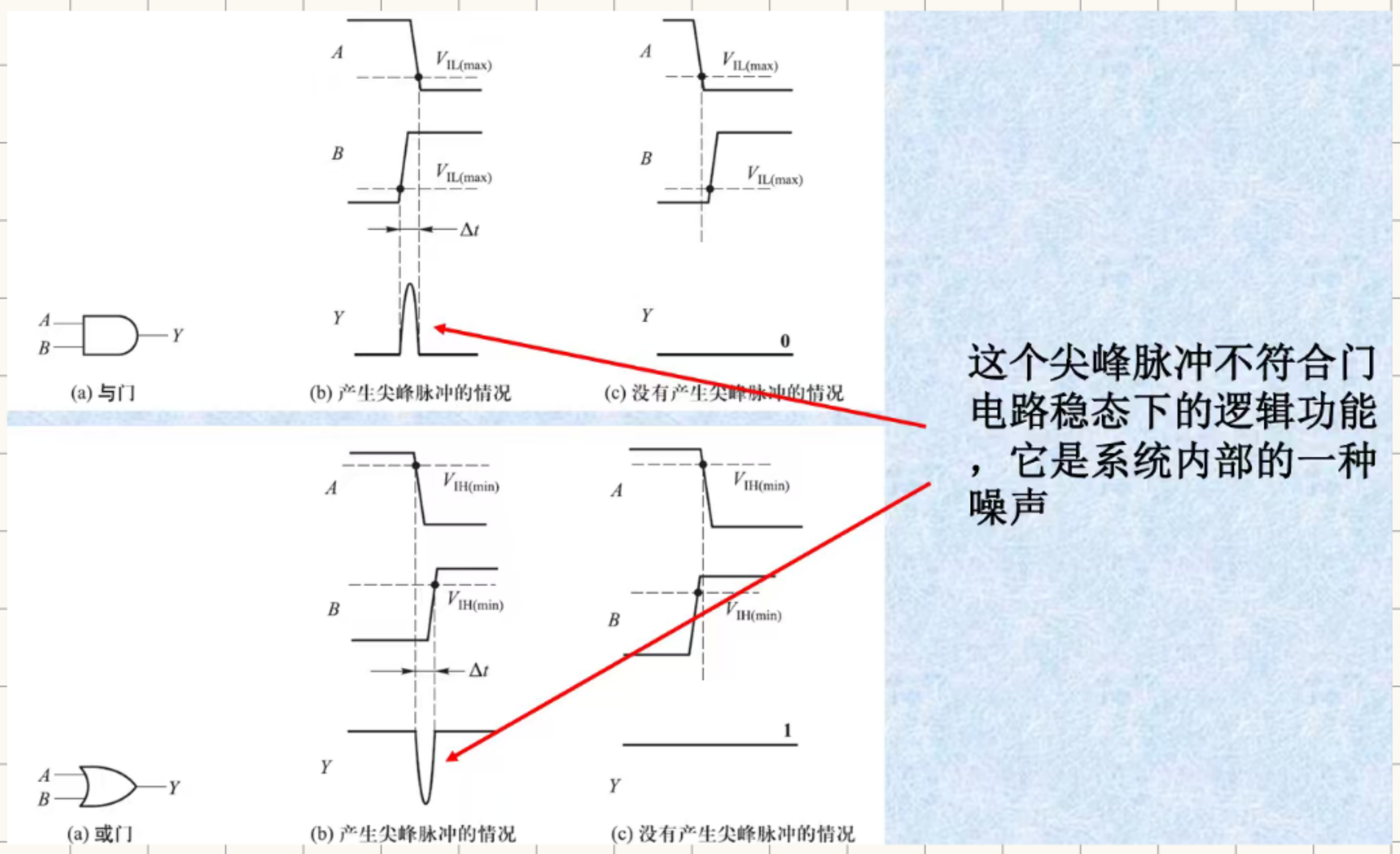
(1) 找输出输出变量及关系

(2) 列真值表

(3) 求各个输出最简与-或表达式

(4) 用逻辑门画出组合逻辑电路

3.1.3 竞争冒险



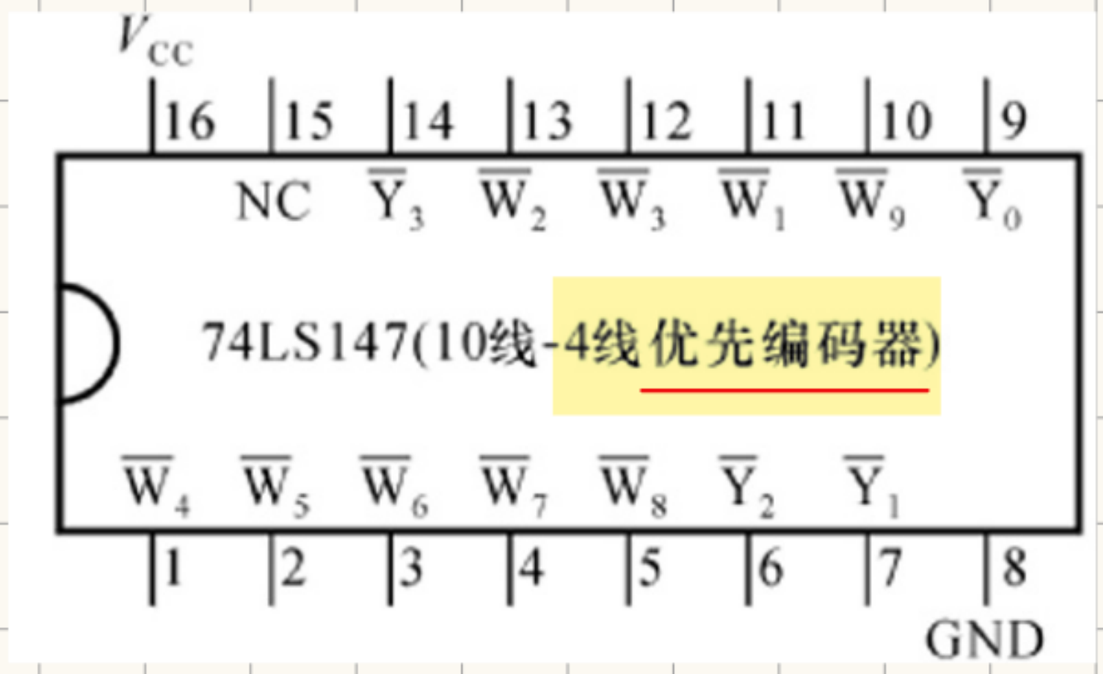
这个尖峰脉冲不符合门电路稳态下的逻辑功能，它是系统内部的一种噪声

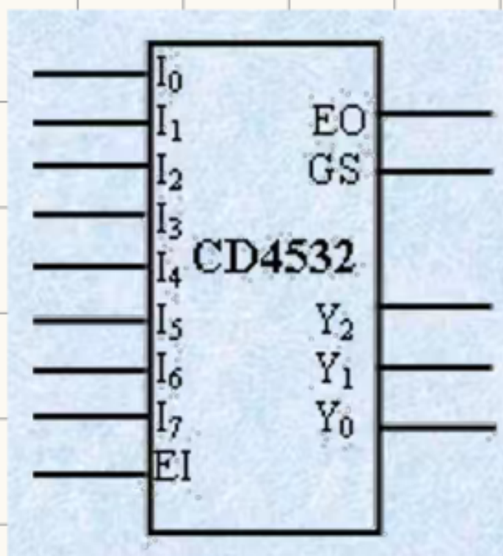
判别方法：

通过化简，赋值，若出现 $Y = A + \bar{A}$ 或 $Y = A \cdot \bar{A}$，则存在竞争冒险

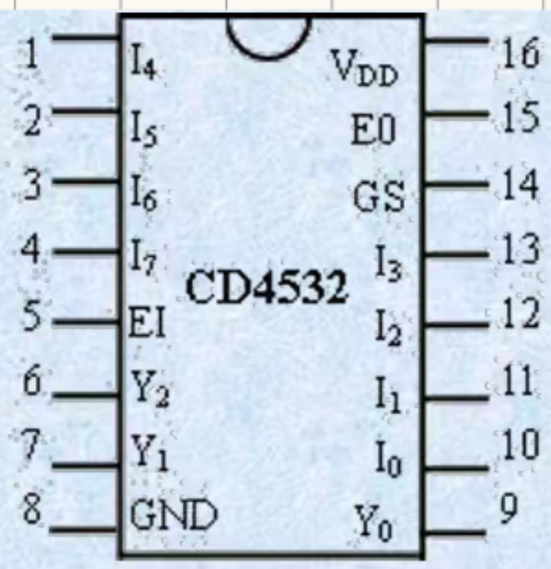
3.2 编码器和译码器

3.2.1 编码器





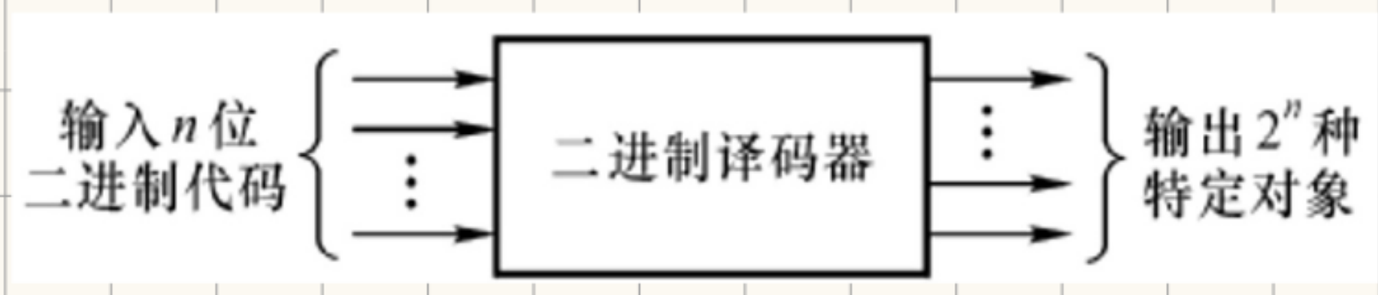
(a) 简化逻辑符号

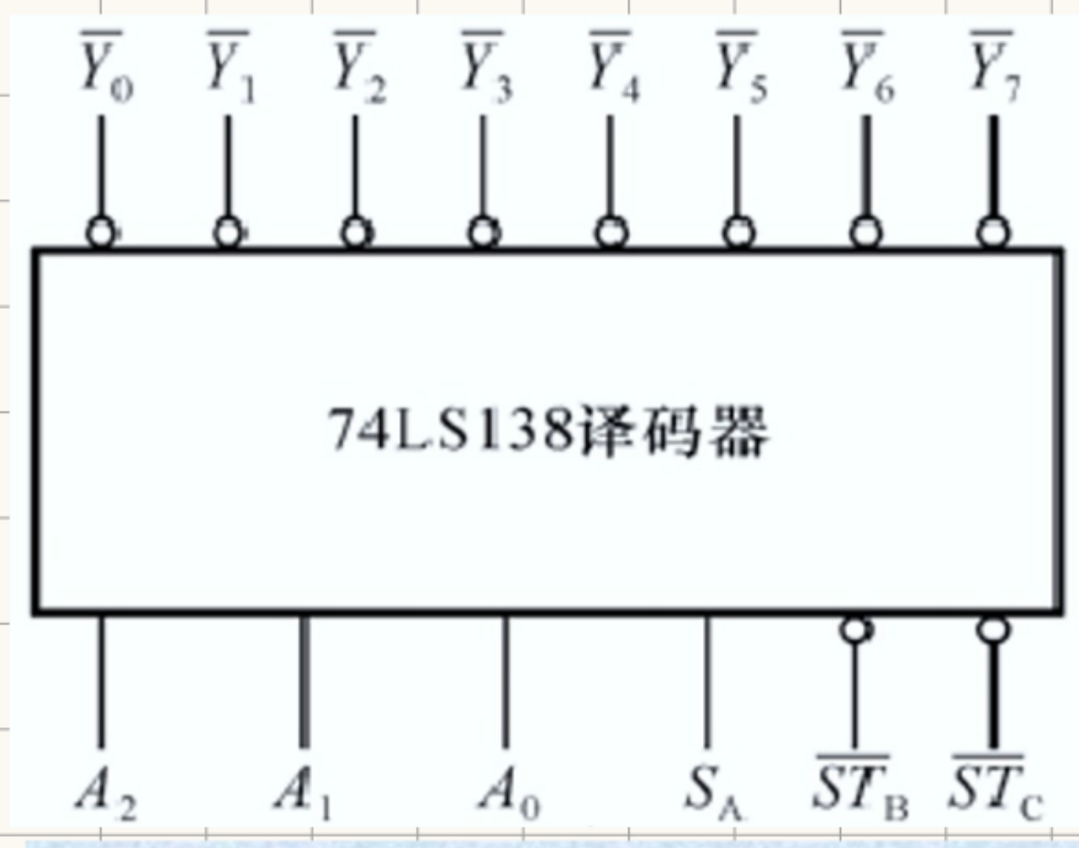


(b) 引脚排列图

编码器输入									代码和控制输出				
EI	I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	Y_2	Y_1	Y_0	GS	EO
0	×	×	×	×	×	×	×	×	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	1
1	1	×	×	×	×	×	×	×	1	1	1	1	0
1	0	1	×	×	×	×	×	×	1	1	0	1	0
1	0	0	1	×	×	×	×	×	1	0	1	1	0
1	0	0	0	1	×	×	×	×	1	0	0	1	0
1	0	0	0	0	1	×	×	×	0	1	1	1	0
1	0	0	0	0	0	1	×	×	0	1	0	1	0
1	0	0	0	0	0	0	1	×	0	0	1	1	0
1	0	0	0	0	0	0	0	1	0	0	0	1	0

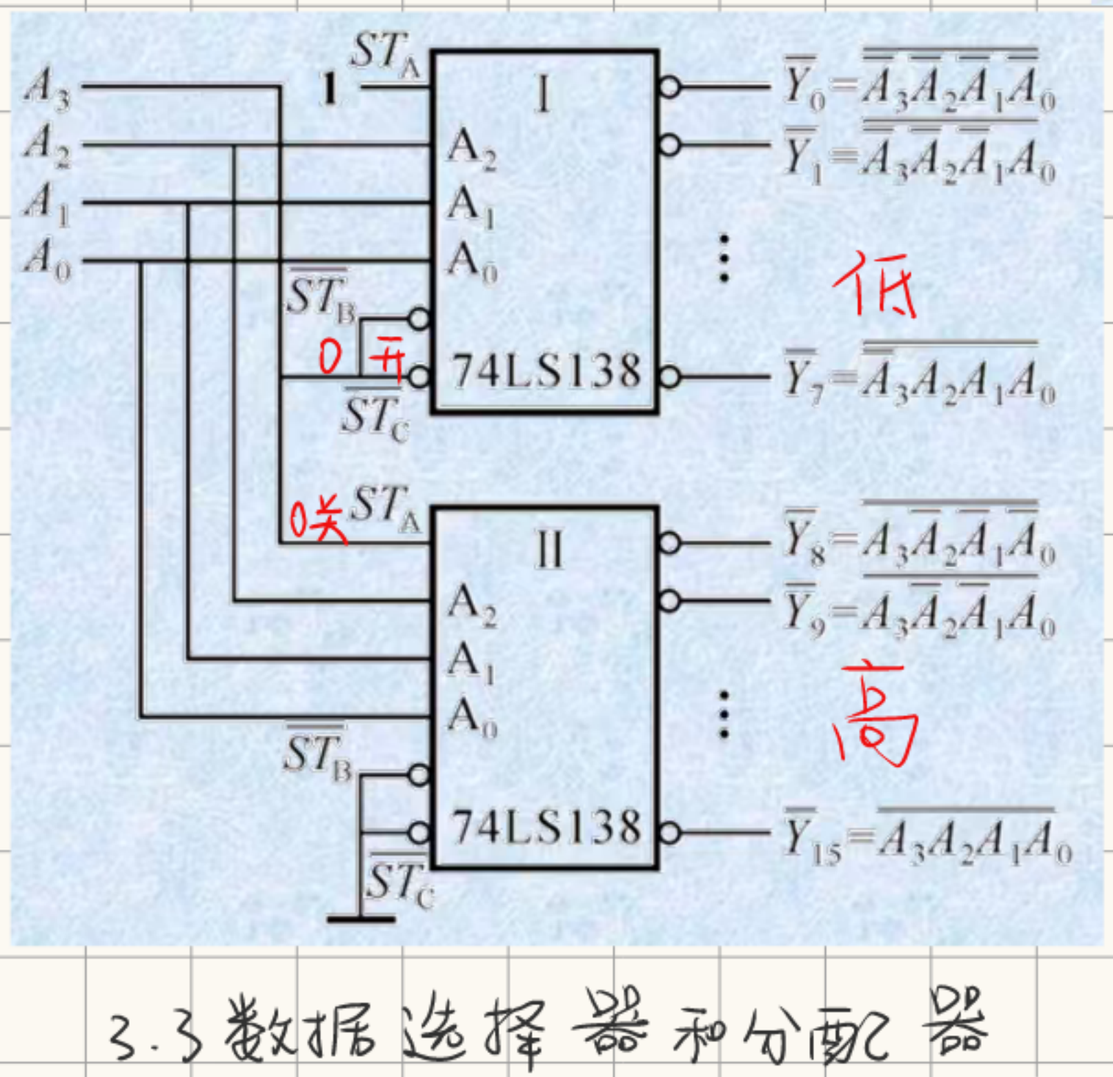
3.2.2 译码器





输入					输出							
ST_A	$\overline{ST_B}+\overline{ST_C}$	A_2	A_1	A_0	$\overline{Y}_0$	$\overline{Y}_1$	$\overline{Y}_2$	$\overline{Y}_3$	$\overline{Y}_4$	$\overline{Y}_5$	$\overline{Y}_6$	$\overline{Y}_7$
0	×	×	×	×	1	1	1	1	1	1	1	1
×	1	×	×	×	1	1	1	1	1	1	1	1
1	0	0	0	0	0	1	1	1	1	1	1	1
1	0	0	0	1	1	0	1	1	1	1	1	1
1	0	0	1	0	1	1	0	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1	1	1
1	0	1	0	0	1	1	1	1	0	1	1	1
1	0	1	0	1	1	1	1	1	1	0	1	1
1	0	1	1	0	1	1	1	1	1	1	0	1
1	0	1	1	1	1	1	1	1	1	1	1	0

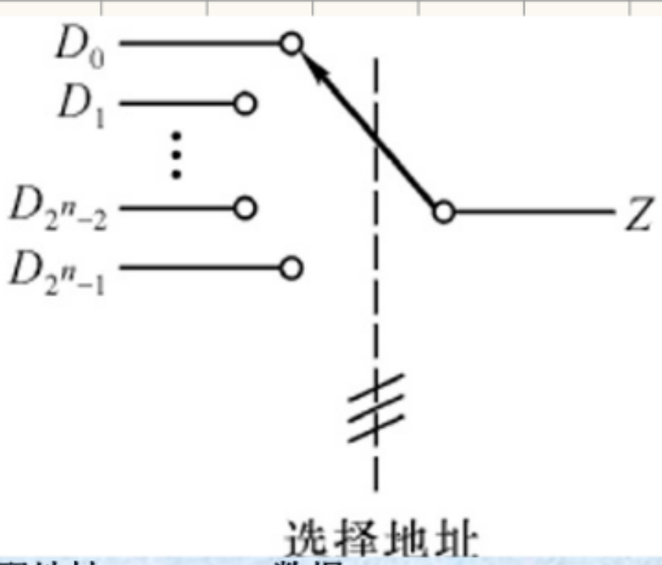
都为0才开始工作

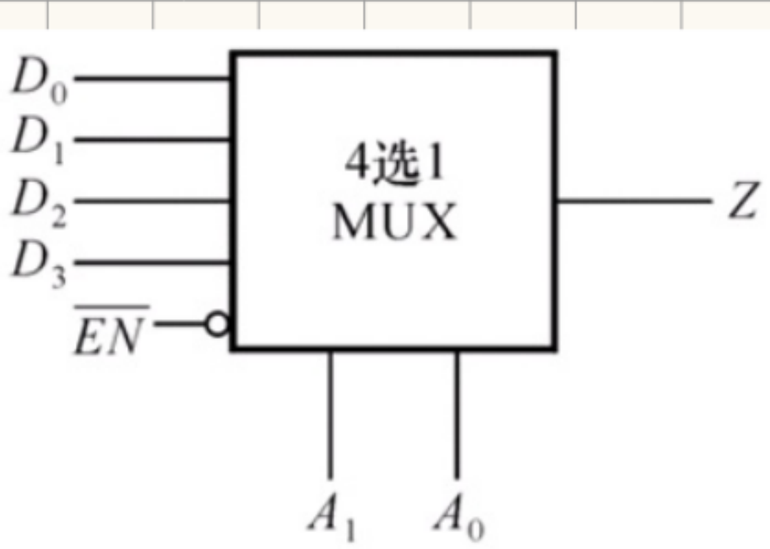


A_3 控制上下哪个芯片工作

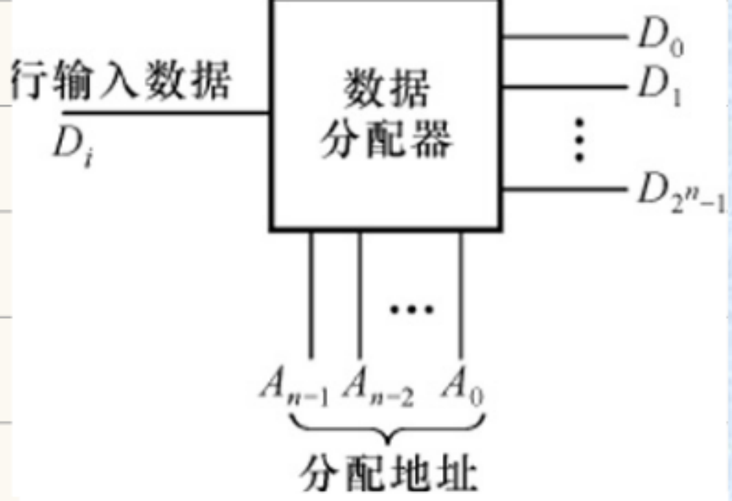
3.3 数据选择器和分配器

$Z=D_3A_1A_0+D_2A_1\overline{A}_0+D_1\overline{A}_1A_0+D_0\overline{A}_1\overline{A}_0$

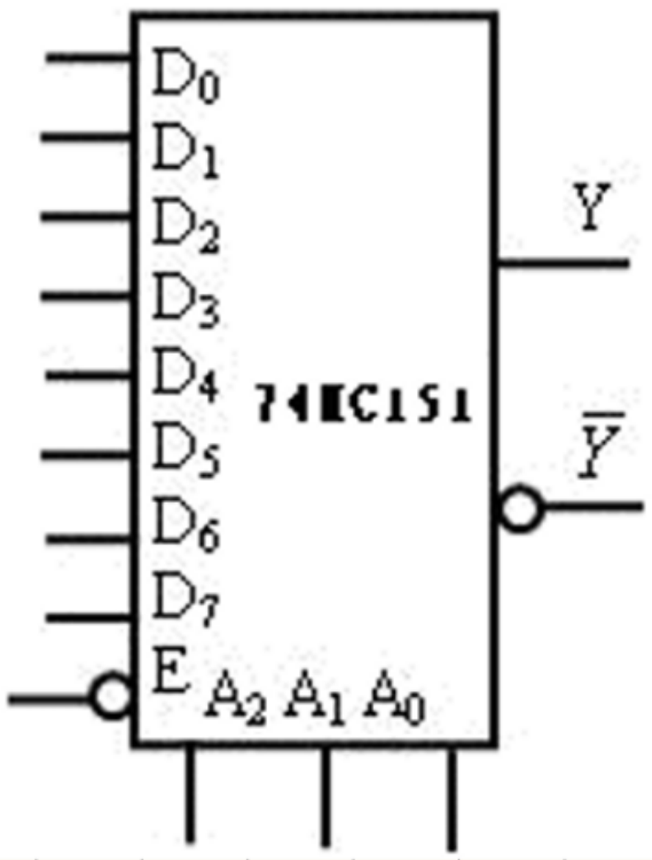




分配地址	数据	输出通道
A_1	A_0	
0	0	$Y_0=D$
0	1	$Y_1=D$
1	0	$Y_2=D$
1	1	$Y_3=D$



分配地址		数据	输出通道			
A_1	A_0	D_i	$\overline{D}_3$	$\overline{D}_2$	$\overline{D}_1$	$\overline{D}_0$
0	0	D_i	1	1	1	D_i
0	1	D_i	1	1	D_i	1
1	0	D_i	1	D_i	1	1
1	1	D_i	D_i	1	1	1

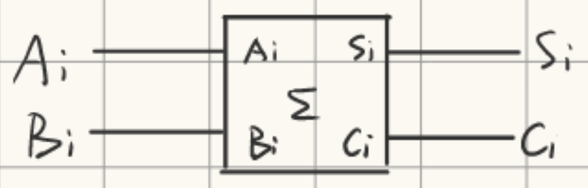


输入				输出	
使能输入	选择地址输入			原码输出	反码输出
$\overline{E}$	A_2	A_1	A_0	Y	$\overline{Y}$
1	×	×	×	0	1
0	0	0	0	D_0	$\overline{D}_0$
0	0	0	1	D_1	$\overline{D}_1$
0	0	1	0	D_2	$\overline{D}_2$
0	0	1	1	D_3	$\overline{D}_3$
0	1	0	0	D_4	$\overline{D}_4$
0	1	0	1	D_5	$\overline{D}_5$
0	1	1	0	D_6	$\overline{D}_6$
0	1	1	1	D_7	$\overline{D}_7$

3.4 加法器与数值比较器

3.4.1 加法器

① 半加器

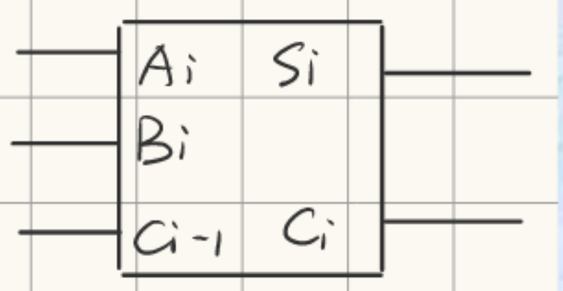


输入		输出	
被加数 A_i	加数 B_i	半加和 S_i	进位 C_i
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$S_i = A_i \oplus B_i$

$C_i = A_i B_i$

② 全加器

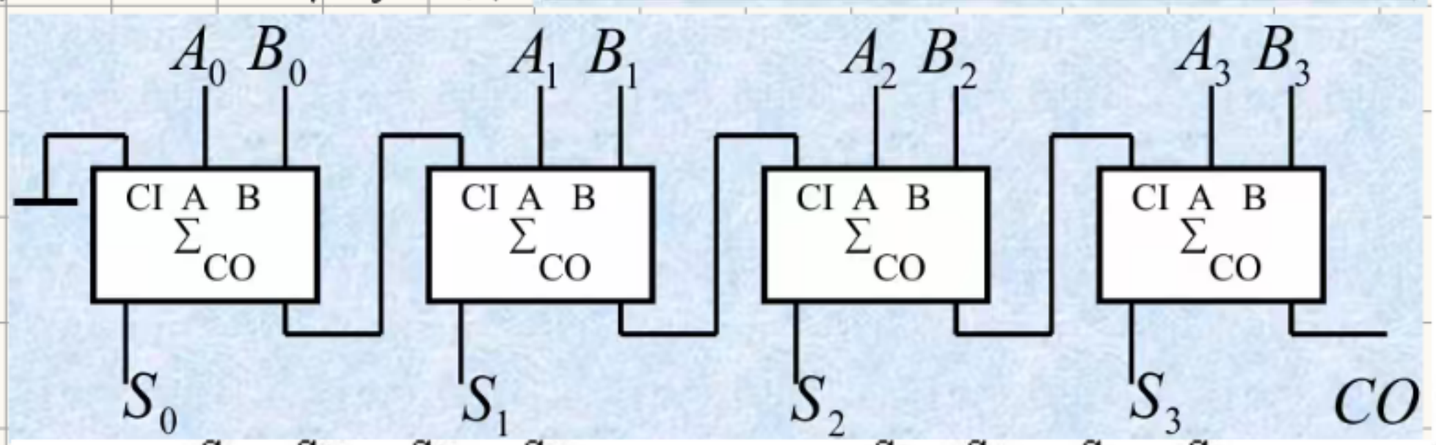


全加器输入			结果输出	
A_i	B_i	C_{i-1}	S_i	C_i
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

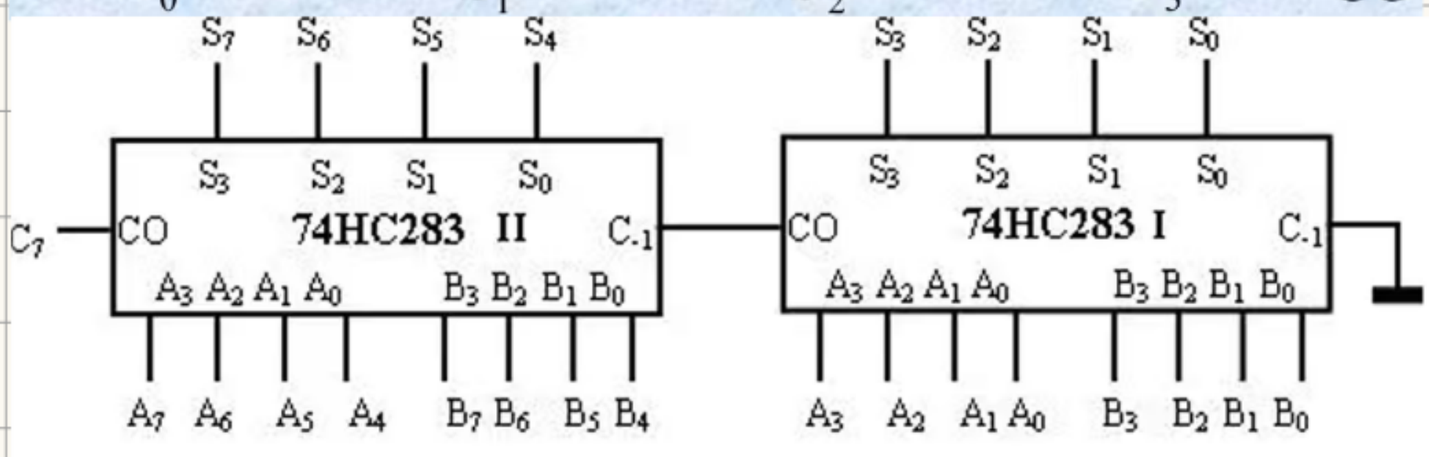
$S_i = A_i \oplus B_i \oplus C_i$

$C_i = A_i B_i + (A_i \oplus B_i) C_{i-1}$

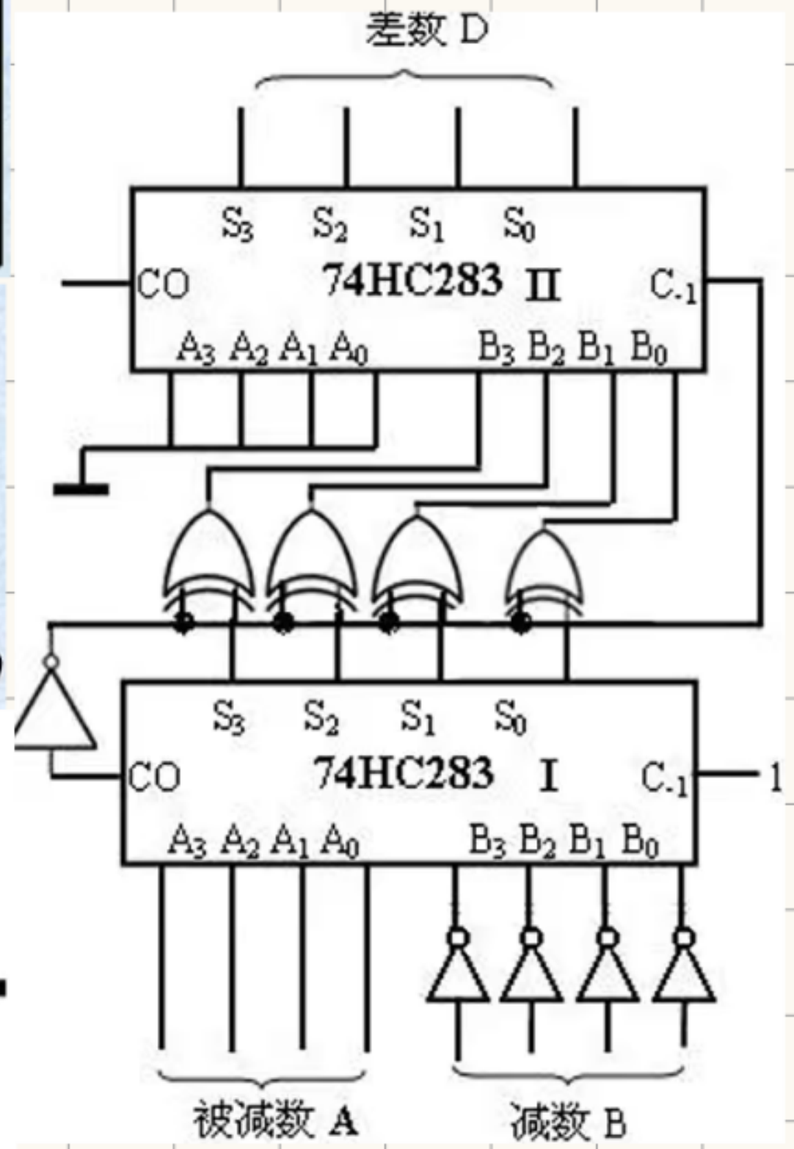
③ 多位二进制串行加法



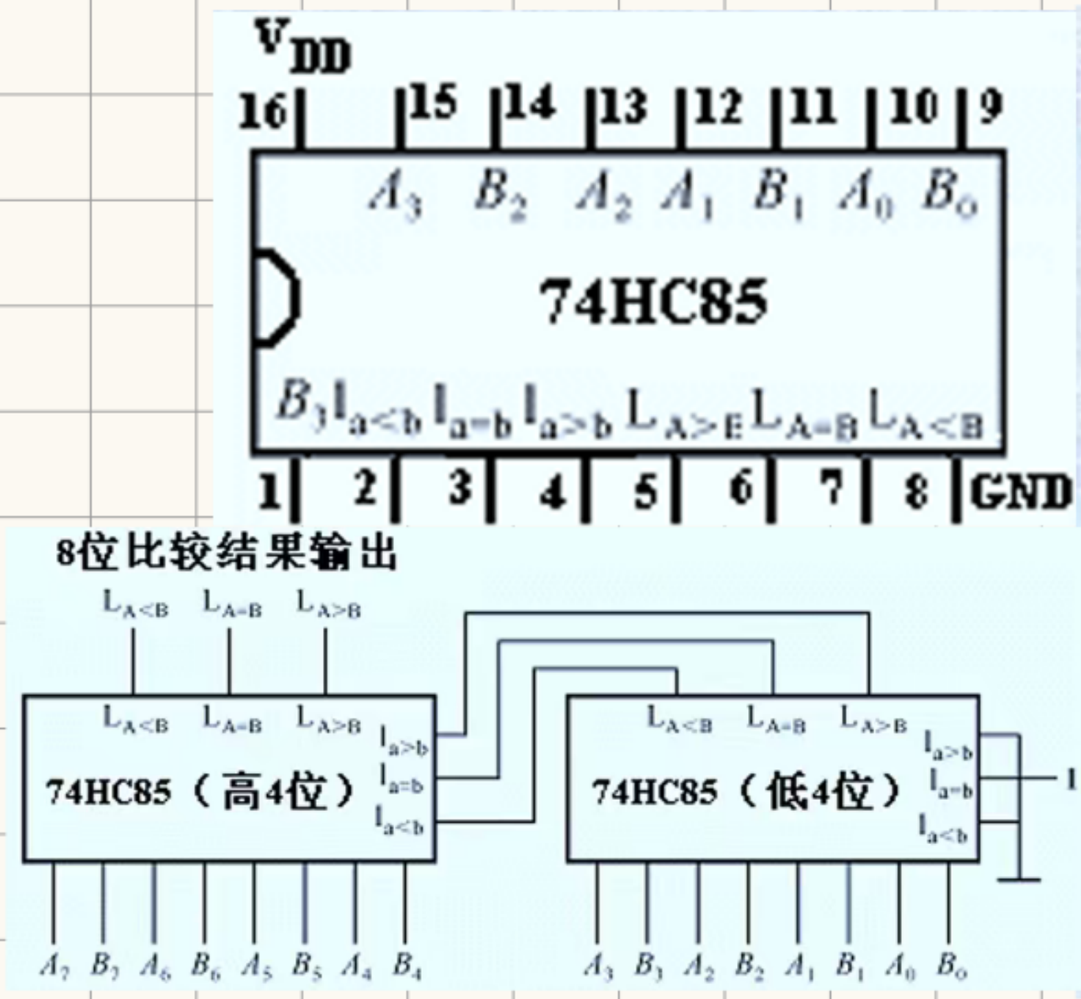
中规模加法



中规模减法计数器



3.4.2 数值比较器



本四位输入				低位结果输入			比较结果输出		
A_3,B_3	A_2,B_2	A_1,B_1	A_0,B_0	$l_{a>b}$	$l_{a<b}$	$l_{a=b}$	$L_{A>B}$	$L_{A<B}$	$L_{A=B}$
G_3	×	×	×	×	×	×	1	0	0
L_3	×	×	×	×	×	×	0	1	0
E_3	G_2	×	×	×	×	×	1	0	0
E_3	L_2	×	×	×	×	×	0	1	0
E_3	E_2	G_1	×	×	×	×	1	0	0
E_3	E_2	L_1	×	×	×	×	0	1	0
E_3	E_2	E_1	G_0	×	×	×	1	0	0
E_3	E_2	E_1	L_3	×	×	×	0	1	0
E_3	E_2	E_1	E_0	1	0	0	1	0	0
E_3	E_2	E_1	E_0	0	1	0	0	1	0
E_3	E_2	E_1	E_0	0	0	1	0	0	1

3.5 奇偶校验器

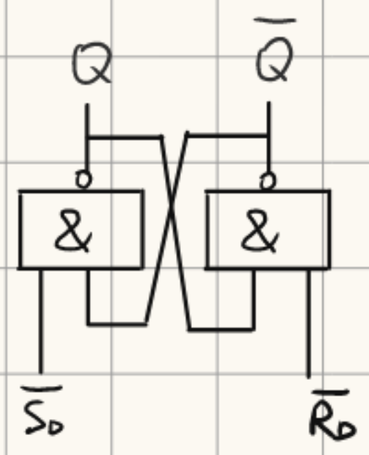
奇校验：末尾补一位，1位数为奇数　　10101011 —— 10101011 0

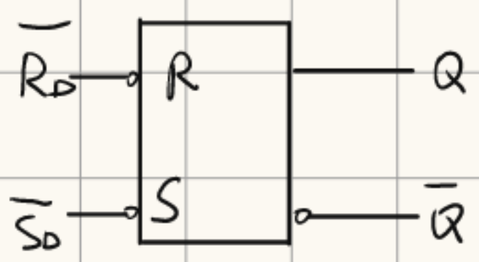
偶校验：末尾补一位，1位数为偶数　　10101011 —— 10101011 1

第四章 触发器及应用

4.1 触发器

4.1.1 基本RS 触发器





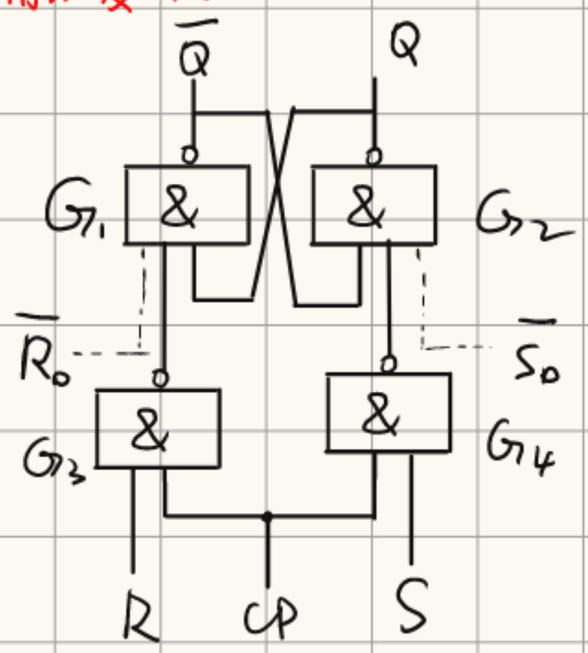
$\bar{R}_D$	$\bar{S}_D$	R_D	S_D	Q	$\bar{Q}$	
0	0	1	1	1	1	禁用
0	1	1	0	0	1	清零
1	0	0	1	1	0	置1
1	1	0	0	不变		保持

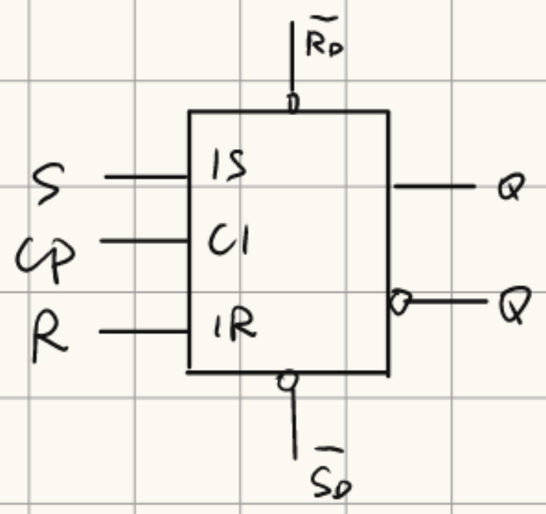
记忆 R：reset R_D为1 清零 $\bar{R}_D=0$

S：set S_D为1 置1 $\bar{S}_D=0$

4.1.2 同步触发器 (CP=1)

① 边沿触发 RS

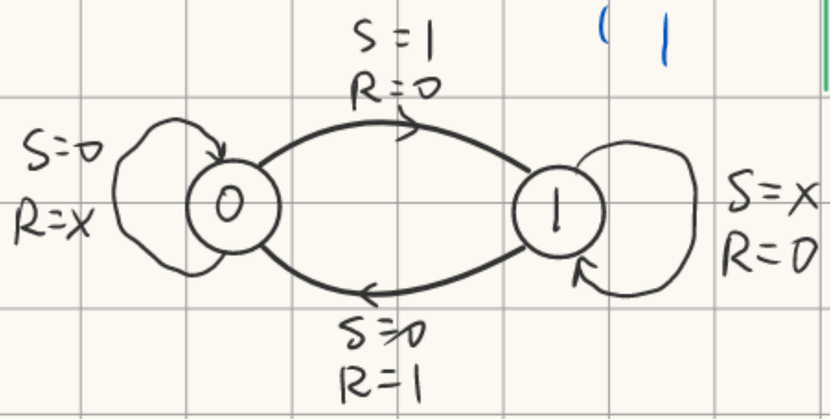


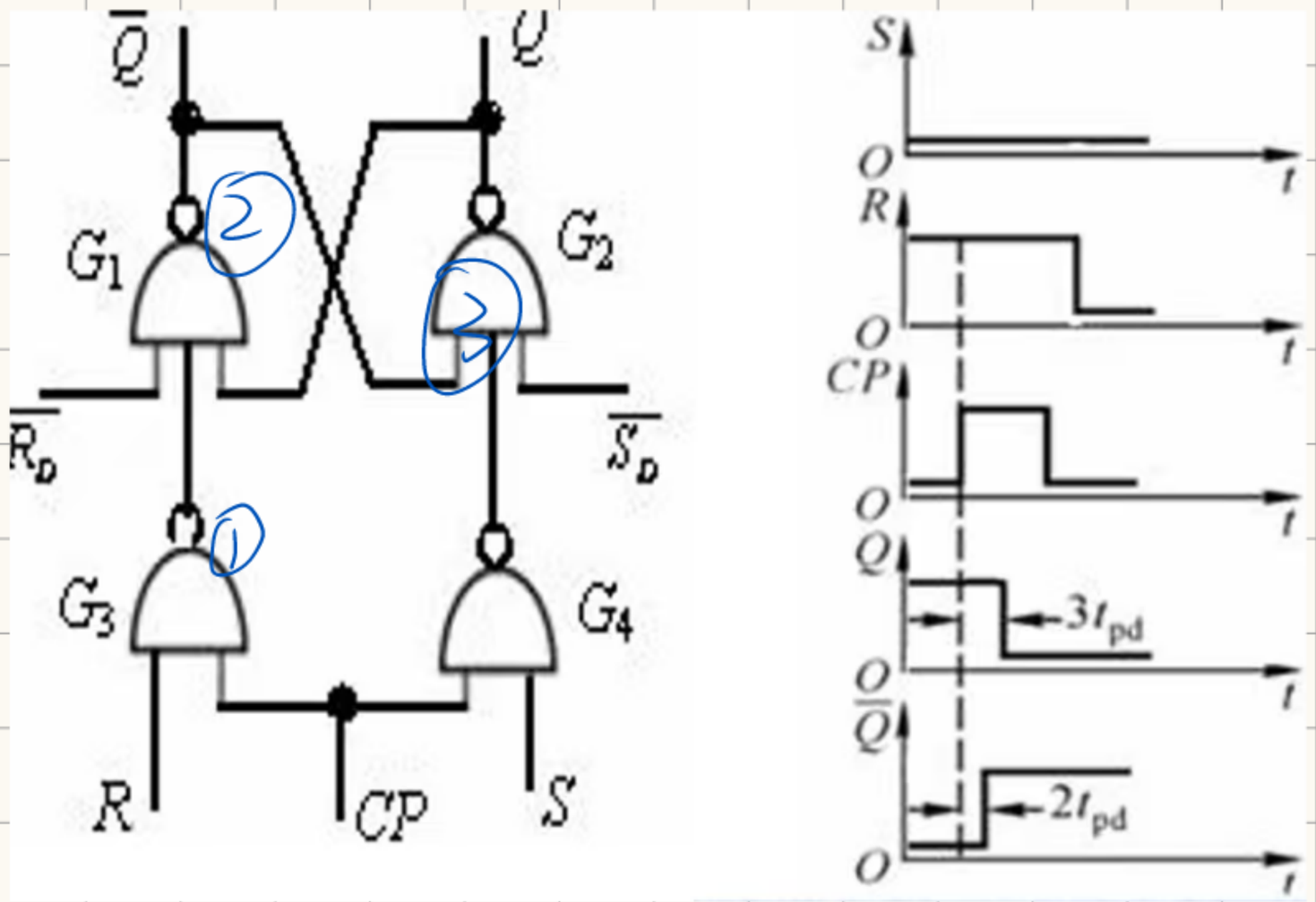


$\bar{R}_D$	$\bar{S}_D$	R	S	Q	Q^{n+1}	说明
0	1	×	×	×	×	异步清零
1	0	×	×	×	×	异步置1
1	1	0	0	0	0	$Q^{n+1}=Q^n$
1	1	0	0	1	1	
1	1	0	1	0	1	$Q^{n+1}=1$
1	1	0	1	1	1	
1	1	1	0	0	0	$Q^{n+1}=0$
1	1	1	0	1	0	
1	1	1	1	0	×	禁用
1	1	1	1	1	×	

Q^{n+1} / R \ SQ^n	00	01	11	10
0	0	1	1	1
1	0	0	×	×

$$\begin{cases} Q^{n+1}=S+\bar{R}Q^n \\ RS=0 \end{cases}$$

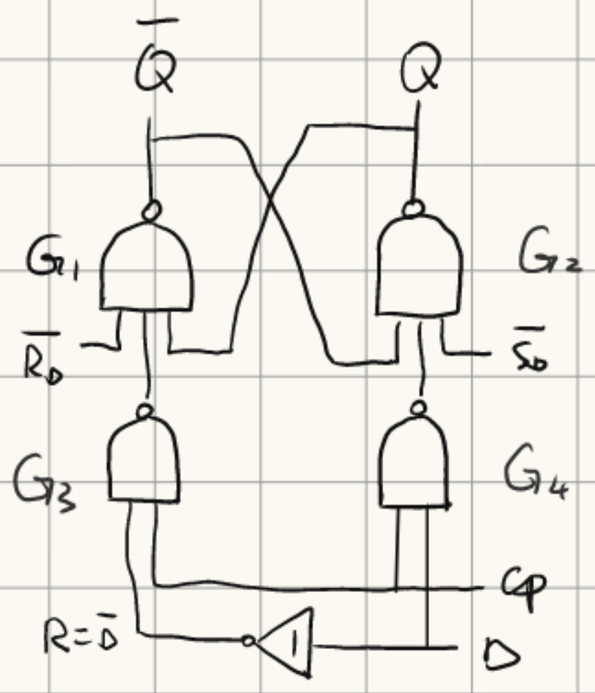


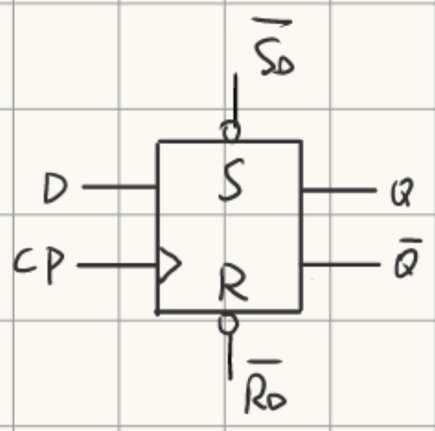


时钟高电平持续时间

应大于 $3t_{pd}$

② 主从型D触发器





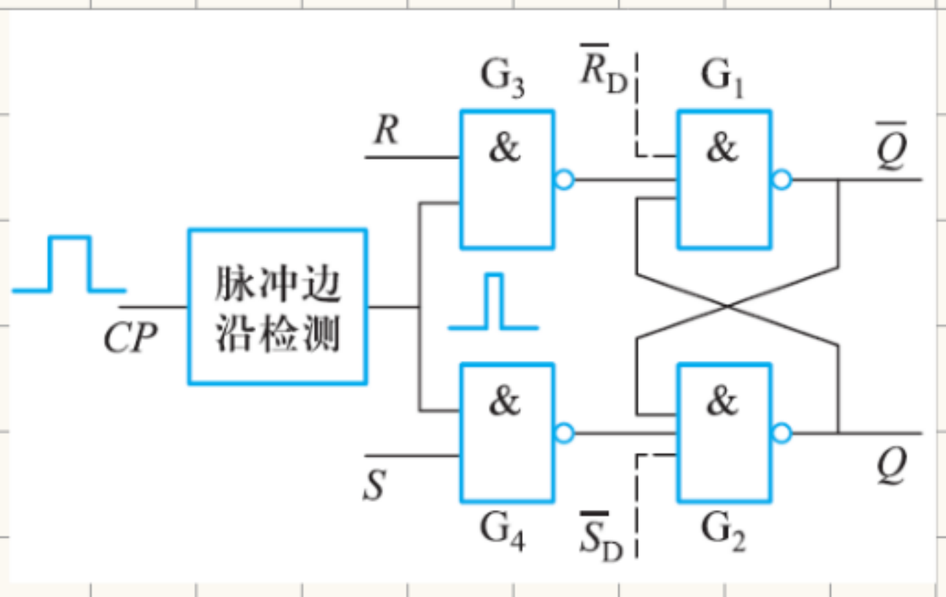
时钟上升沿有效

$Q^{n+1}=D$

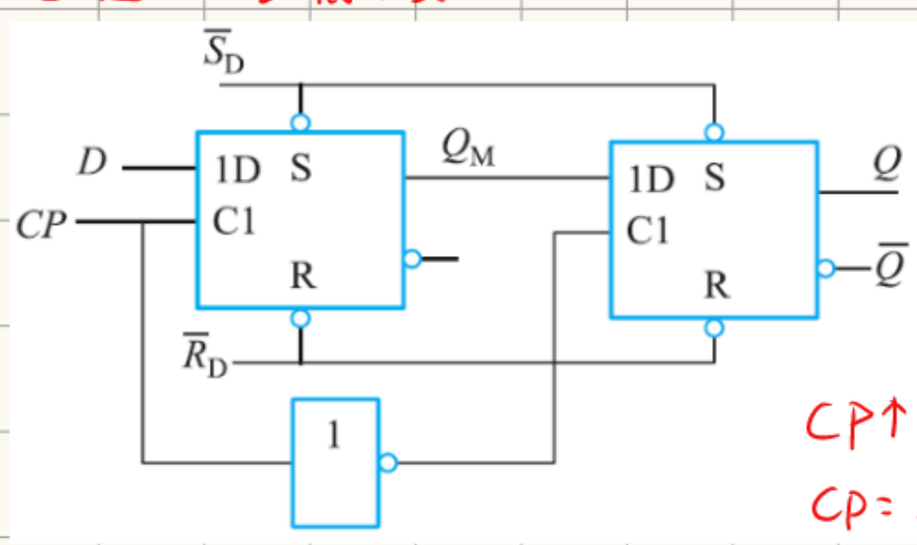
输入	初态	次态	
D	Q^n	Q^{n+1}	
0	0	0	置0
	1	0	
1	0	1	置1
	1	1	

4.1.3 边沿触发器 (CP上升或下降沿)

① 边沿RS

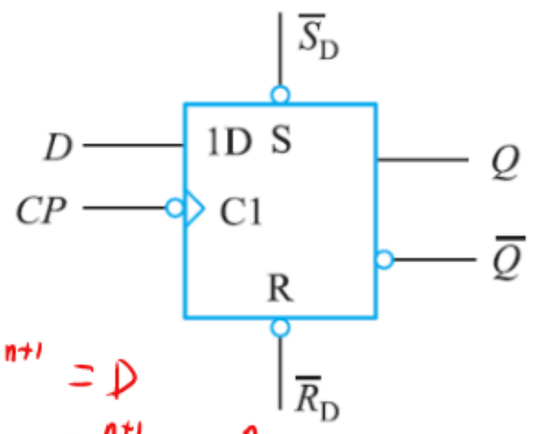


② 边沿D触发

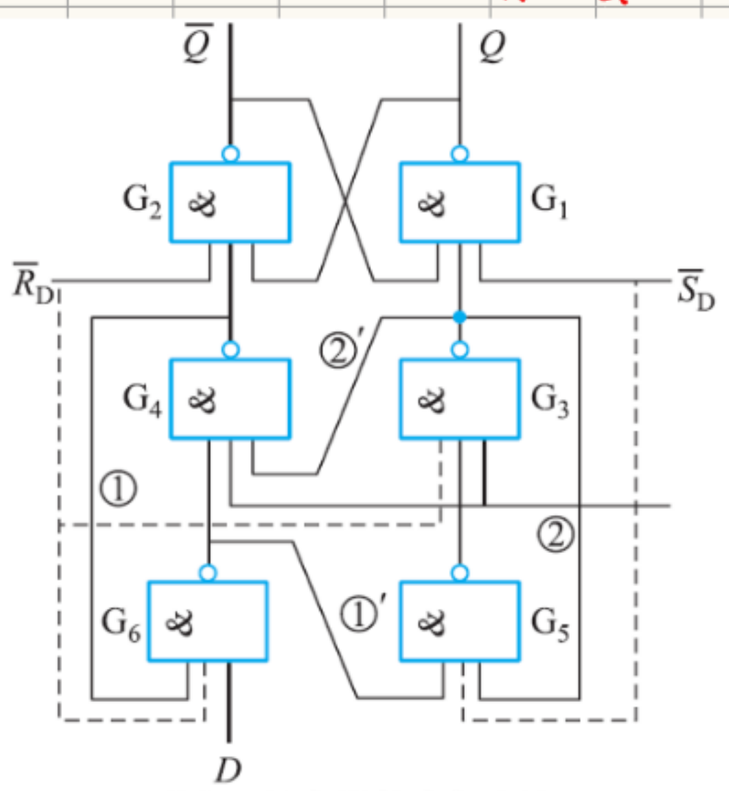


CP↑ $Q^{n+1}=D$

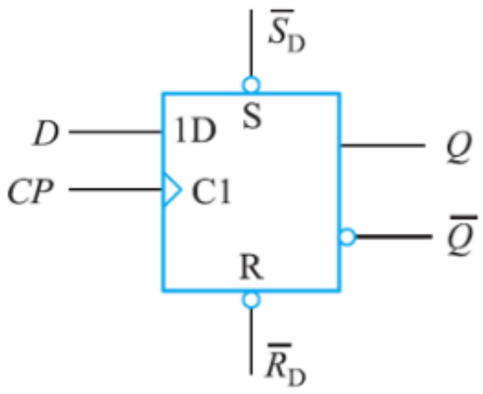
CP=其他 $Q^{n+1}=Q^n$



③ 维持阻塞D触发

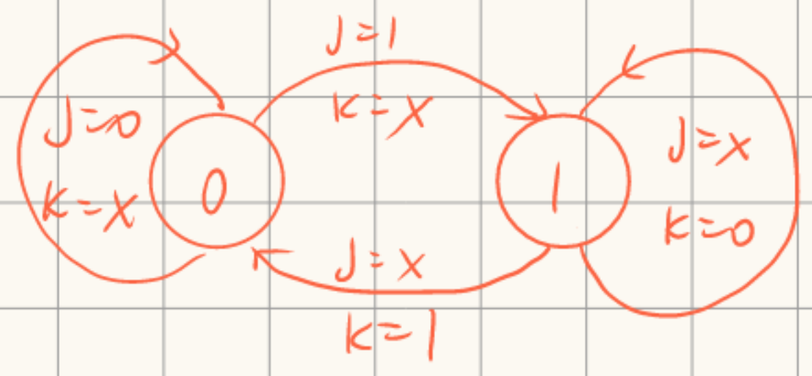


(a) 维阻D触发器的内部电路

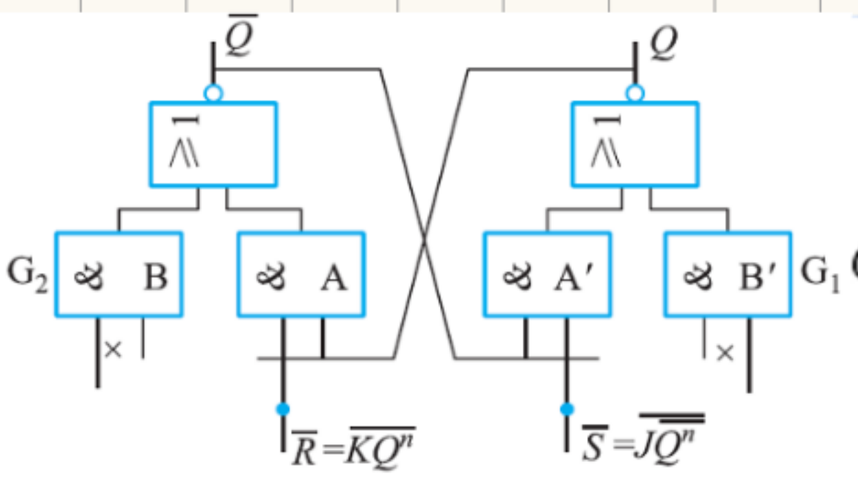


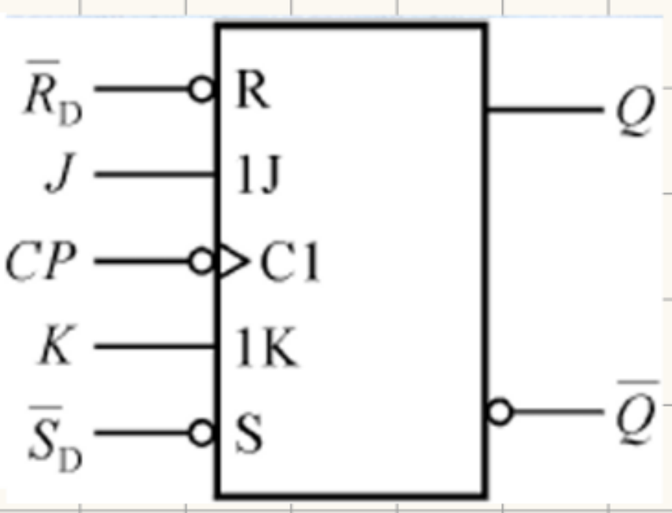
(b) 逻辑符号

一定要看清楚 触发器 初始状态



④ 负边沿JK触发器



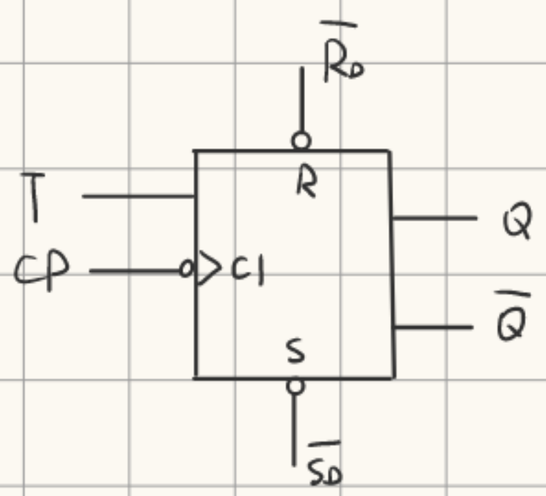


$Q^{n+1}=J\overline{Q^n}+\overline{K}Q^n$

J	K	Q^n	Q^{n+1}	次态
0	0	0	0	$Q^{n+1}=Q$ 保持
0	0	1	1	
0	1	0	0	$Q^{n+1}=0$ 置0
0	1	1	0	
1	0	0	1	$Q^{n+1}=1$ 置1
1	0	1	1	
1	1	0	1	$Q^{n+1}=\overline{Q^n}$ 翻转
1	1	1	0	

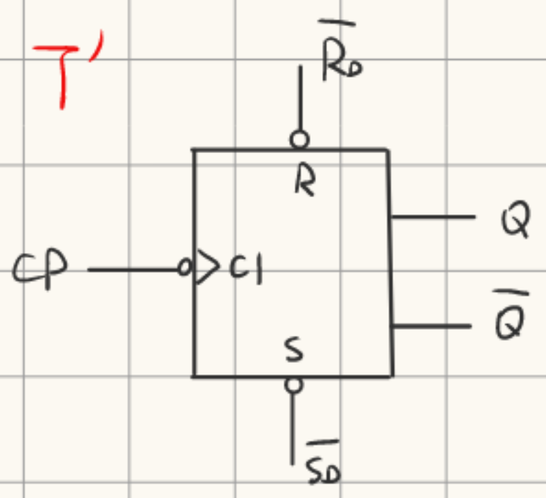
⑤ T触发器

(将) K接一起，功能 翻转/保持)



T	Q^n	Q^{n+1}	
0	0	0	保持不变
	1	1	
1	0	1	翻转
	1	0	

T'



$Q^{n+1} = \overline{Q}^n$ (下降沿翻转)

4.2 触发器的应用

4.2.1 寄存器、移位寄存器

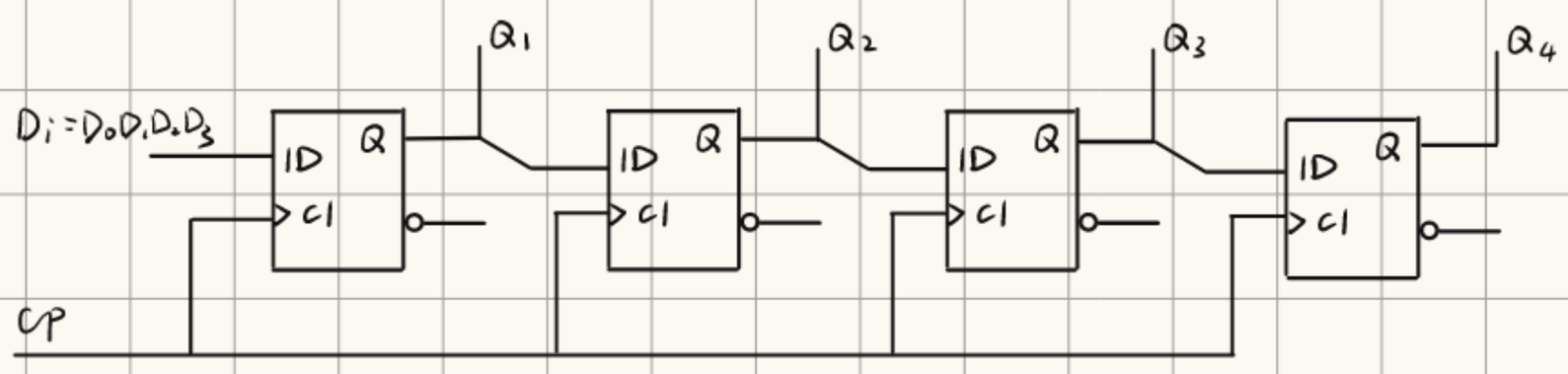
① 清零

② 读取信息

③ 锁存信息

右移 ÷2

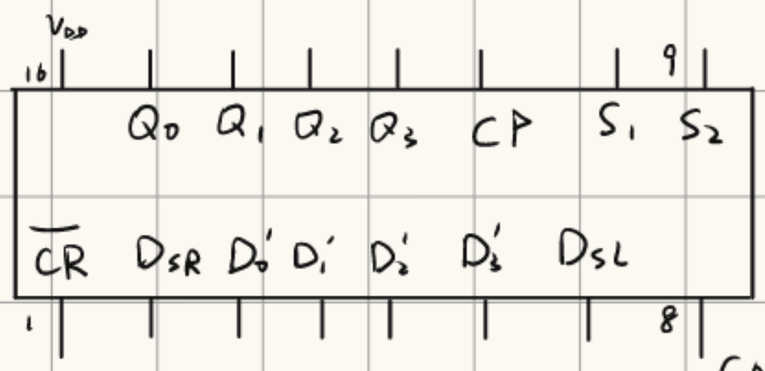
左移 ×2



每次只移一位 ☆

集成移位寄存器

(1) 双向移位



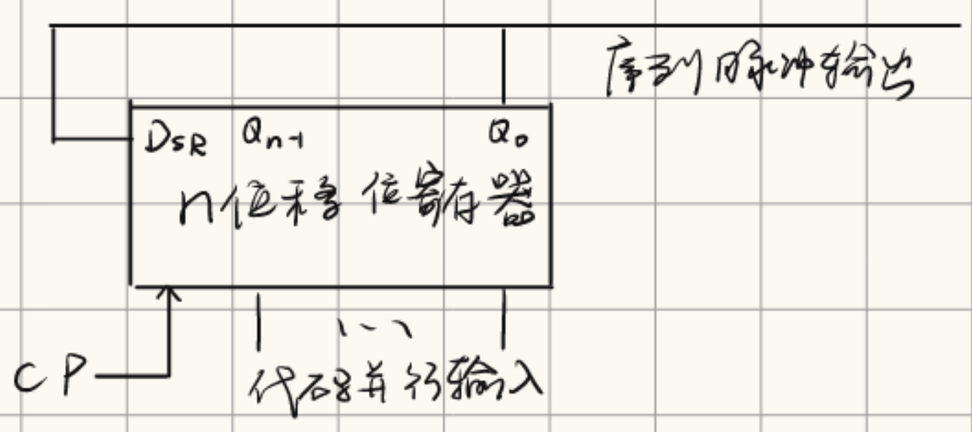
$\overline{CR}$	CP	S_2S_1	D_{SR}	D_{SL}	$D_3D_2D_1D_0$	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	功能
0	×	××	×	×	××××	0	0	0	0	清零
1	0	××	×	×	××××	Q_3	Q_2	Q_1	Q_0	异步保持
1	↑	11	×	×	DCBA	D	C	B	A	并行输入
1	↑	00	×	×	××××	Q_3	Q_2	Q_1	Q_0	同步保持
1	↑	01	D	×	××××	D	Q_3	Q_2	Q_1	右移(高→低)
1	↑	10	×	D	××××	Q_3	Q_2	Q_1	D	左移(低→高)

$$1D_3 = \overline{S_2}\,\overline{S_1}\,Q_3^n + \overline{S_2}S_1 D_{SR} + S_2\overline{S_1}\,Q_2^n + S_2 S_1 D_3'$$

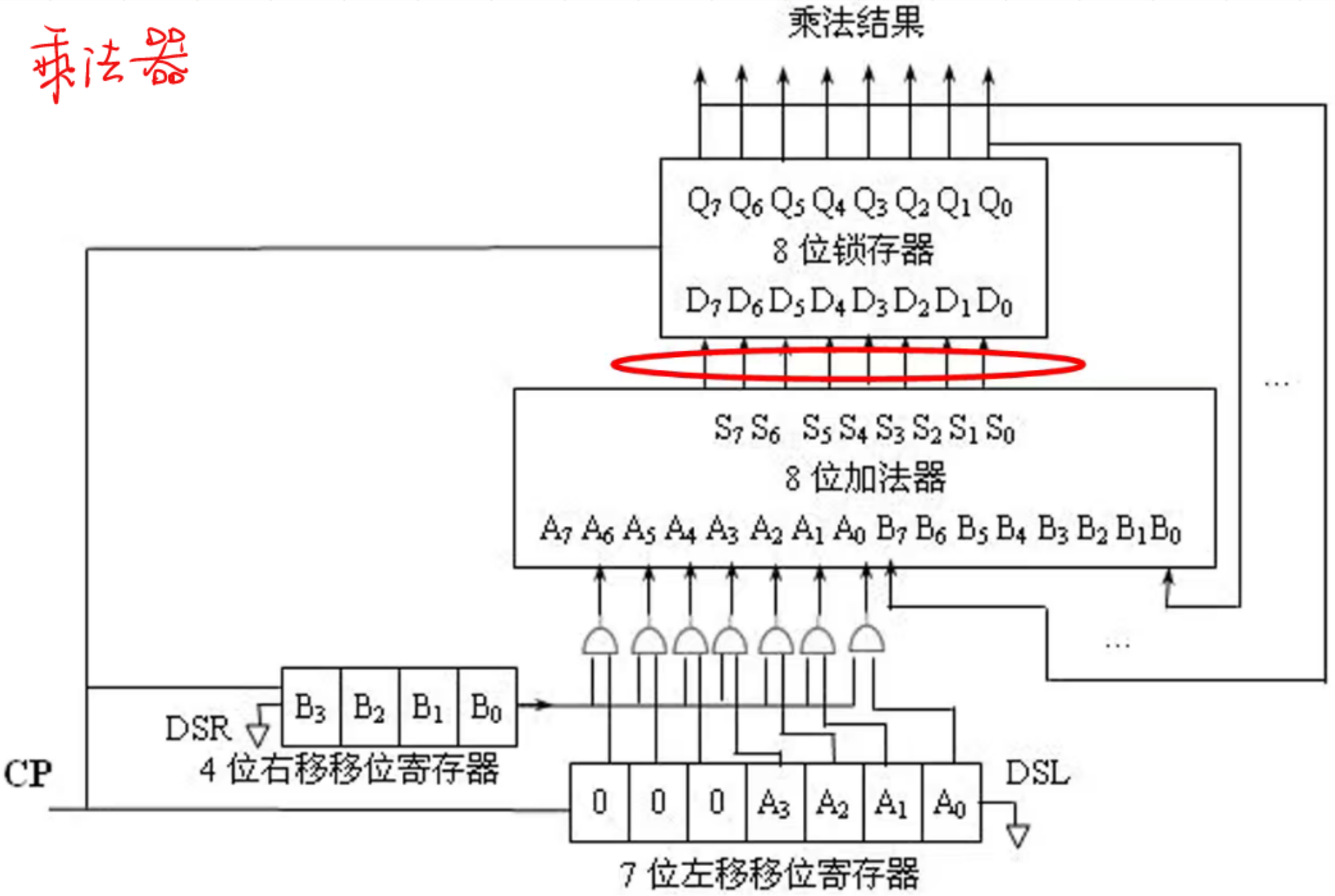
$$1D_2 = \overline{S_2}\,\overline{S_1}\,Q_2^n + \overline{S_2}S_1 Q_3^n + S_2\overline{S_1}\,Q_1^n + S_2 S_1 D_2'$$

$$1D_1 = \overline{S_2}\,\overline{S_1}\,Q_1^n + \overline{S_2}S_1 Q_2^n + S_2\overline{S_1}\,Q_0^n + S_2 S_1 D_1'$$

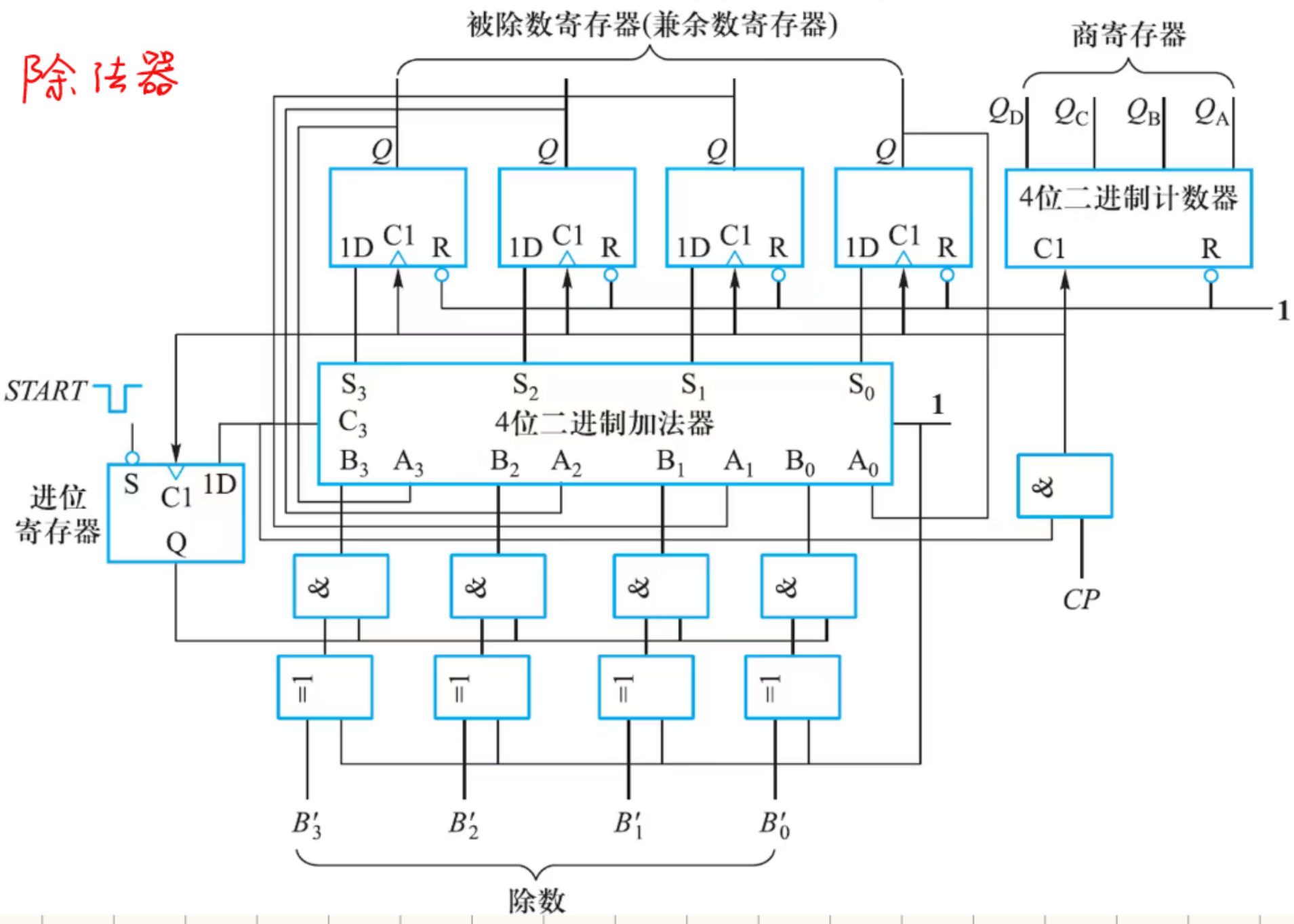
$$1D_0 = \overline{S_2}\,\overline{S_1}\,Q_0^n + \overline{S_2}S_1 Q_1^n + S_2\overline{S_1}\,D_{SL} + S_2 S_1 D_0'$$



乘法器

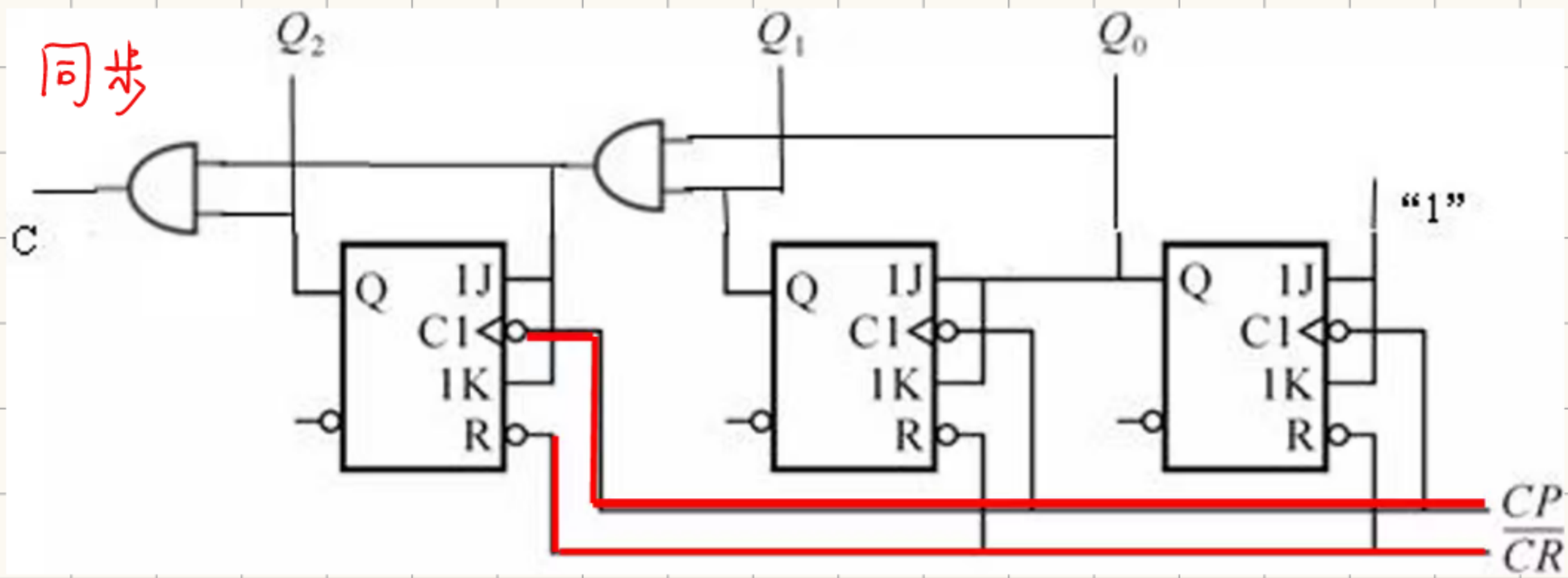


除法器



4.2.2 二进制计数器

同步

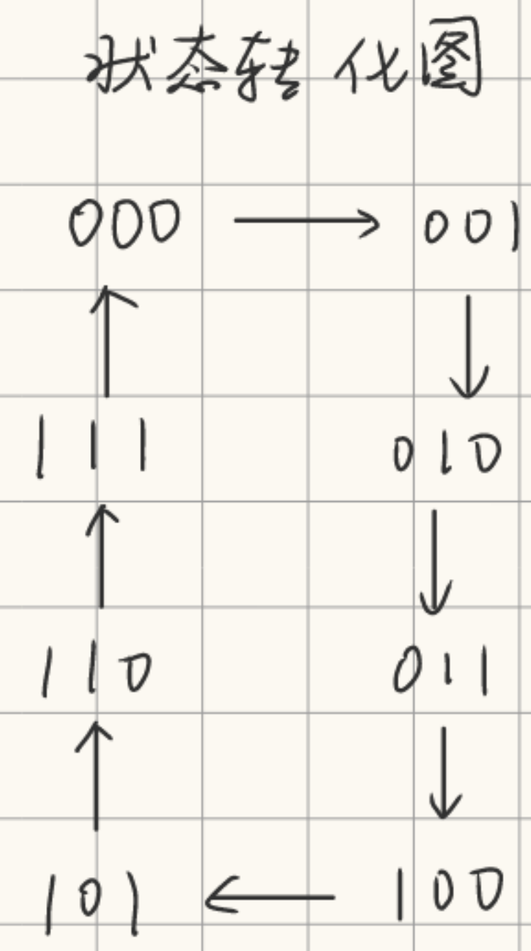


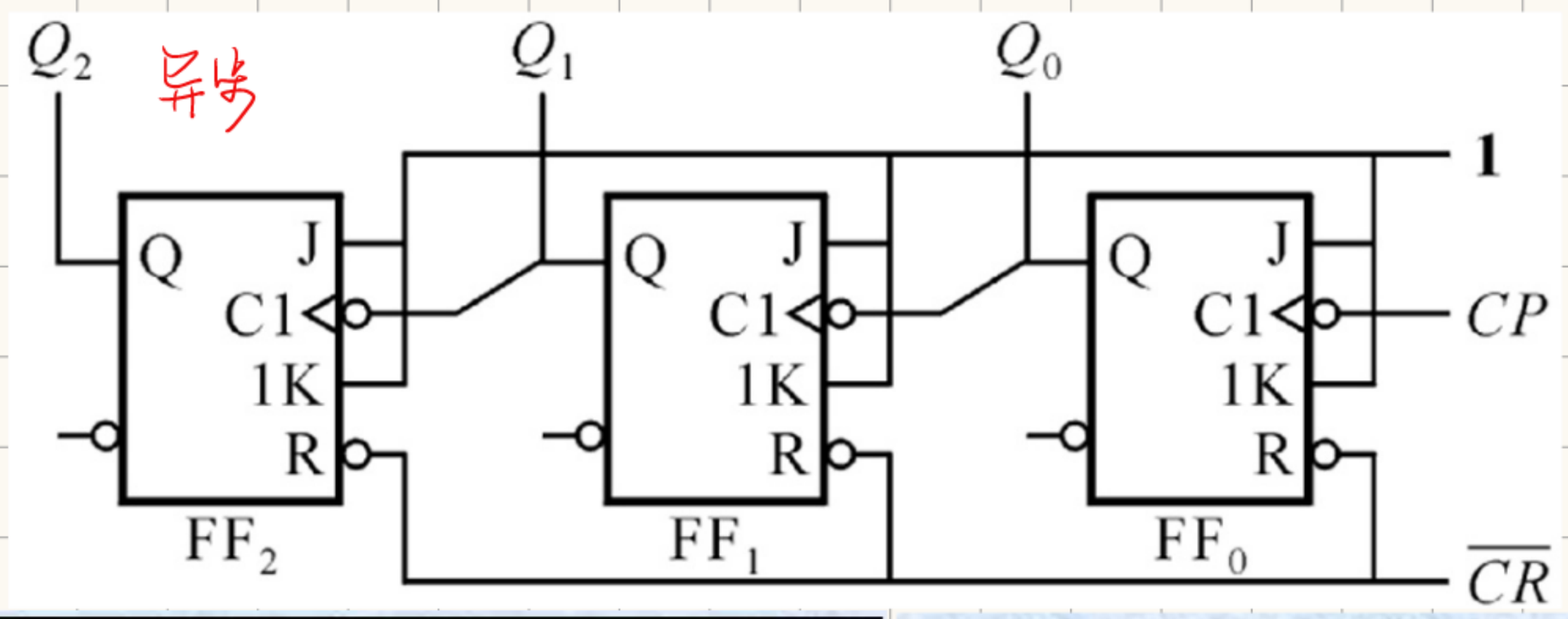
$T_0 = 1$

$T_1 = Q_0^n$

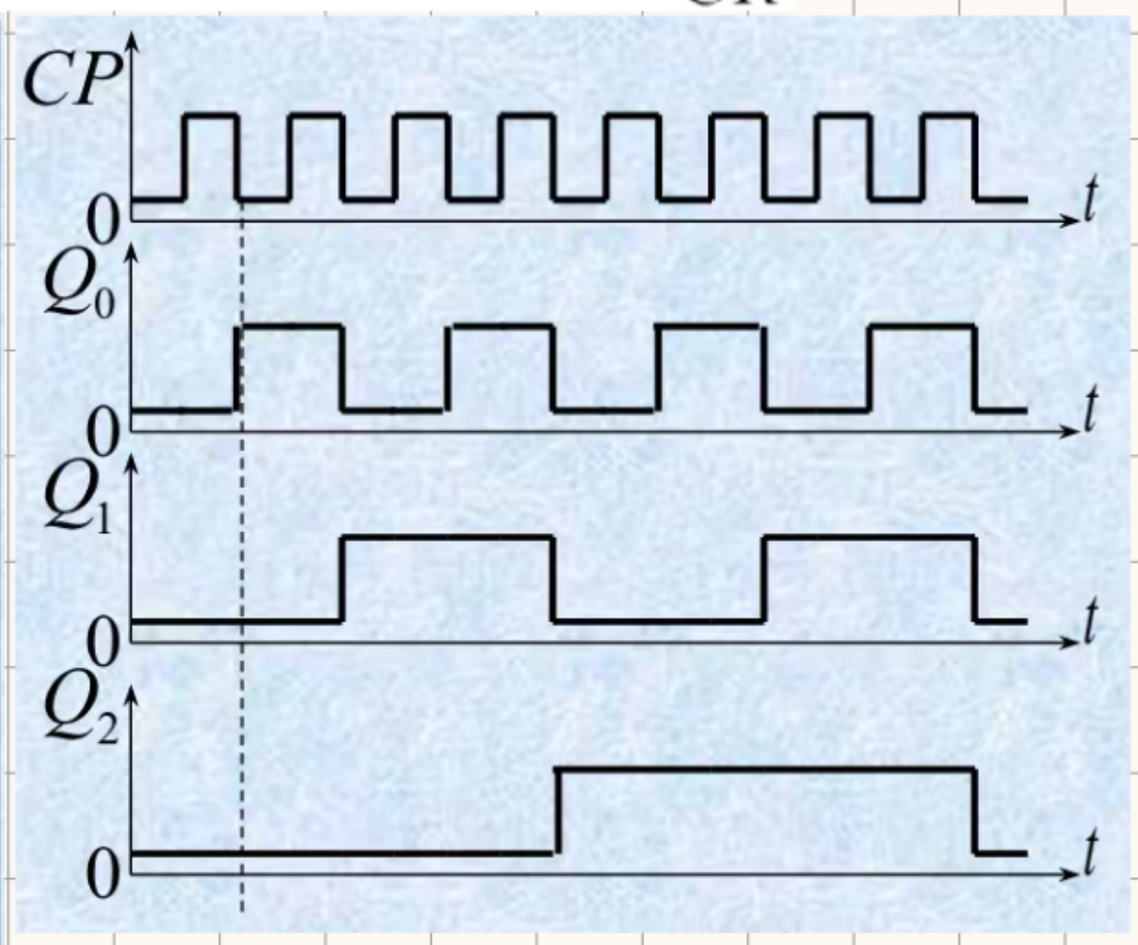
$T_2 = Q_1^n Q_0^n$

状态顺序	现态			T端状态			次态		
	Q2	Q1	Q0	T2	T1	T0	Q2	Q1	Q0
0	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**
1	**0**	**0**	**1**	0	**1**	**1**	**0**	**1**	**0**
2	0	1	0	0	0	1	0	1	1
3	0	1	1	1	1	1	1	0	0
4	1	0	0	0	0	1	1	0	1
5	1	0	1	0	1	1	1	1	0
6	1	1	0	0	0	1	1	1	1
7	1	1	1	1	1	1	0	0	0



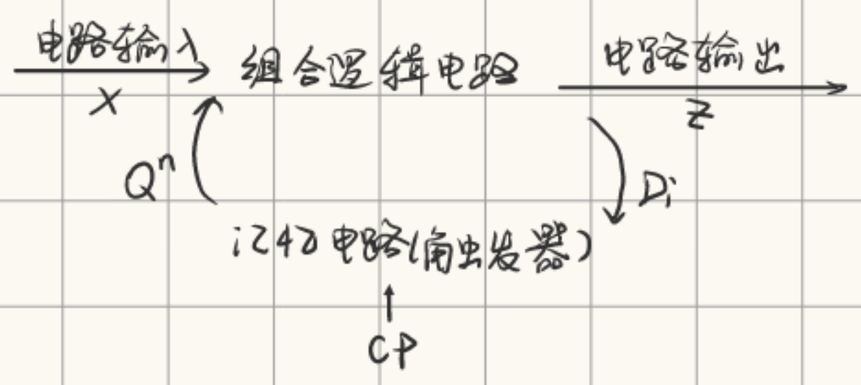


CP顺序	触发器状态		
	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0



第五章 时序逻辑电路

5.1 基本结构功能



触发器驱动方程 $D_i = F_1(X, Q^n)$

触发器次态方程 $Q^{n+1} = F_2(D, Q^n)$

电路输出方程 $Z = F_3(X, Q^n)$

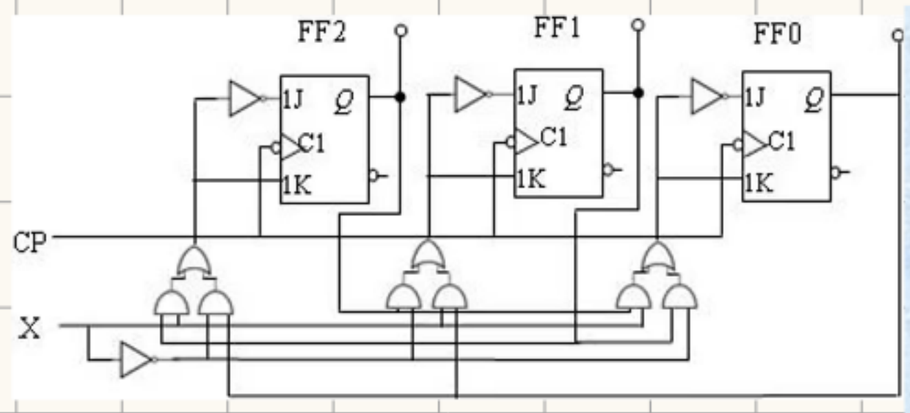
驱动 $\begin{cases} J=1 \\ K=1 \end{cases}$ 特性 $Q^{n+1} = J\overline{Q^n} + \overline{K}Q^n$ 状态 $Q^{n+1} = \overline{Q^n}$

5.2 时序逻辑电路分析

同步：每个触发器 CP 连一起，同时受触发

异步：每个触发器 CP 不连一起

5.2.1 同步时序电路



$$Q^{n+1} = J\overline{Q^n} + \overline{K}Q^n \quad (1)$$

$$Q_2^{n+1} = f(Q_2^n, Q_1^n, Q_0^n)$$

$$Q_1^{n+1} = f(Q_2^n, Q_1^n, Q_0^n)$$

$$Q_0^{n+1} = f(Q_2^n, Q_1^n, Q_0^n) \quad (2)$$

$$J_2 = \overline{\overline{X}Q_0 + XQ_1}, K_2 = \overline{J}_2$$

$$J_1 = \overline{\overline{X}Q_2 + XQ_0}, K_1 = \overline{J}_1$$

$$J_0 = \overline{\overline{X}Q_1 + XQ_2}, K_0 = \overline{J}_0$$

分析步骤

(1) 列 驱动方程

(2) 代入 得状态方程

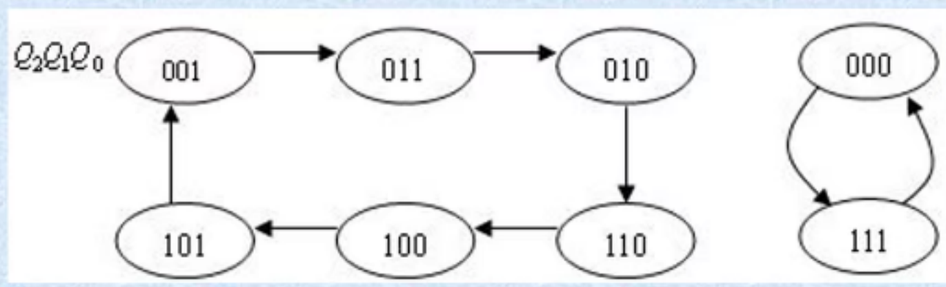
(3) 求得状态转移图

(4) 得时序电路逻辑功能

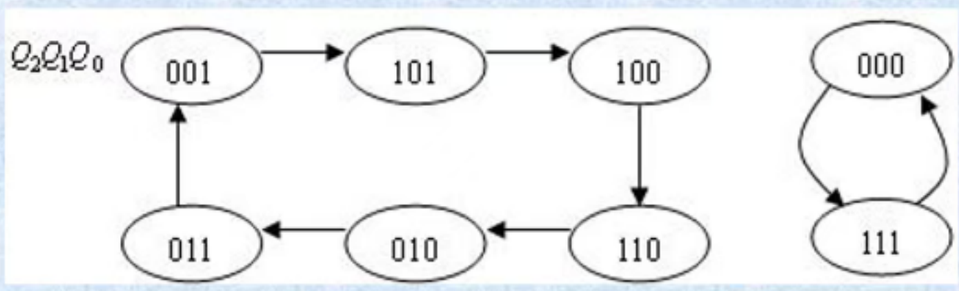
④ **特殊编码的六进制计数器**

不能自启动的时序电路 ③

X=0时状态转移图：



X=1时状态转移图：



5.2.2 异步时序电路

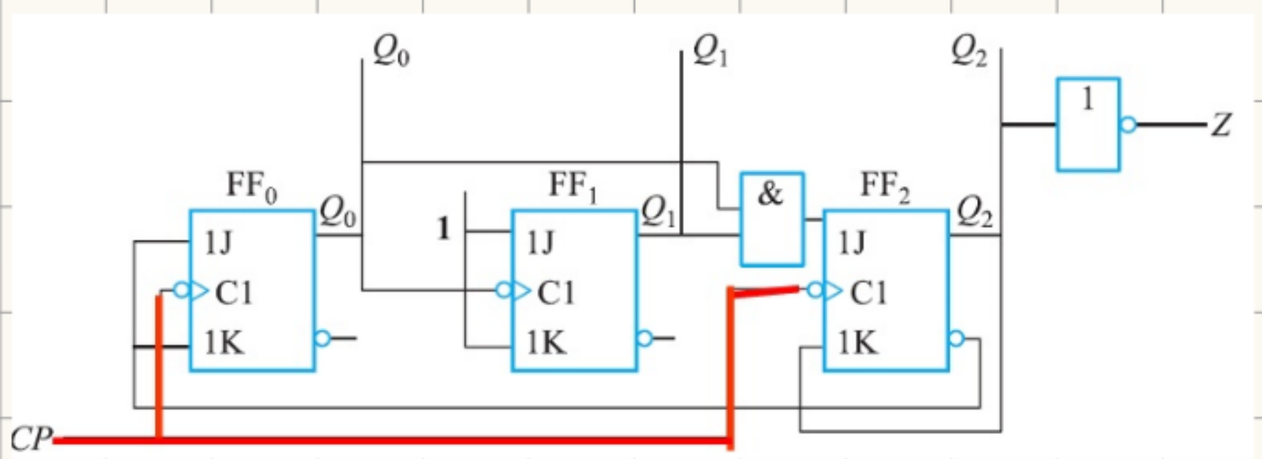
(1) 写 驱动方程，

特性方程，CP方程

(2) 代入 驱动方程 求状态方程

(3) 初态 → 次态

(4) 列出 真值转化表、状态转化图，画时序图，得电路逻辑方程



(1) 触发器的驱动方程 → $1J_0 = 1K_0 = \overline{Q_2^n}$；$1J_1 = 1K_1 = 1$；$1J_2 = Q_1^n Q_0^n$　$1K_2 = Q_2^n$

特别注意CP方程！

$$Q_2^{n+1} = J_2\overline{Q_2^n} + \overline{K_2}Q_{2.}^n = \overline{Q_2^n}Q_1^nQ_0^n \qquad CP_2 = CP$$

$$Q_1^{n+1} = J_1\overline{Q_1^n} + \overline{K_1}Q_1^n = \overline{Q_1^n} \qquad CP_1 = Q_0^n$$

$$Q_0^{n+1} = J_0\overline{Q_0^n} + \overline{K_0}Q_0^n = \overline{Q_2^n}\,\overline{Q_0^n} + Q_2^nQ_0^n \qquad CP_0 = CP$$

输出方程：$Z = \overline{Q_2}$

设定初态，依次求出次态：

$$Q_2^{n+1} = J_2\overline{Q_2^n} + \overline{K_2}Q_{2.}^n = \overline{Q_2^n}Q_1^nQ_0^n \qquad CP_2 = CP$$

$$Q_1^{n+1} = J_1\overline{Q_1^n} + \overline{K_1}Q_1^n = \overline{Q_1^n} \qquad CP_1 = Q_0^n$$

$$Q_0^{n+1} = J_0\overline{Q_0^n} + \overline{K_0}Q_0^n = \overline{Q_2^n}\,\overline{Q_0^n} + Q_2^nQ_0^n \qquad CP_0 = CP$$

$Q_2^nQ_1^nQ_0^n = 000 \rightarrow 001 \rightarrow \mathbf{010} \rightarrow \mathbf{011} \rightarrow \mathbf{100} \rightarrow \mathbf{000}$

$Q_2^nQ_1^nQ_0^n = 101 \rightarrow 001$

$Q_2^nQ_1^nQ_0^n = 110 \rightarrow \mathbf{010}$

$Q_2^nQ_1^nQ_0^n = 111 \rightarrow \mathbf{011}$

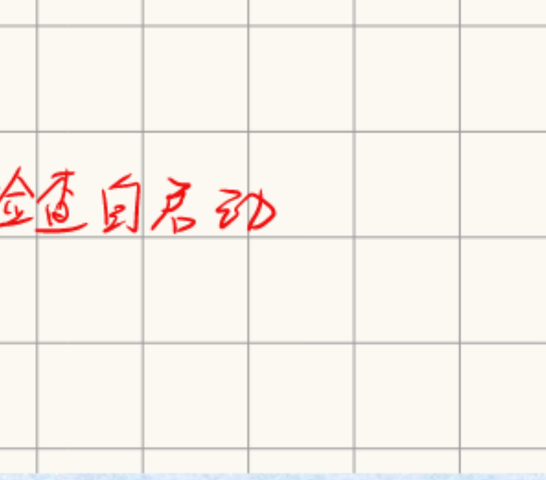
电路的状态转换规律已非常清楚了

这样可以更方便简洁地列出状态转换真值表

状态真值表　上升沿翻转

CP 有效沿	Q_2	Q_1	Q_0	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	Z
1	0	0	0	0	0	1	1
2	0	0	1	0	1	0	1
3	0	1	0	0	1	1	1
4	0	1	1	1	0	0	1
5	1	0	0	0	0	0	0
6	1	0	1	0	0	1	0
7	1	1	0	0	1	0	0
8	1	1	1	0	1	1	0

(3) 状态转移图



421BCD编码的异步五进制加法计数器

能够自启动的时序逻辑电路

5.3 同步时序电路设计

(1) 画状态转化图，二进制编码；选触发器

(2) 列真值表

(3) 现态和输入为变量，求输入逻辑函数，输出函数，检查自启动

(4) 画电路图

[例] 用下降沿触发的JK触发器，设计一个同步的按8421编码计数的十进制减法计数器。

解：题目已明确告诉计数编码、触发器等，所以，实际上已告诉了状态转换图

①

直接做第二步：列出状态真值表（即I/O关系）

②

CP	初态 Q^n				次态 Q^{n+1}				$Q^n \rightarrow Q^{n+1}$ 对JK要求				输出 B
	Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0	J_3K_3	J_2K_2	J_1K_1	J_0K_0	
0	0	0	0	0	1	0	0	1	1×	0×	0×	1×	1
1	1	0	0	1	1	0	0	0	×0	0×	0×	×1	0
2	1	0	0	0	0	1	1	1	×1	1×	1×	1×	0
3	0	1	1	1	0	1	1	0	0×	×0	×0	×1	0
4	0	1	1	0	0	1	0	1	0×	×0	×1	1×	0
5	0	1	0	1	0	1	0	0	0×	×0	0×	×1	0
6	0	1	0	0	0	0	1	1	0×	×1	1×	1×	0
7	0	0	1	1	0	0	1	0	0×	0×	×0	×1	0
8	0	0	1	0	0	0	0	1	0×	0×	×1	1×	0
9	0	0	0	1	0	0	0	0	0×	0×	0×	×1	0

求J、K和B的函数式：（用卡诺图求，1010至1111转移结果在设计并没有要求，可将这六个状态都当作无关项处理，从而简化电路设计）

③

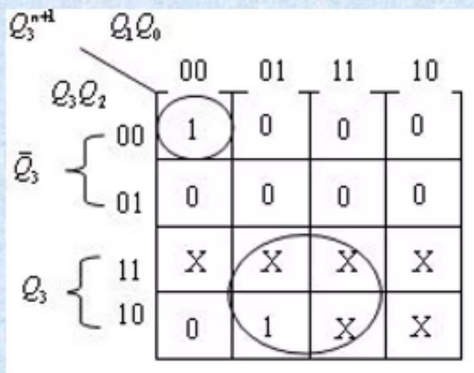
$$\begin{cases} J_3 = \overline{Q_2^n Q_1^n Q_0^n} \\ K_3 = \overline{Q_0^n} \end{cases} \qquad \begin{cases} J_1 = Q_2^n\overline{Q_0^n} + Q_3^n\overline{Q_0^n} = \overline{\overline{Q_3^n}\,\overline{Q_2^n}\,\overline{Q_0^n}} \\ K_1 = \overline{Q_0^n} \end{cases}$$

$$\begin{cases} J_2 = Q_3^n\overline{Q_0^n} \\ K_3 = \overline{Q_1^nQ_0^n} \end{cases} \qquad \begin{cases} J_0 = 1 \\ K_0 = 1 \end{cases} \qquad B = \overline{Q_3^nQ_2^nQ_1^nQ_0^n}$$

另一种方法是由状态真值表直接得到各触发器的次态方程

$$Q_i^{n+1} = f_i(Q_3, Q_2, Q_1, Q_0), i = 0,1,2,3$$

然后通过和 $Q_i^{n+1} = J_i\overline{Q_i} + \overline{K_i}Q_i$ 比对得到 J_i, K_i 逻辑函数关系



$$Q_3^{n+1} = \overline{Q_3}\,\overline{Q_2}\overline{Q_1}\overline{Q_0} + Q_3Q_0$$

可得：

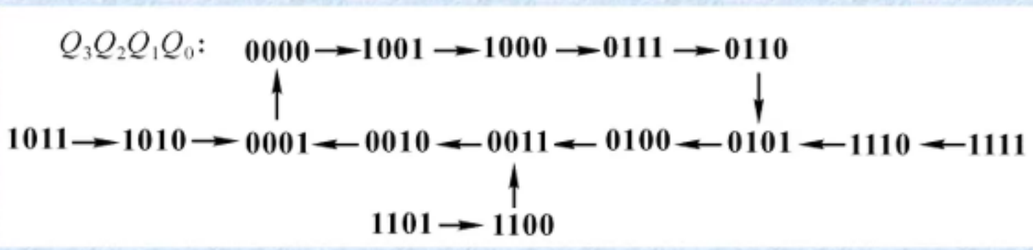
$$\begin{cases} J_3 = \overline{Q_2^n}\,\overline{Q_1^n}\,\overline{Q_0^n} \\ K_3 = \overline{Q_0^n} \end{cases}$$

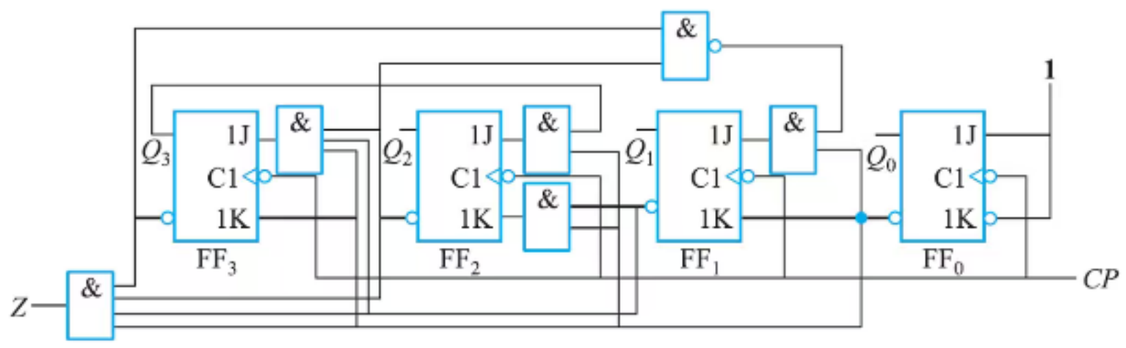
设计时利用了六个无关项，完成后必须检查电路能否自启动

8421 BCD编码的十进制减法计数器

4

得到实际状态转移图：





满足设计要求

5.4中规模集成计数器

集成计数器 74HC161 功能表

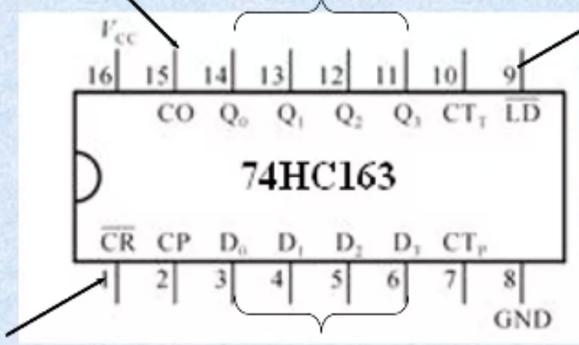
输入									触发器状态			
CP	$\overline{CR}$	$\overline{LD}$	CT_P	CT_T	D_3	D_2	D_1	D_0	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}
x	0	x	x	x	x	x	x	x	0	0	0	0
↑	1	0	x	x	A_3	A_2	A_1	A_0	A_3	A_2	A_1	A_0
↑	1	1	1	1	x	x	x	x	四位二进制加法计数			
↑	1	1	0	x	x	x	x	x	保持			
x	1	1	x	0	x	x	x	x	不变			

进位输出端

$CO = Q_3Q_2Q_1Q_0$

四位二进制数输出

同步置数控制端



CT_P、CT_T用于功能扩展控制

同步总清零（总复位端）

四位数据输入端

同步清零，同步置数

输入									触发器状态			
CP	$\overline{CR}$	$\overline{LD}$	CT_P	CT_T	D_3	D_2	D_1	D_0	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}
↑	0	x	x	x	x	x	x	x	0	0	0	0
↑	1	0	x	x	A_3	A_2	A_1	A_0	A_3	A_2	A_1	A_0
↑	1	1	1	1	x	x	x	x	四位二进制加法计数			
↑	1	1	0	x	x	x	x	x	同步保持			
x	1	1	x	0	x	x	x	x	异步保持			

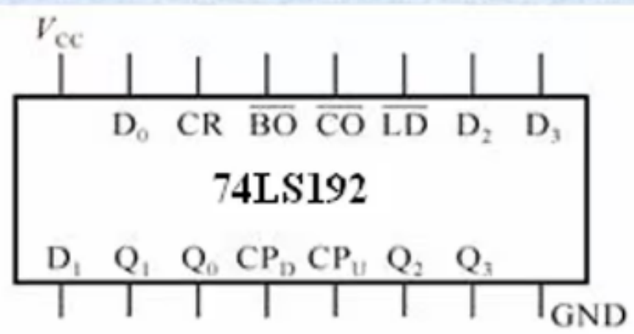
反馈清零法/置数法

◆十进制可逆计数器74LS192

74LS192 功能表

输入								触发器状态			
CR	$\overline{LD}$	CP_U	CP_D	D_3	D_2	D_1	D_0	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}
1	x	x	x	x	x	x	x	0	0	0	0
0	0	x	x	A_3	A_2	A_1	A_0	A_3	A_2	A_1	A_0
0	1	↑	1	x	x	x	x	8421 十进制加法计数			
0	1	1	↑	x	x	x	x	8421 十进制减法计数			
0	1	1	1	x	x	x	x	保持			

$$\begin{cases} \overline{CO} = \overline{Q_3 Q_0 \overline{CP_U}} \\ \overline{BO} = \overline{\overline{Q_3}\,\overline{Q_2}\,\overline{Q_1}\,\overline{Q_0}\,\overline{CP_D}} \end{cases}$$



区别：

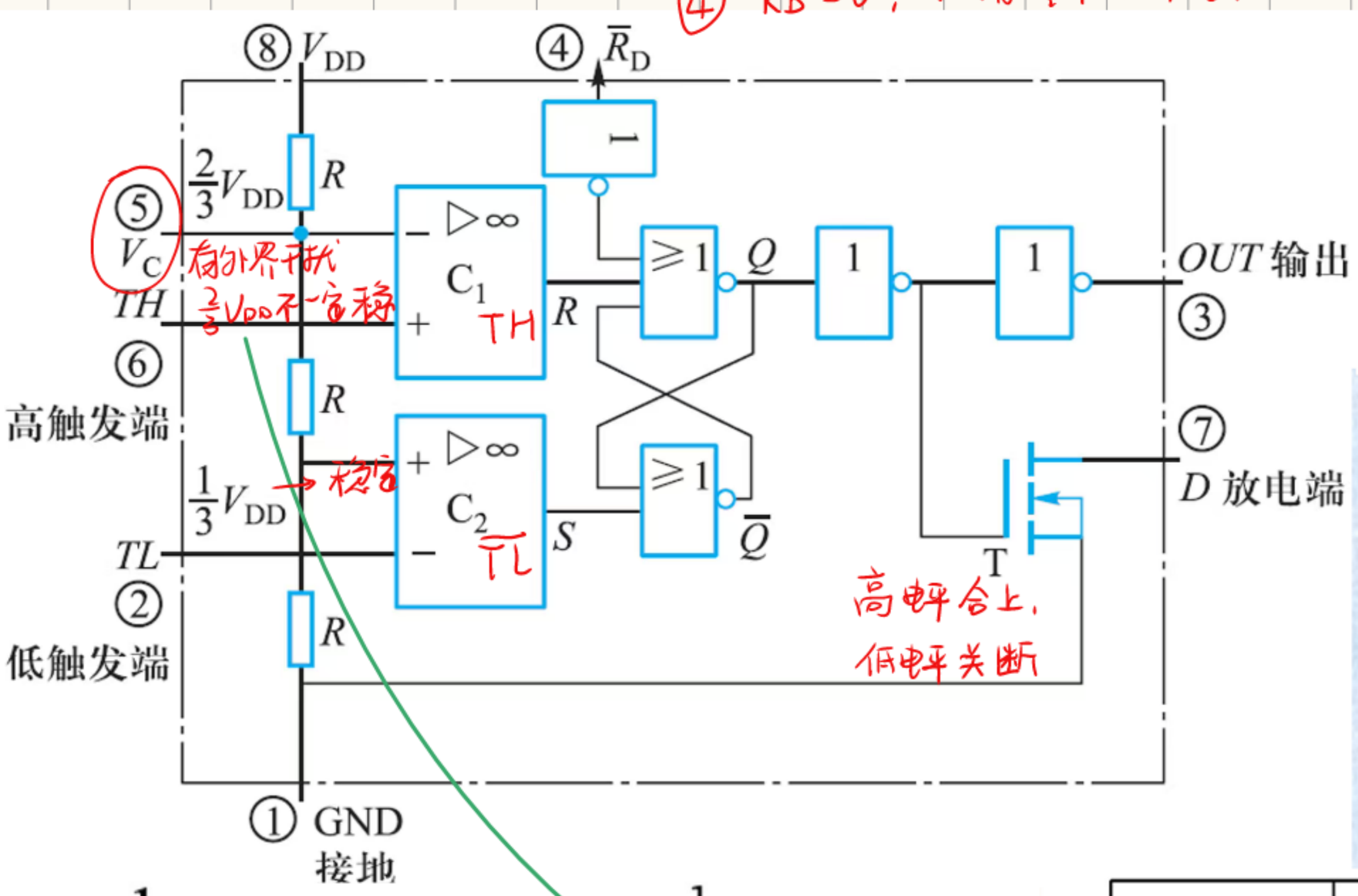
74HC217

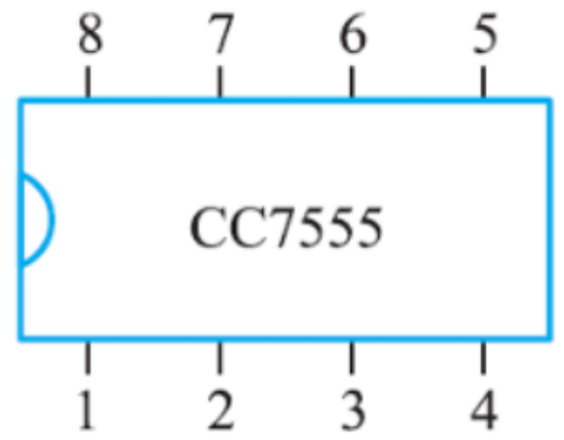
$$\begin{cases} \overline{CO} = \overline{Q_3 Q_0 CP_U} \\ \overline{BO} = \overline{Q_3 Q_2 Q_1 Q_0 CP_D} \end{cases}$$

第六章 555集成定时器

6.1 基本结构和工作特性

④ $\overline{R_D}=0$，将清零，输出0



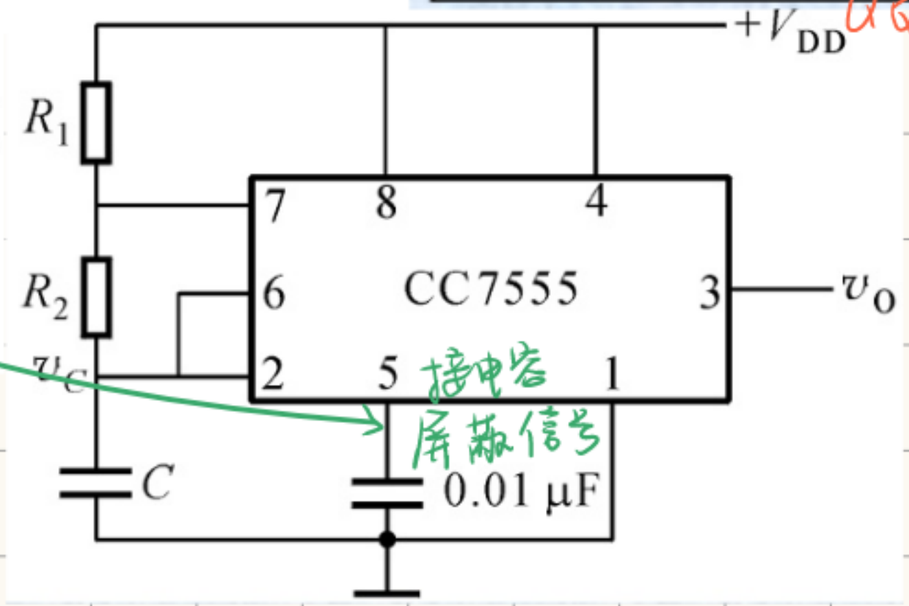


R	S	Q
0	0	保持
0	1	1
1	0	0
1	1	禁用

置1
置0
Q Q都0 矛盾

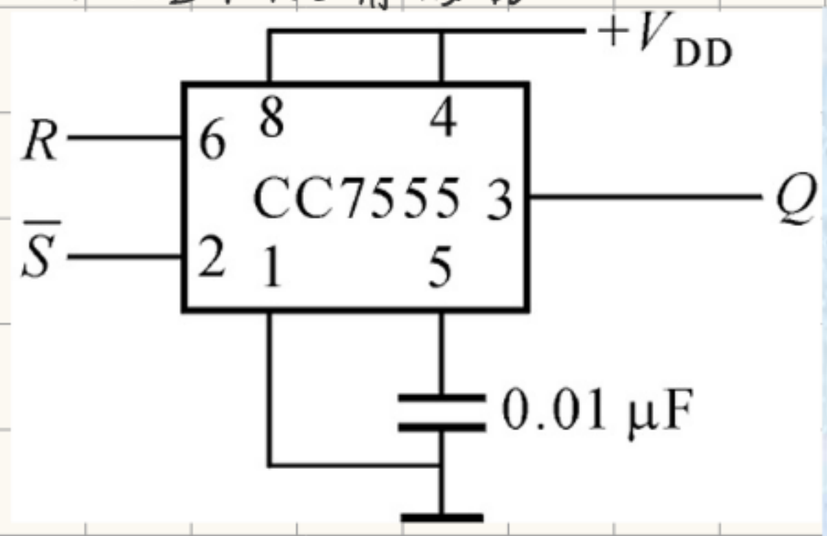
$$V_{TL}>\frac{1}{3}V_{DD}\ \text{时，S=0} \qquad V_{TL}<\frac{1}{3}V_{DD}\ \text{时，S=1}$$

$$V_{TH}>\frac{2}{3}V_{DD}\ \text{时，R=1} \qquad V_{TH}<\frac{2}{3}V_{DD}\ \text{时，R=0}$$



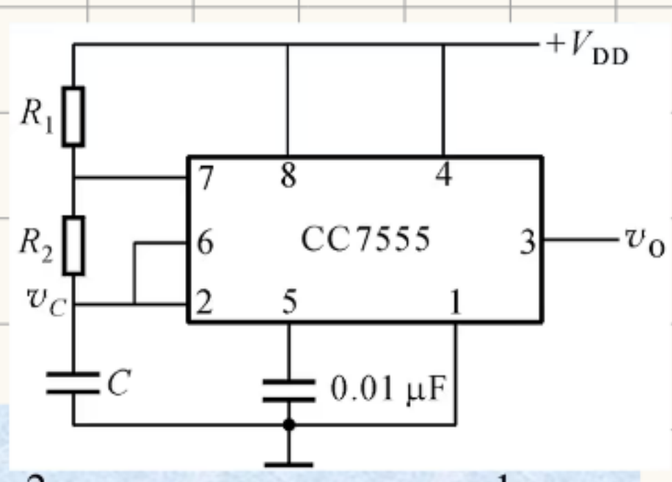
6.2 555定时器典型应用

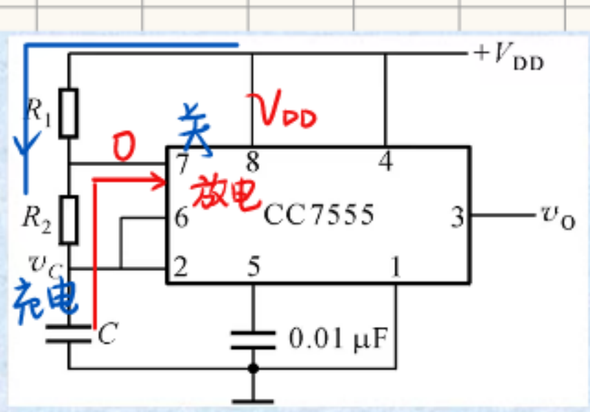
6.2.1 构成基本RS触发器

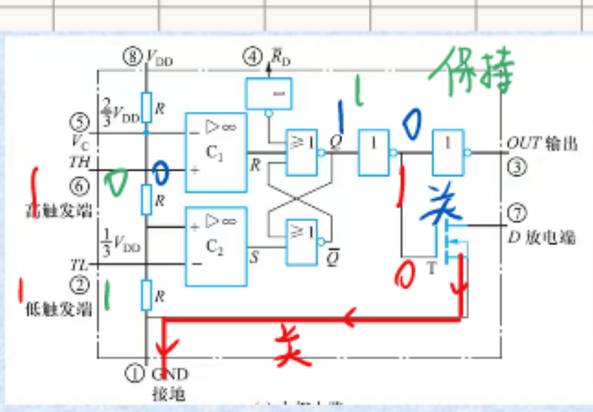


外触发信号		内部RS		
R	$\overline{S}$	R	S	Q
0	0	0	1	1
0	1	0	0	保持
1	× 0 不允许	1	1	禁用
1	1	1	0	0

6.2.2 构成多谐振荡器







$$T=T_1+T_2$$

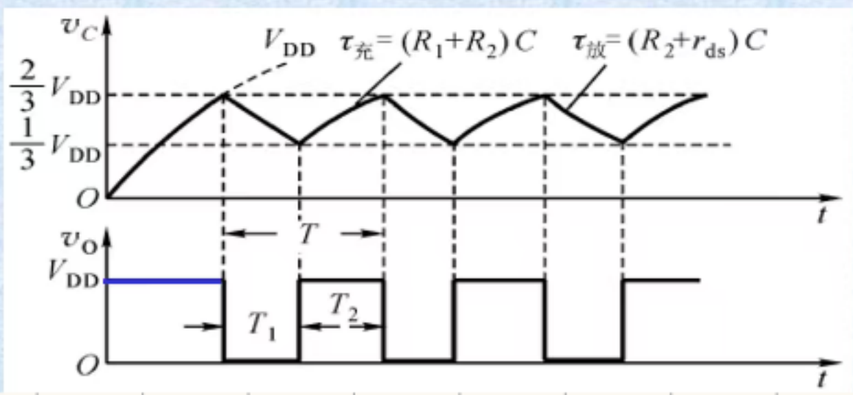
$$=R_2C\ln\frac{0-\frac{2}{3}V_{DD}}{0-\frac{1}{3}V_{DD}}+(R_1+R_2)C\ln\frac{V_{DD}-\frac{1}{3}V_{DD}}{V_{DD}-\frac{2}{3}V_{DD}}$$

$$=(R_1+2R_2)C\ln 2$$

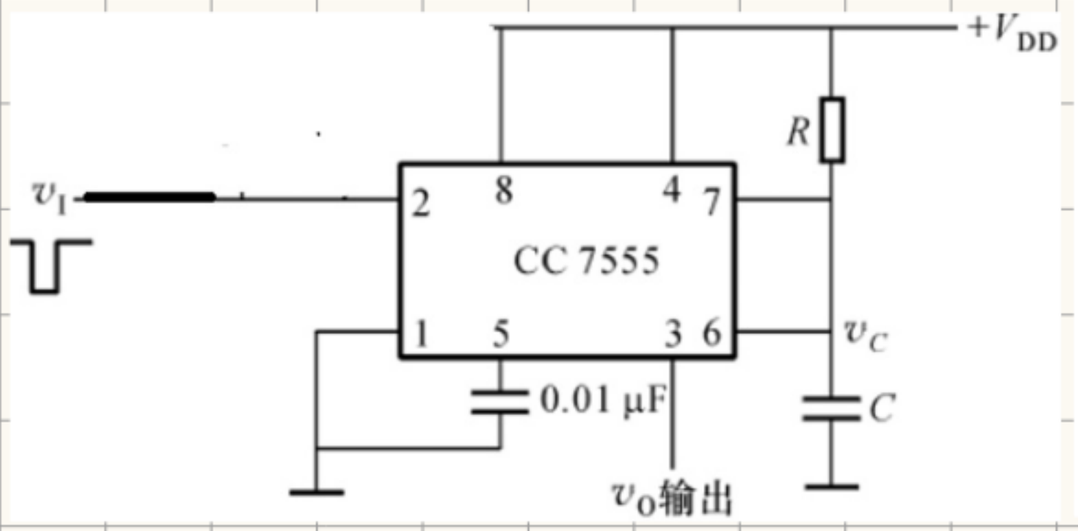
该公式始终大于50%

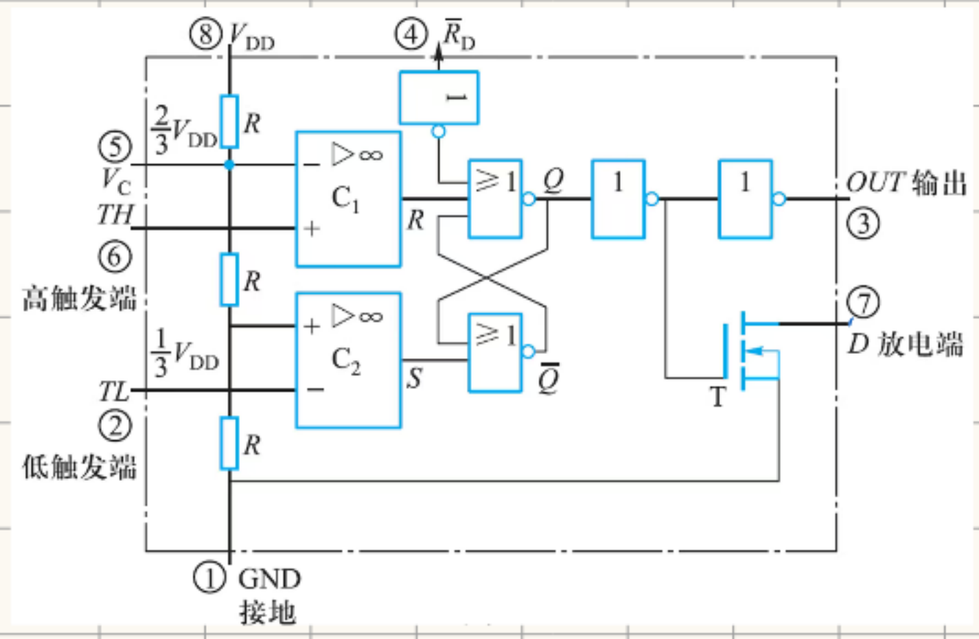
$$\text{占空比}\quad D=\frac{T_2}{T}=\frac{R_1+R_2}{R_1+2R_2}$$

➢ 合上电源瞬间，C上的电压不能突变，引脚2、6都为0，输出为高电平，内部场效应管截止



6.2.3 构成单稳态触发器





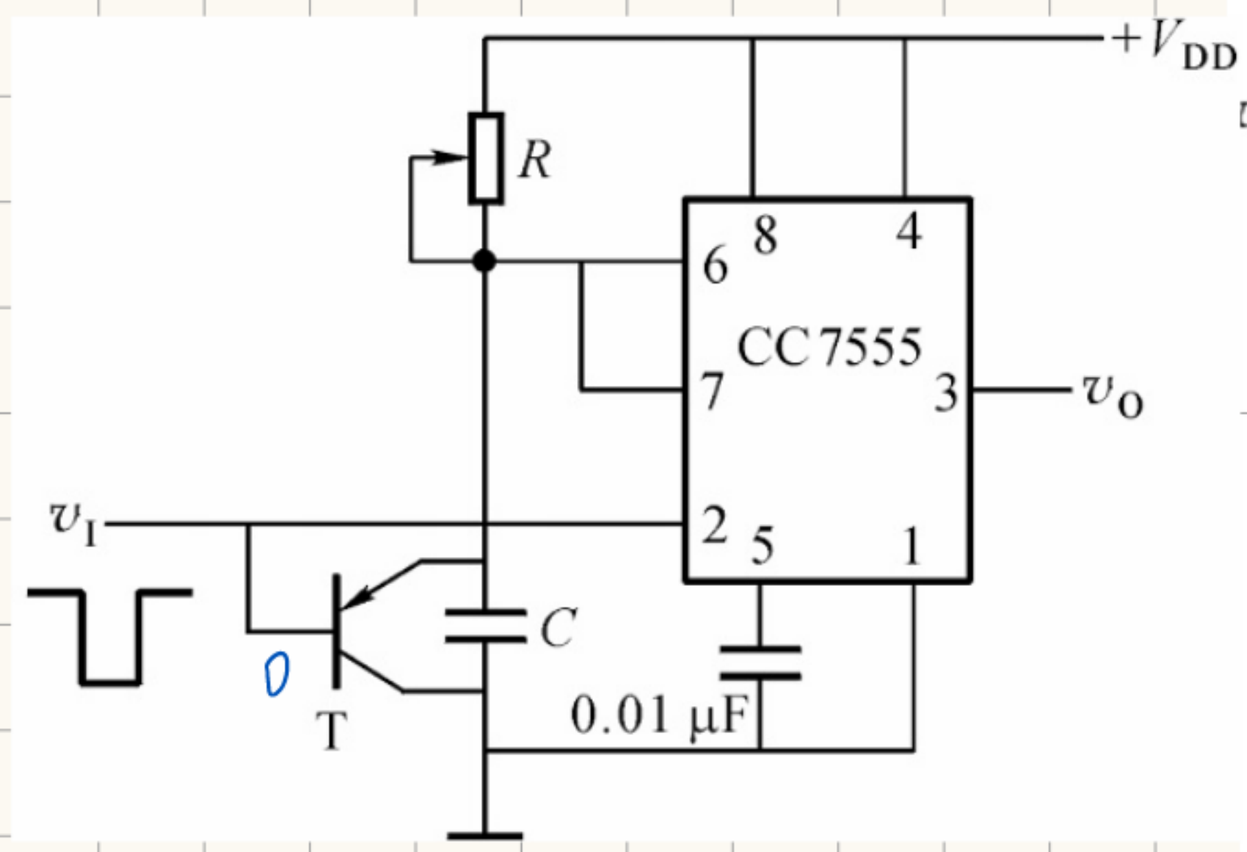
① 无信号. 2 高电平 5 0 . 31 脚和6,7 低电平 R0 输出低电平

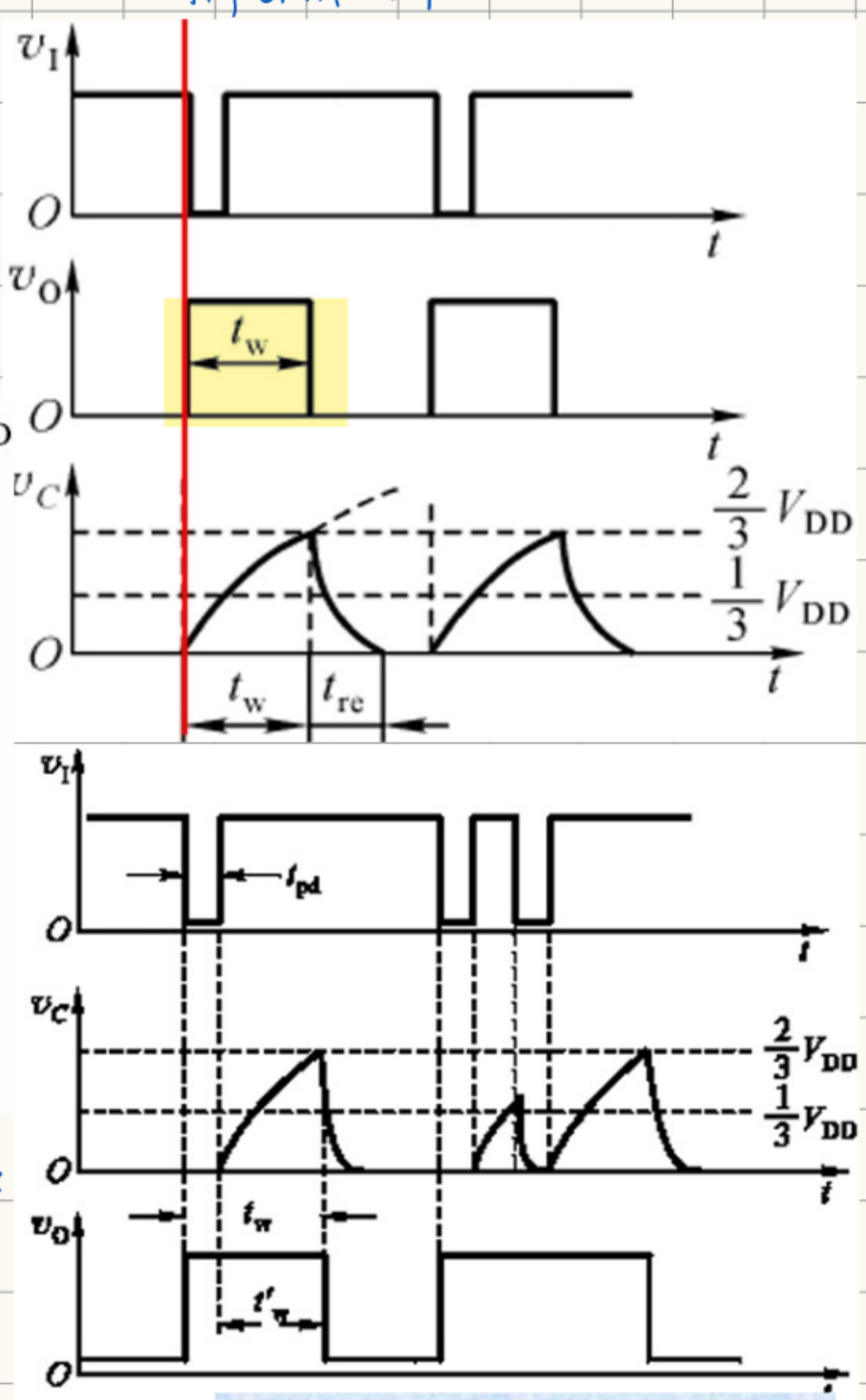
② 加入触发 2 低电平, out 跳变高电平

⇒ 7 高电平 开始充电, 开关 断开, 保持

③ 达到 $\frac{2}{3}V_{DD}$. 6 才翻转, 因此有一段稳定时间

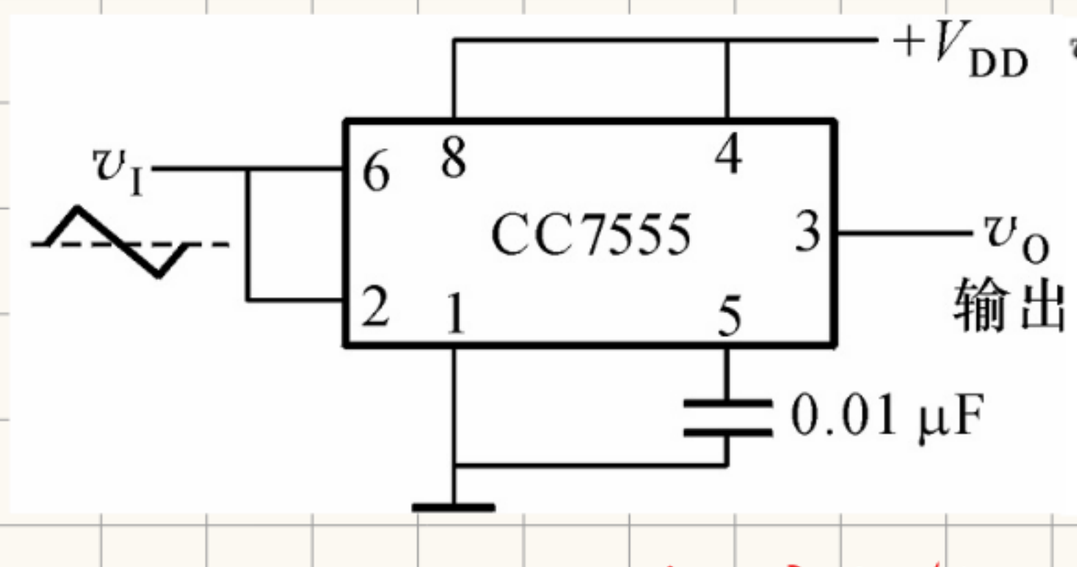
可重触发的单稳态触发器

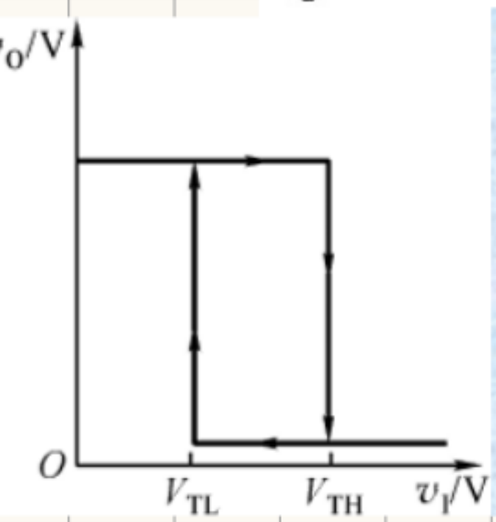


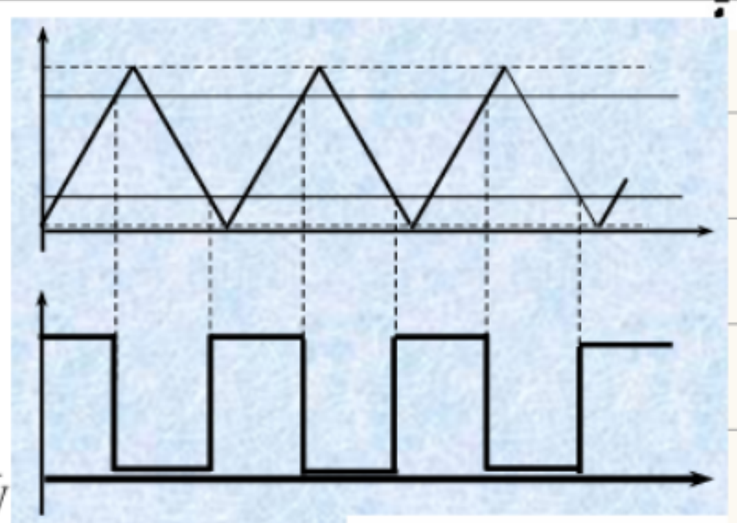


按下 C 迅速放电, 可再次触发, V_C 下降

6.2.4 构成斯密特触发器

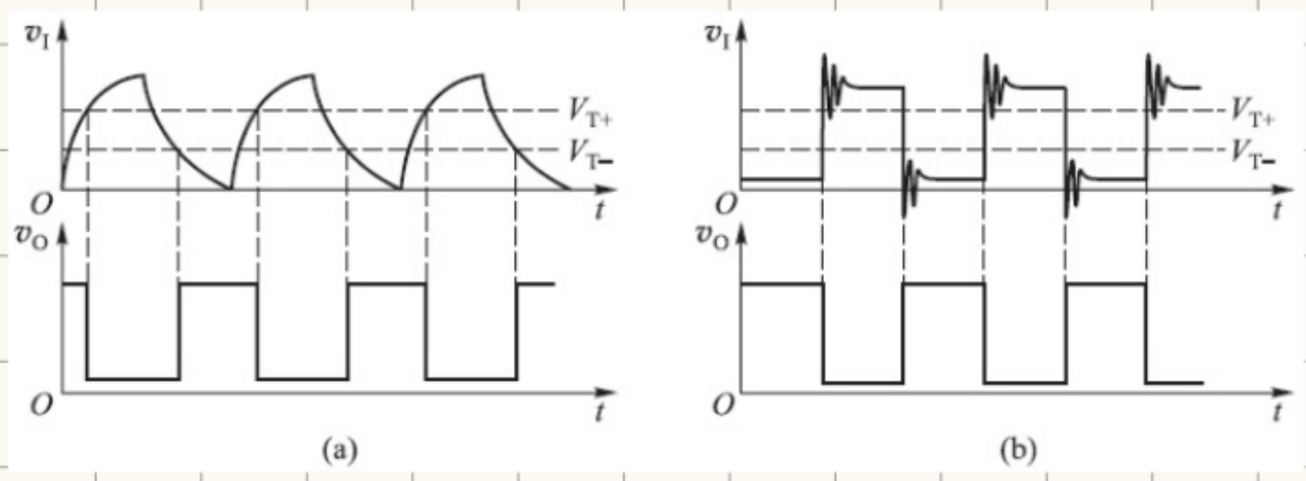


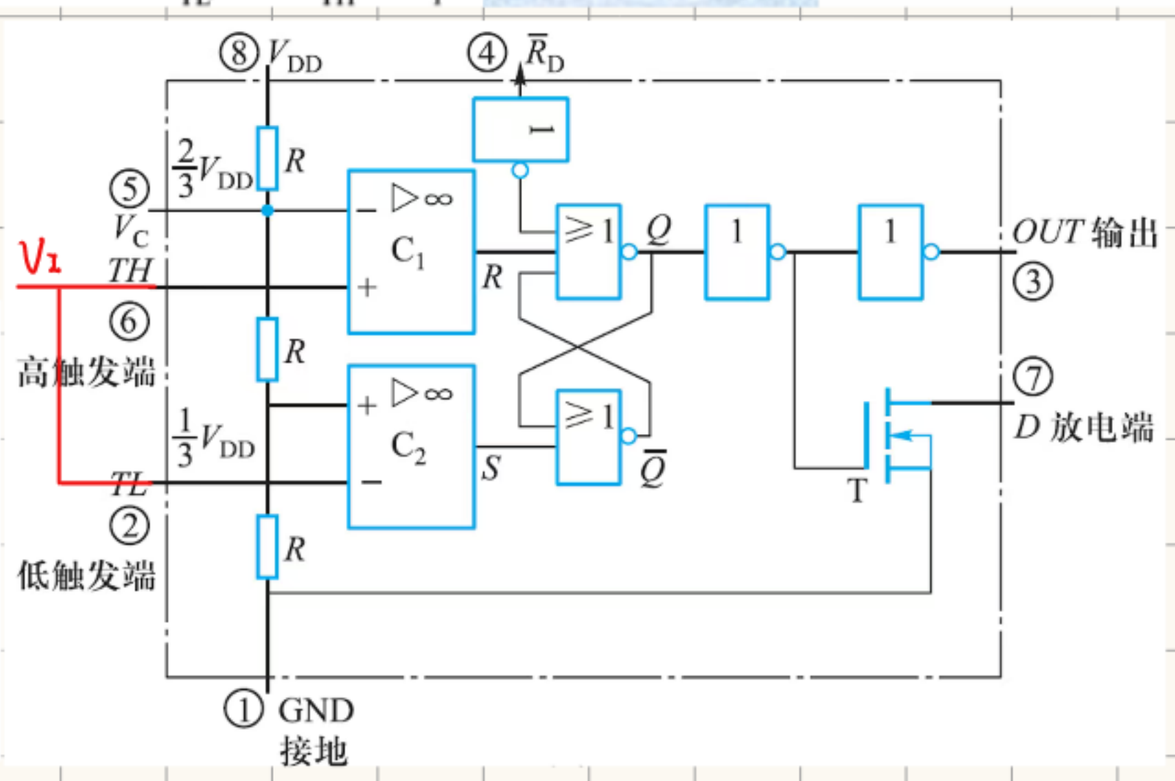




V_1 大于 $\frac{2}{3}V_{CC}$ 输出 0

V_1 小于 $\frac{2}{3}V_{CC}$ 输出 1





第七章 存储器

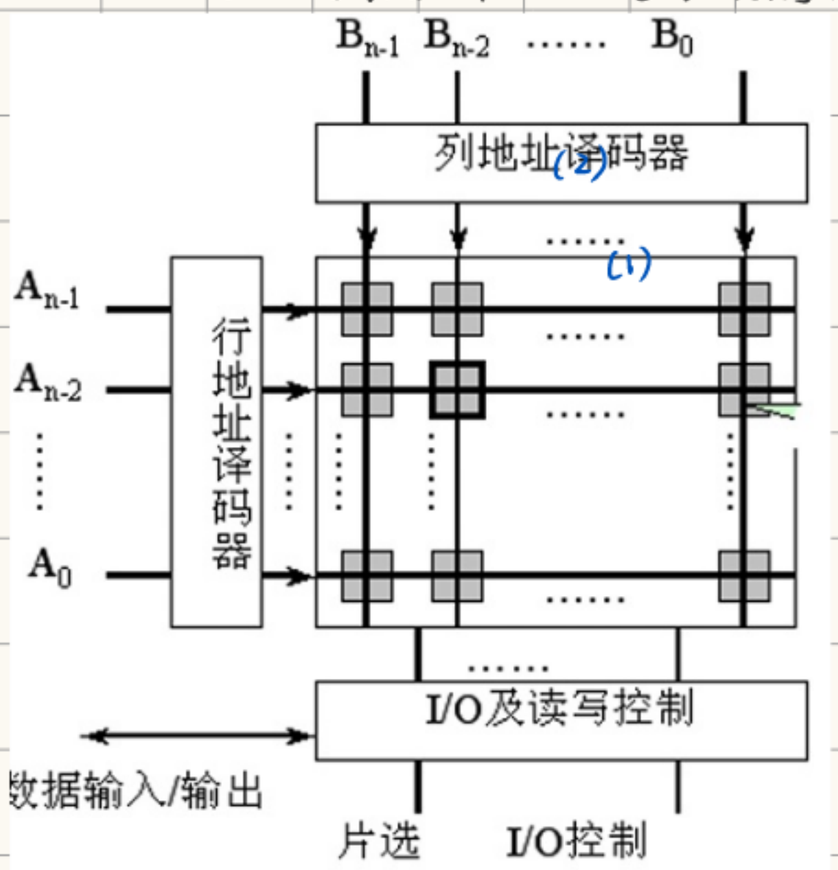
7.1 存储器的分类

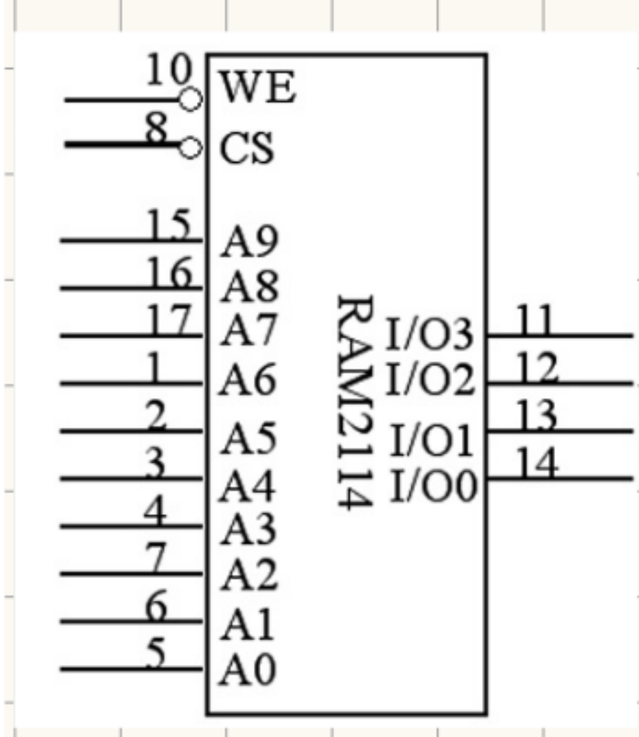
① 随机存储器 RAM { 动态 DRAM / 静态 SRAM

② 只读存储器 ROM { 不可改写 —— 掩膜 ROM ; 可改写 —— PROM { EPROM / EEPROM / 闪存

RAM一般结构

(1) 存储体　(2) 行列地址译码器　(3) 读/写控制及I/O缓冲器





RAM一般结构

CS=0 选中该片ROM为工作状态

A9-A0 具体想访问的地址

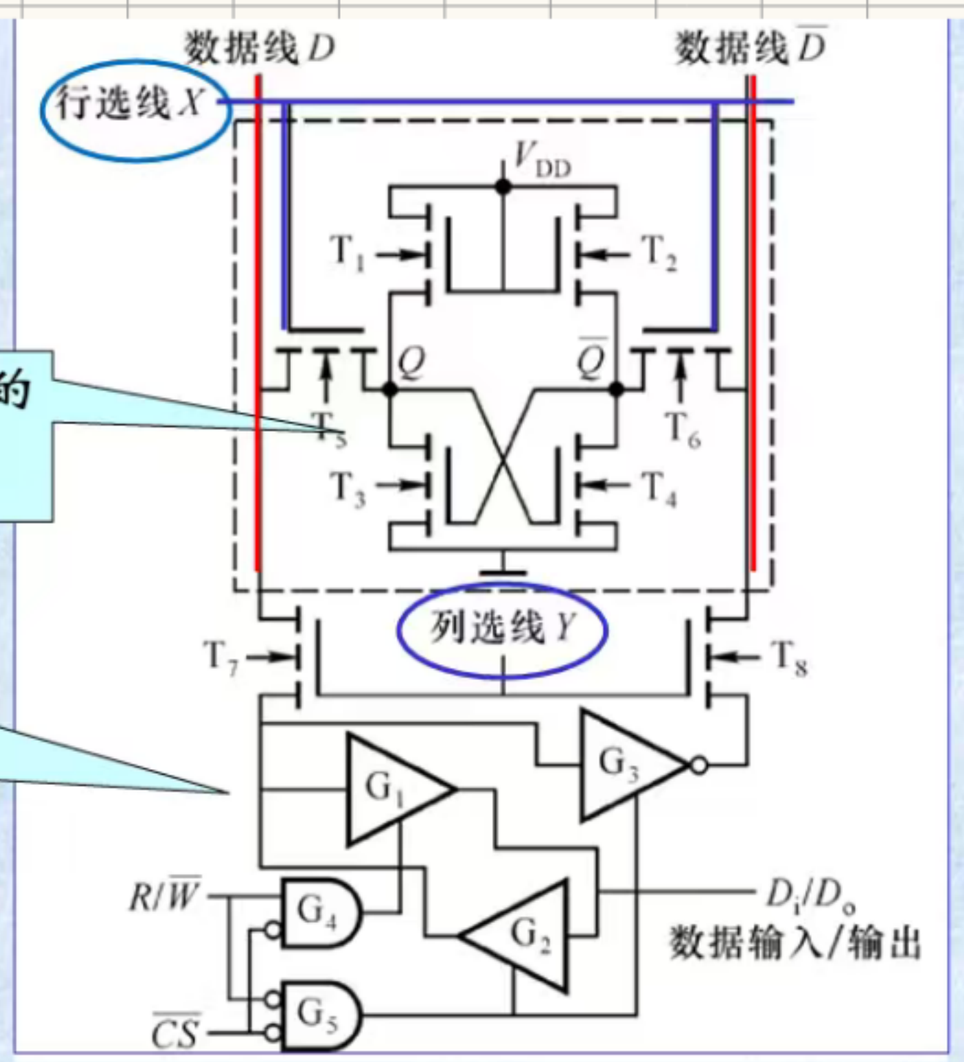
WE=1 读出信息

WE=0 写入信息

7.2 RAM中的存储单元

- 静态存储单元（SRAM）的典型结构

T_5、T_6、T_7、T_8都是门控管，只要栅极高电平，这些管子就工作在可变电阻区，当作开关。



NMOS结构的静态SRAM

写入信息的操作过程

在第一次写入信息之前，存储单元中的信息是随机信息。

假定要写入信息“1”：

◆ 地址码加入，地址有效后，相对应的行选线X和列选线Y都为高电平，T_5、T_6、T_7、T_8导电。

◆◆ 片选信号$\overline{CS}$有效（低电平）

◆◆◆ 写入信号有效 $R/\overline{W}=0$

这时三态门G_2、G_3为工作态，G_1输出高阻态，信息“1”经G_2、T_7、T_5达到Q端；经G_3反相后信息“0”经T_8、T_6达到$\overline{Q}$。T_4导电，T_3截止，显然，信息“1”已写入了存储单元。

假定要读出“1”：

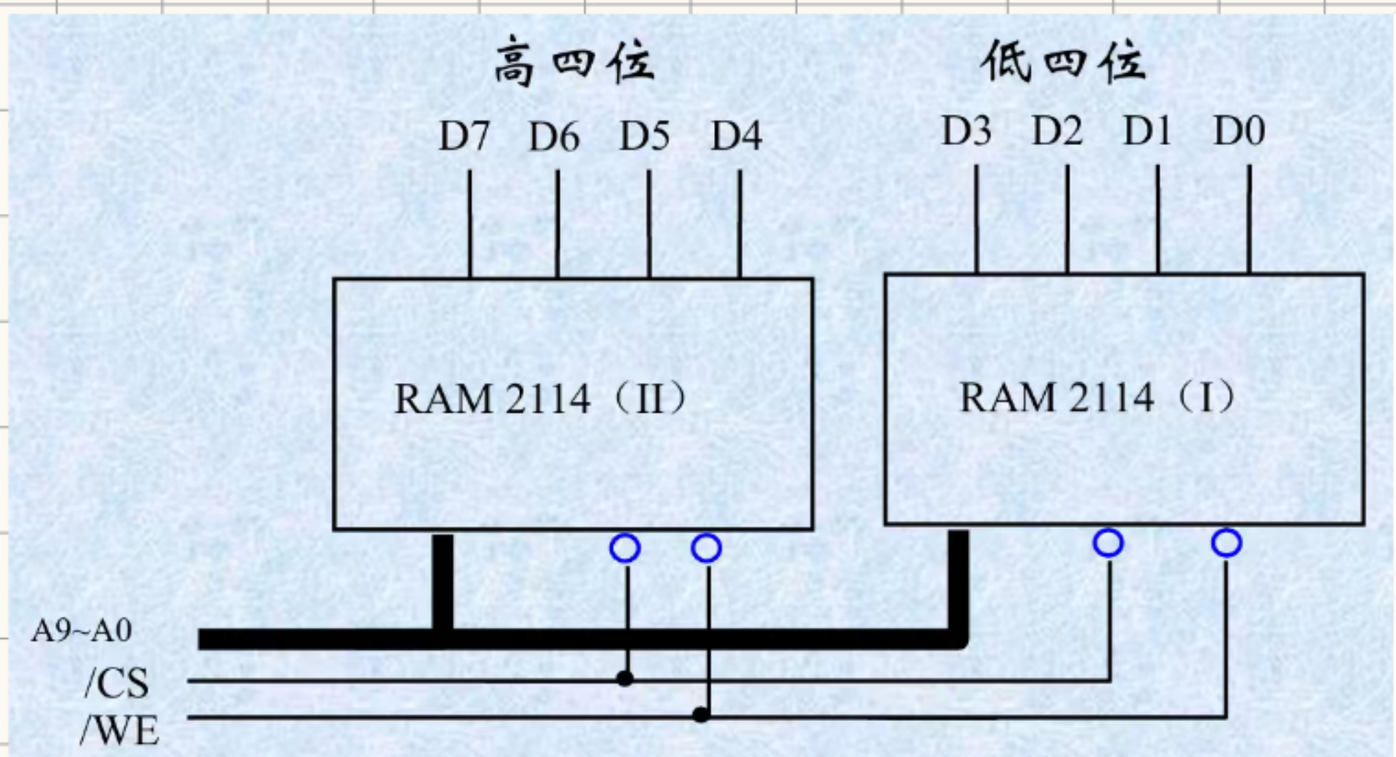
- 访问该地址单元的地址码有效
- 片选有效 $\overline{CS}=0$
- 读操作有效 $R/\overline{W}=1$

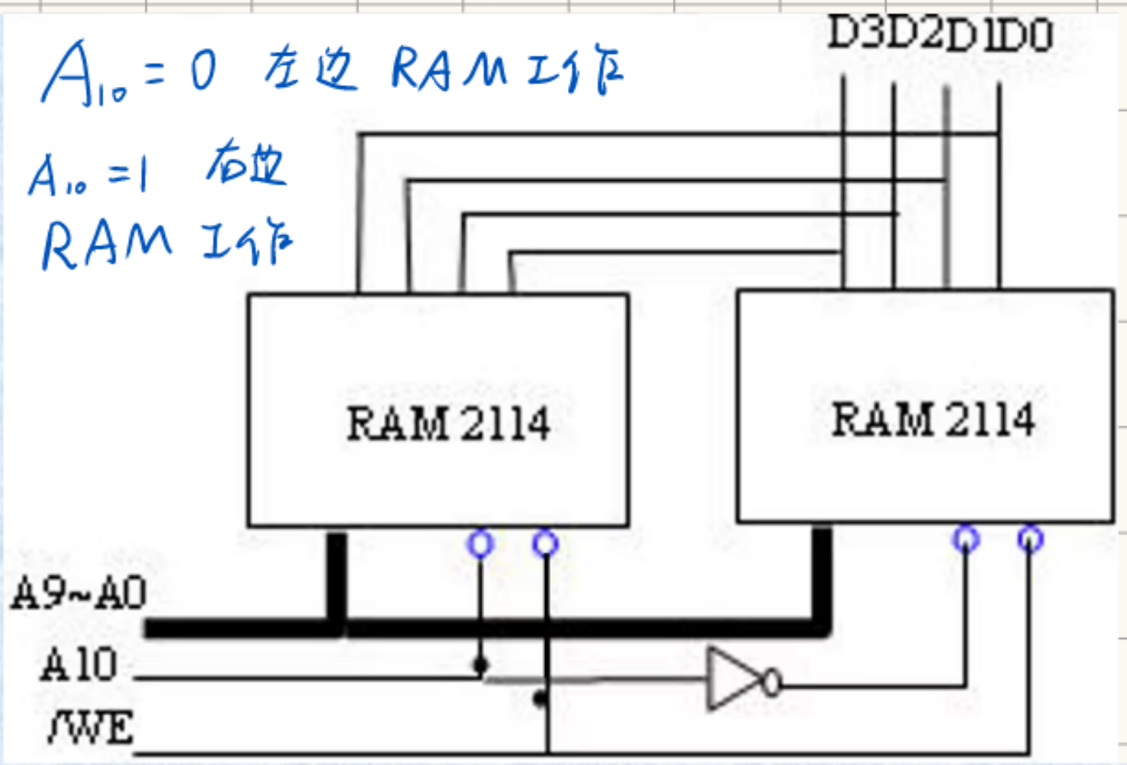
此时：三态门G_1工作态，G_2、G_3高阻态，存储单元中的信息“1”经T_5、T_7、G_1三态门读出。

SRAM（静态）比 DRAM（动态）快 2~5倍

7.2.2 SRAM的拓展

数据位拓展、地址拓展



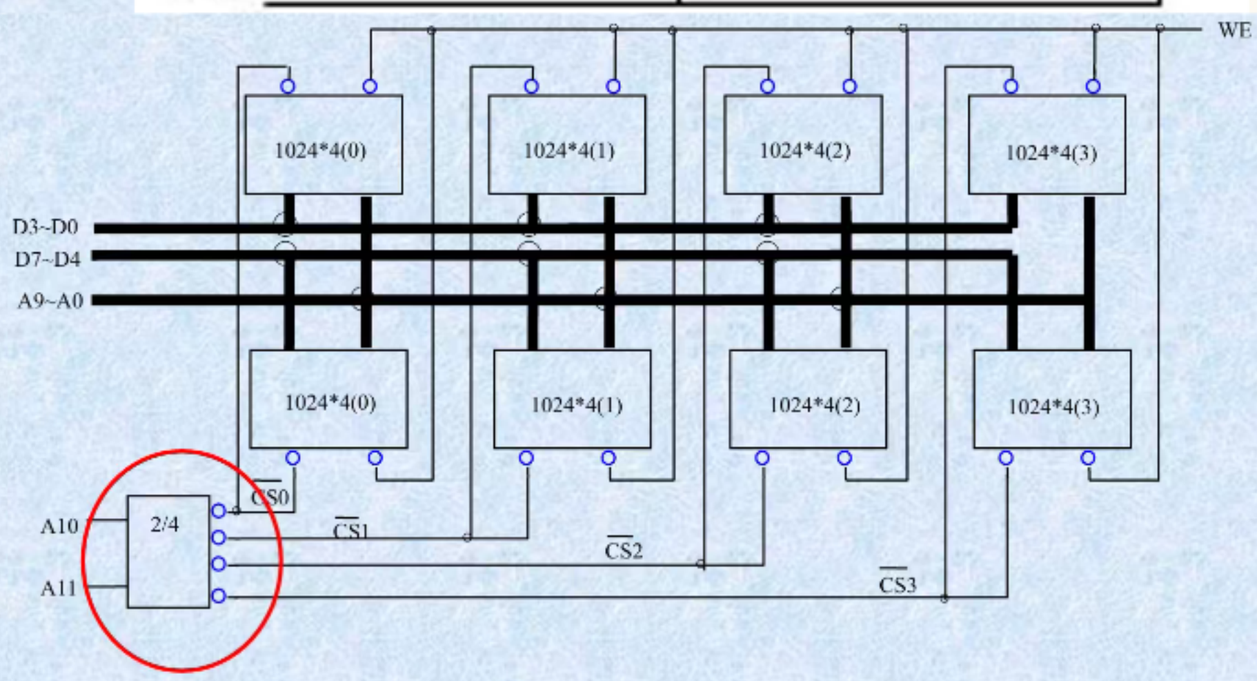


★★★SRAM容量的扩展---字位扩展

地址扩展，数据位扩展

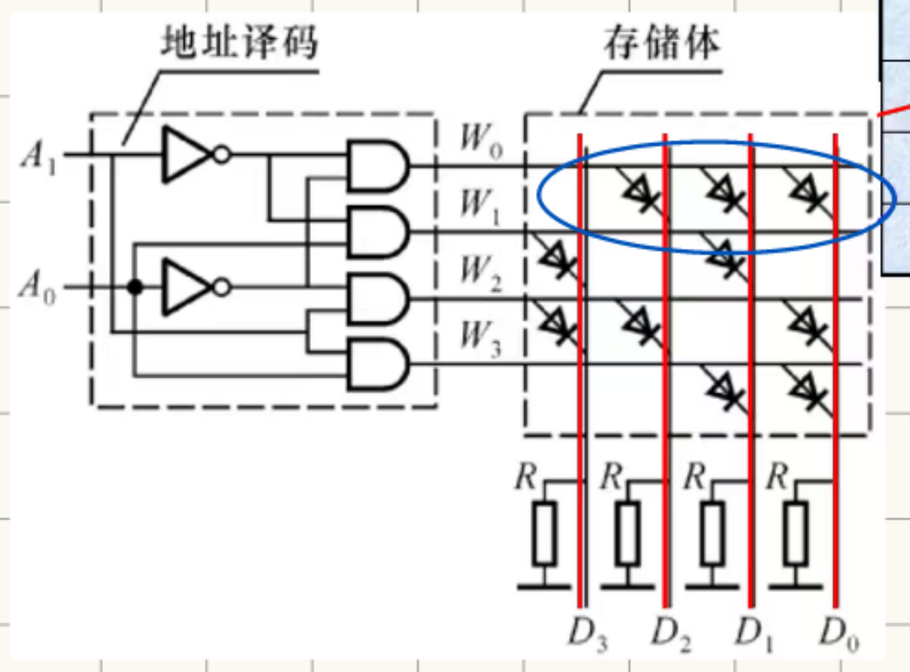
1024×4　　×4　　×2

例 用RAM2114，扩展成容量为4096×8字位（32K）的RAM。　地址4倍　数据2倍

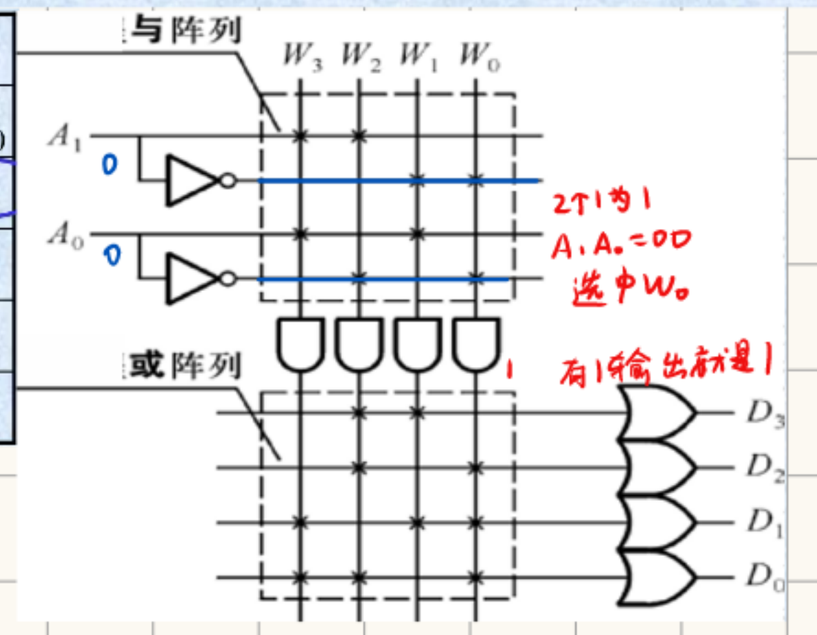


解：4096需要12位地址，而RAM2114只有10位地址，所以需要进行地址扩展，同时应该将一字4位，扩展成一字8位。字的位扩展用前面方法，地址扩展用译码器完成，用8片RAM2114。

7.3 只读存储器ROM



地址		字选线	字输出			
A_1	A_0	W	D_3	D_2	D_1	D_0
0	0	W_0	0	1	1	1
0	1	W_1	1	0	1	0
1	0	W_2	1	1	0	1
1	0	W_3	0	0	1	1



种类　1. 掩膜型只读存储器　（一旦形成无法更改）

2. 一次编程（改写）的只读存储器 PROM　（可编程一次，编完不可更改）

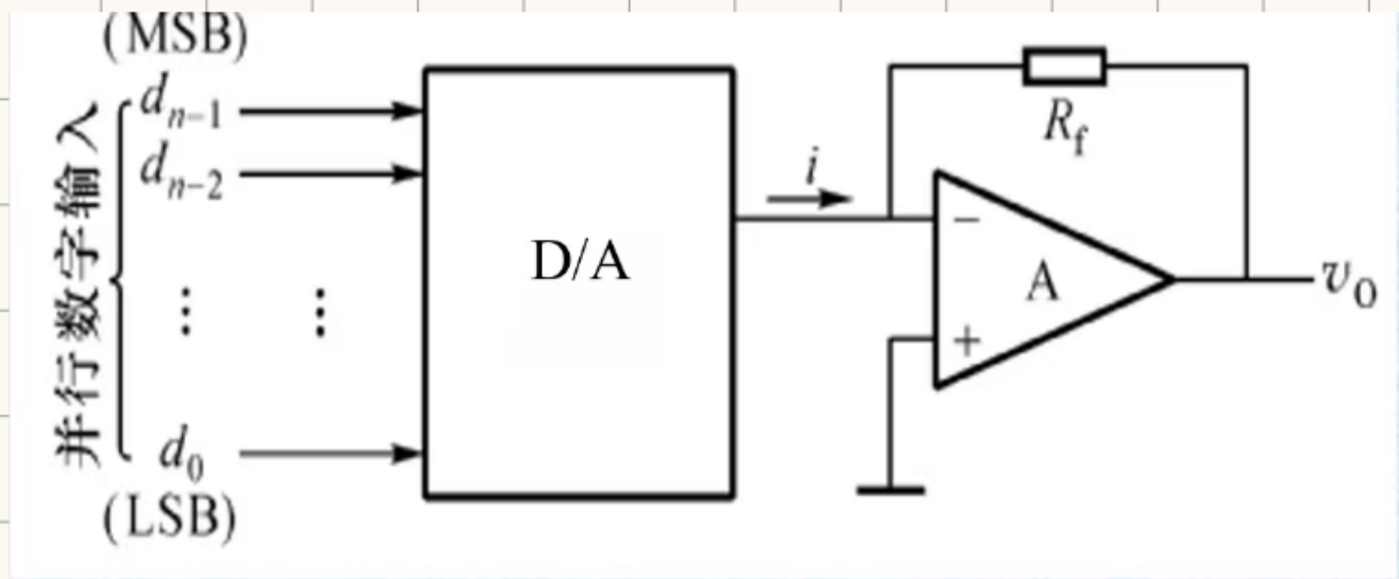
3. EPROM（紫外线可擦除可编程只读存储器）

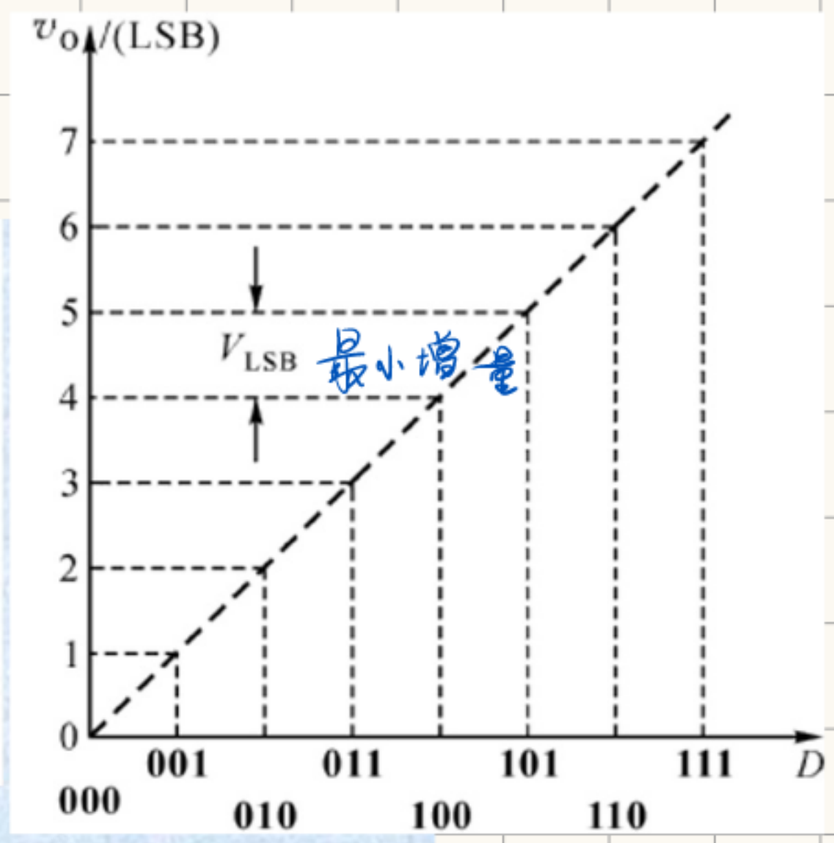
4. E^2PROM（电可擦除可编程只读存储器）

5. 快闪存技术（FLASH Memory）U盘

第八章 数模和模数转化电路

8.1 数模转化电路

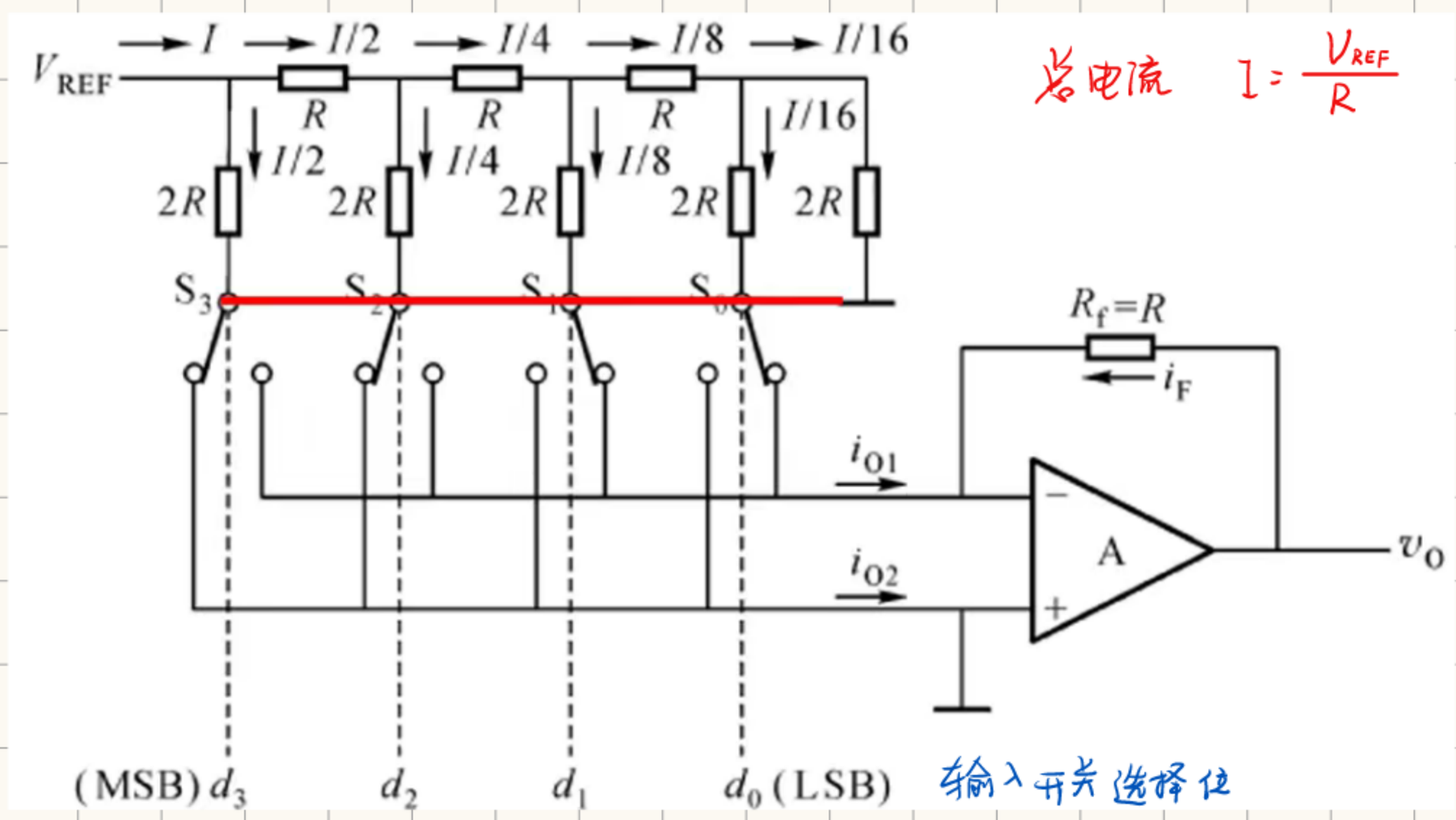




$$i = K_I D_n = K_I(d_{n-1}\cdot 2^{n-1} + d_{n-2}\cdot 2^{n-2} + \cdots + d_1\cdot 2^1 + d_0\cdot 2^0) = K_I\sum_{i=0}^{n-1} d_i\cdot 2^i$$

$$v_o = -iR_f = -K_I\cdot R_f\cdot D_n = -K_I\sum_{i=0}^{n-1} d_i\cdot 2^i$$

8.1.2 倒T形电阻网络D/A转化器



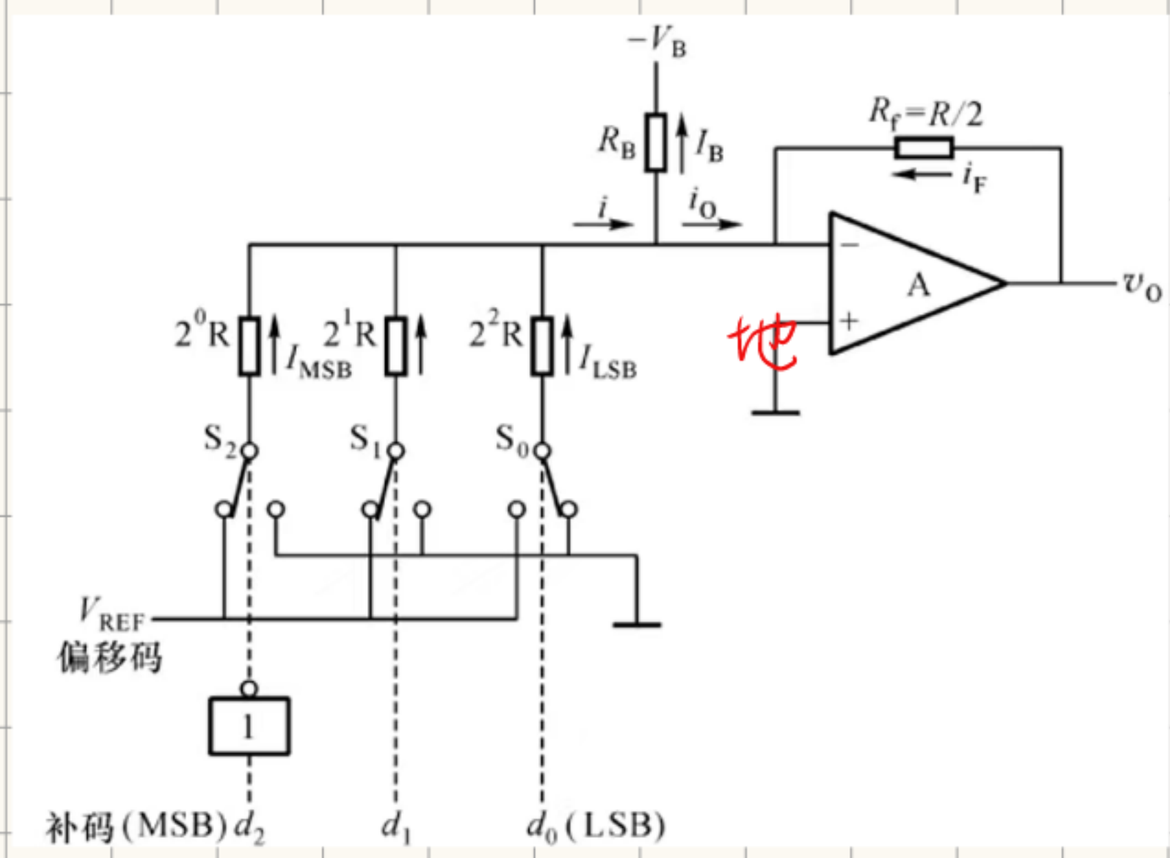
总电流 $I = \frac{V_{REF}}{R}$

输入开关选择位

$$V_o = -i_{o1}R_f = -\left(\frac{1}{2}\cdot d_3 + \frac{1}{4}d_2 + \frac{1}{8}d_1 + \frac{1}{16}d_0\right)R_f$$

$$= -\frac{V_{REF}}{R}\cdot\frac{R_f}{2^4}\left(d_3\cdot 2^3 + d_2\cdot 2^2 + d_1\cdot 2^1 + d_0\cdot 2^0\right)$$

8.1.3 D/A转换器双极性输出



$I_B = I_{MSB} = \frac{V_B}{R_B}$，使输出模拟电压为0

双极性时，中间位置0

$I_B = I_{MSB} = 2^{n-1}\ I_{LSB} = \frac{2^{n-1}}{2^n-1} I_{max}$

往上翻 2^{n-1} 位.

$$i_o = i - I_B = i - \frac{2^{n-1}}{2^n-1} I_{max}$$

$$V_o = -\left(i - \frac{2^{n-1}}{2^n-1}\right) I_{max} R_f$$

8.1.4 D/A 转换器主要技术指标与应用要点

1. 分辨率

$$\frac{V_{LSB}}{V_m} = \frac{1}{2^n - 1}$$

位数越多，分辨率越高

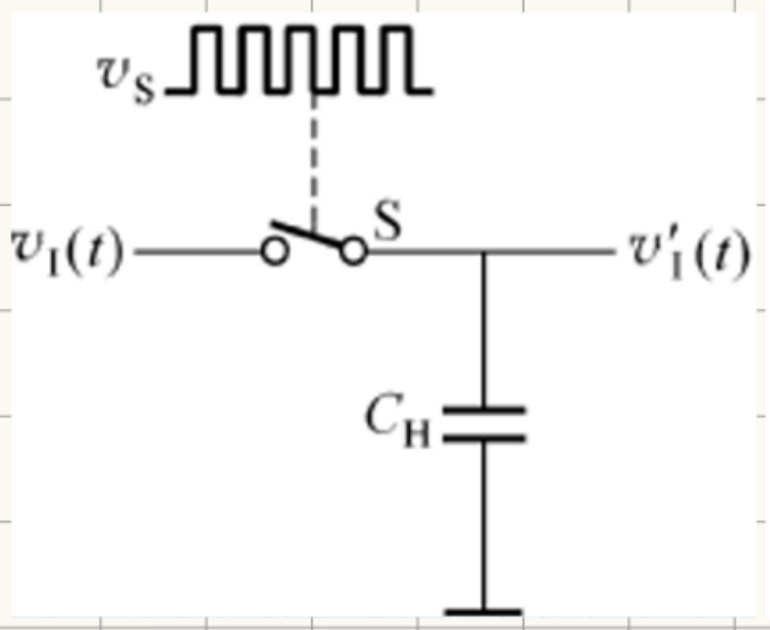
2. 转换精度

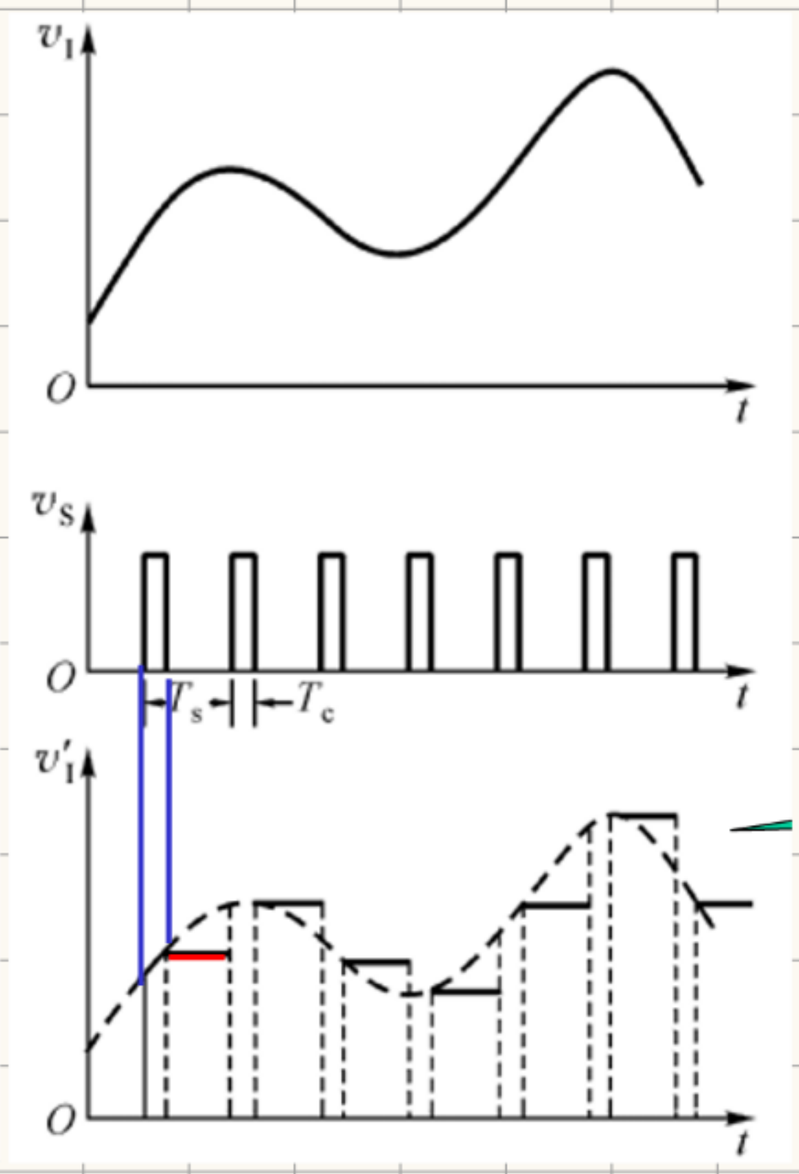
3. 非线性误差

4. 转换速率/建立时间

8.2 模数转化电路

1. 采样定理和 采样保持电路





2. 量化和编码

(1) 舍尾取整法：

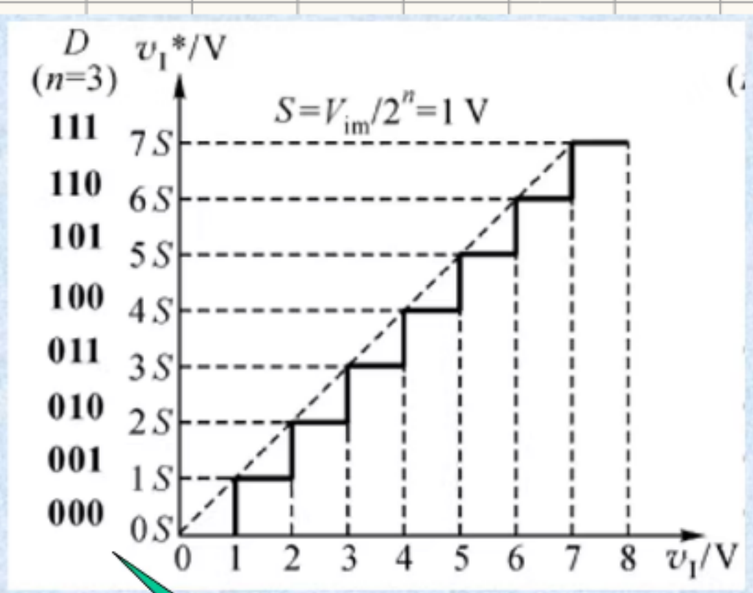
舍去不足一个量化单位的尾部，取其整数。当S表示为量化单位时，即有：

$$(K-1)S \le v_I < KS$$

取量化值 $v_I^* = (K-1)S$

如当S=1V时，$v_I = 3.8V$时，量化值$v_I^* = 3V$;

$v_I = 5V$时，量化值$v_I^* = 5V$.



舍尾取整量化特性

量化单位 $S = \frac{V_{im}}{2^n}$ V_{im}采保后的最大值电压

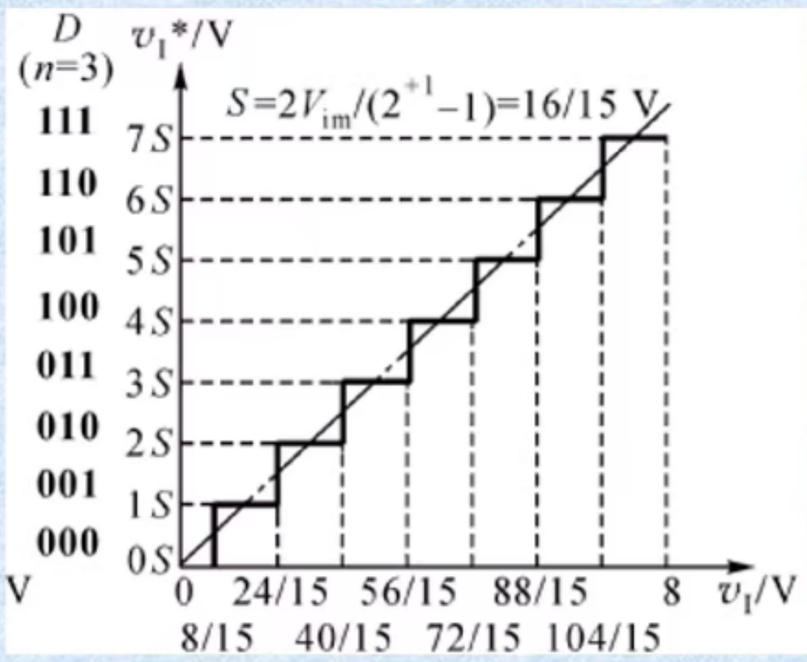
(2) 四舍五入量化法：

大于S/2量化单位的尾部归整，舍去小于S/2量化单位的尾部。

如当S=1V时

$v_I = 3.6V$时，量化值$v_I^* = 4V$;

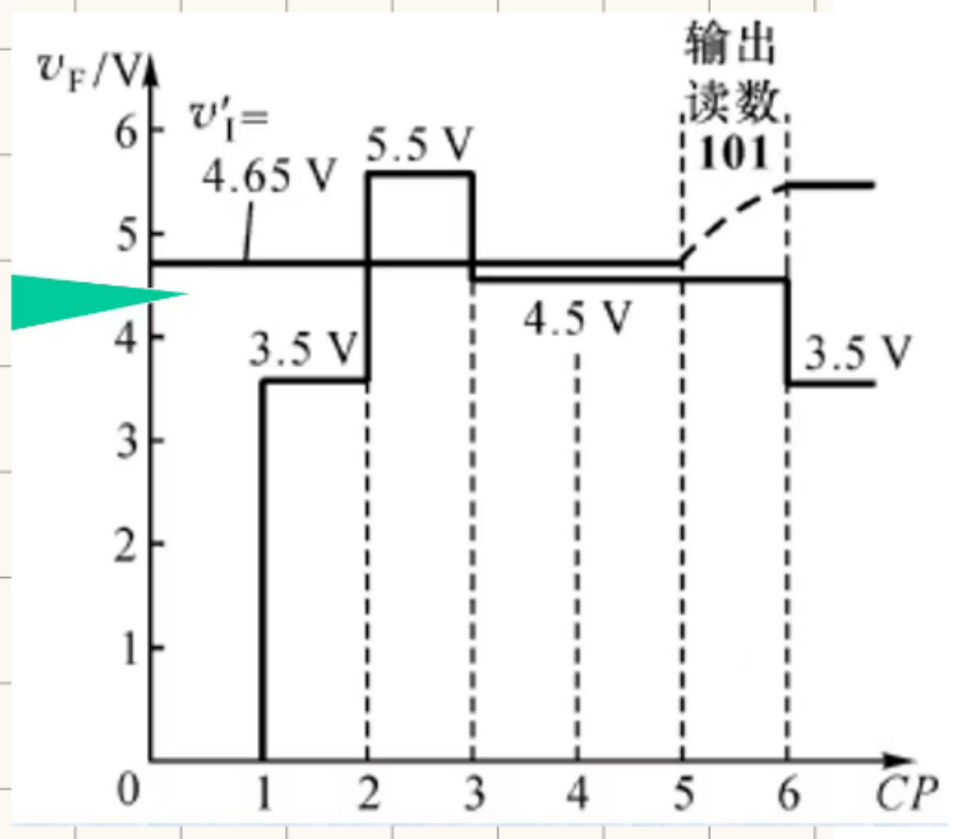
$v_I = 5.4V$时，量化值$v_I^* = 5V$.



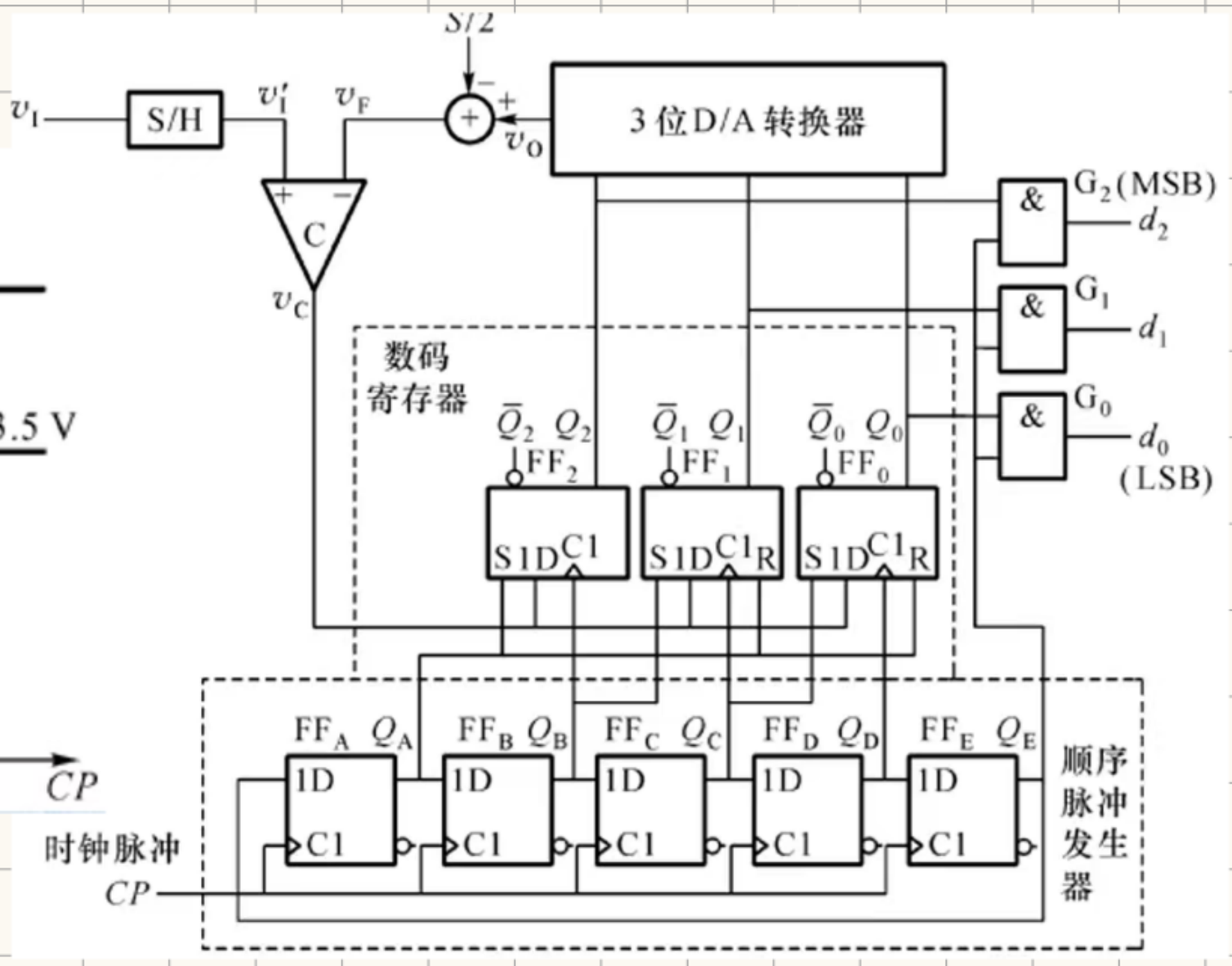
四舍五入量化转移特性

量化单位 $S = \frac{V_{im}}{2^n - \frac{1}{2}} = \frac{2V_{im}}{2^{n+1} - 1}$

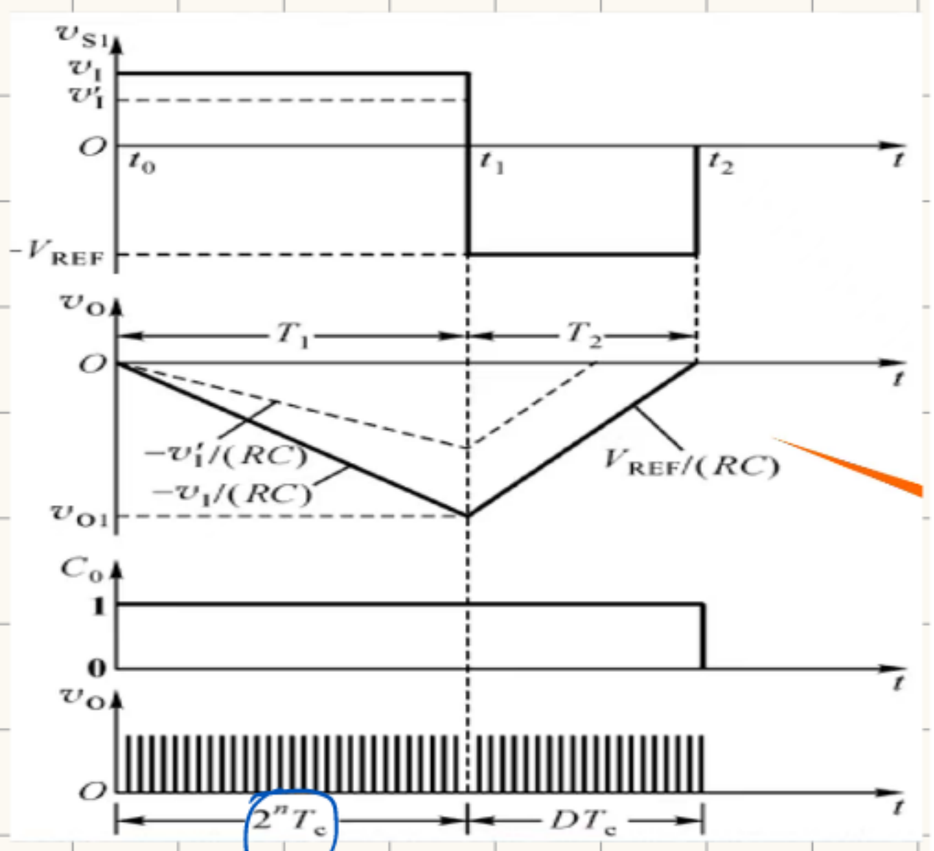
8.2.2 逐次逼近型 A/D 转换型

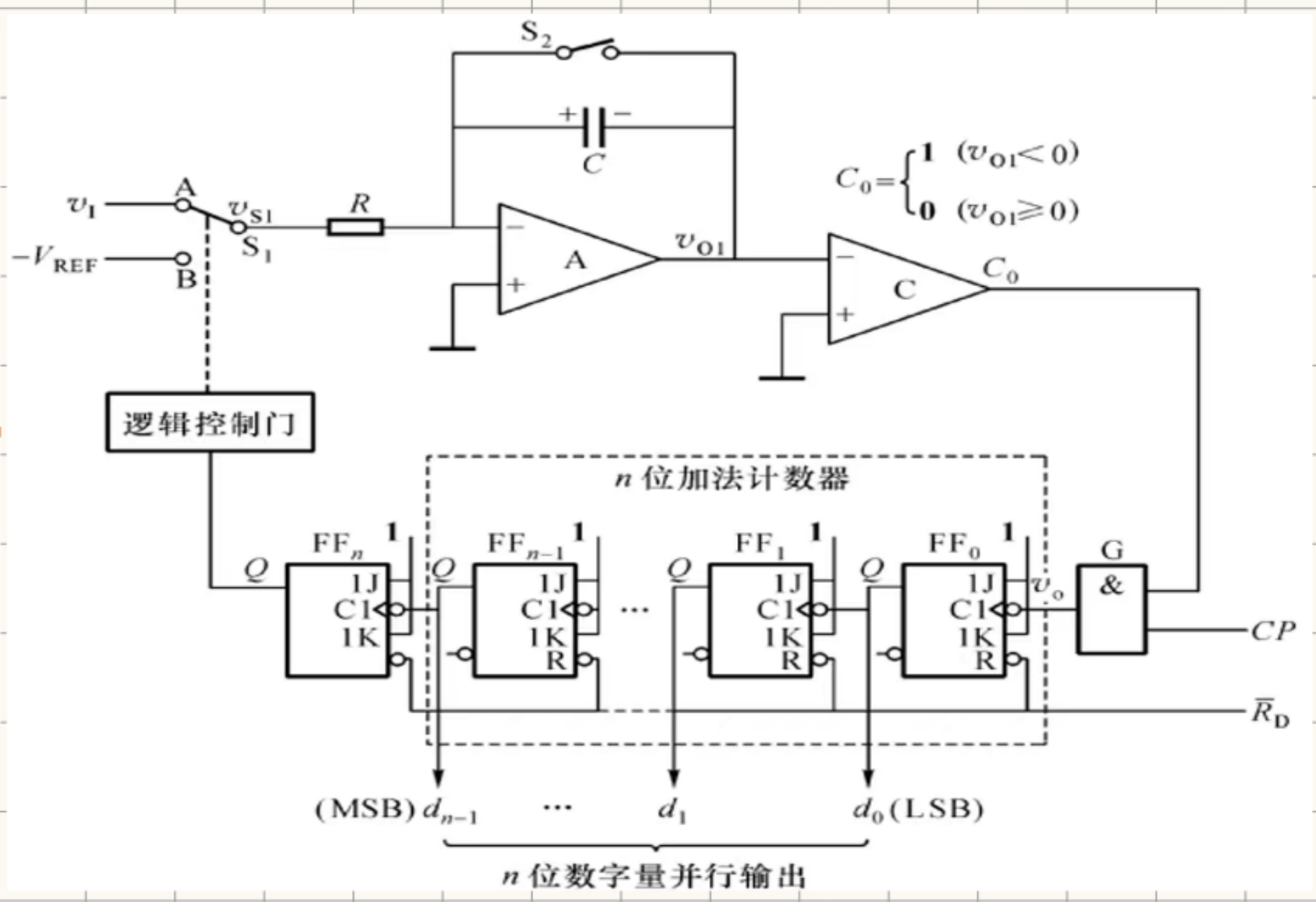


(n+2) 个时钟周期



8.2.3 双积分型 A/D 转化器





模拟信号(要测量的)固定时间

D 是用已知 V_{REF} 得到的时间

$$D=\frac{T_2}{T_{cp}}=\frac{V_I}{V_{REF}}\cdot 2^n$$

与输入电压 V_I 成正比

$2\times 2^n-1$ 个时钟周期

8.2.4 A/D 转化器的主要技术指标

(1) 位数 n, 满刻划 A V

$\Rightarrow$ 分辨率 $\frac{A}{2^n-1}$

(2) 精度误差